Incompressible Box Transport and Finite Computation

OpenAI (with community contributions)

November 6, 2026

Abstract

We realize finite positive diagonal affine maps of determinant one by effective smooth incompressible flows on neighborhoods of entire closed rational solid boxes. The source and target families are each disjoint, but may overlap each other. The construction uses localized curls, evacuation to storage and obstacle detours. A balanced three-stack recorder then assigns every machine and finite input a smooth Navier–Stokes force with one compact spatial support, periodic after a loading interval, at any fixed positive computable viscosity. The fluid starts at rest, and one fixed particle enters one fixed open cube exactly when the machine halts. Further constructions give fixed torus charts, periodicity from time zero, alternative history guards and bounded or slab observers. Onto slow clocks yield separate decaying forces. Each construction includes an all-time observation proof, effective derivative bounds and a stated pressure comparison class.

1 Introduction

A finite instruction can prescribe a volume-preserving affine map on a box. A fluid realization must do more: it must move every point of that box, preserve incompressibility outside it, and leave the other instructions available. Source boxes and destination boxes can overlap. Moreover, a particle used to report the answer must avoid the observation region throughout every nonhalting transition. We study these requirements together.

Our geometric result is Theorem 3.4. It extends any finite collection of positive diagonal determinant-one maps between separately disjoint rational solid boxes to the time-one map of an effective smooth compactly supported incompressible velocity. The maps hold on neighborhoods of the entire closed boxes. The proof separates centers, changes shapes, evacuates the sources to storage, and delivers boxes along paths avoiding everything still occupied. Positive margins make the construction effective and permit additional observation guards.

The first computational application, Theorem 4.1, uses all three coordinates for symbolic data. Three stacks retain a tape and its execution history. Every local instruction preserves the total length of the prefixes it replaces, so its affine map has determinant one. A fixed particle, initially in a fluid at rest, enters the fixed cube $(-1,2)^3$ exactly when the encoded machine halts. The force is smooth, compactly supported in space and periodic after a one-unit loading interval. Its prescription uses the finite rule table, not the executed computation. This application completes the main construction before we vary initialization, history guards and observers.

The passage from a symbolic instruction to a fluid observation begins with retained history, which makes the instruction injective on its entire domain of tapes, not just along the chosen computation. A rule specifies finite prefixes and leaves their infinite tails free; gapped radix expansions place the codes in separately disjoint closed source and target boxes. Prefix replacement extends to an affine map of each whole box, which smooth incompressible transport then realizes. An additional

geometric bound must exclude unintended visits to the observer between successive rule times. Finally, the prescribed velocity determines a body force, and an energy comparison identifies the resulting solution in the declared class. Each later variant changes one or more of these steps while retaining this separation of tasks.

The remaining constructions answer more specific questions. Two tape coordinates can be thickened into solid boxes, or their planar scale changes can be compensated in the normal direction. A temporary mark, a persistent mark, an explicit frontier and a boundary between occupied and unused history give different partial instruction maps. We prove their inverses on their full domains because this is what separates the target boxes. A bounded observation box permits horizontal motion above it; a slab needs a coordinate bound throughout that motion. Storage and obstacle detours supply further alternatives. Loading once gives eventual periodicity, whereas placing the initial code at the fixed label or repeating a loading branch gives periodicity from time zero. Onto slow clocks give separate decaying forces with derivative norms square-integrable in time.

1.1 Prior work and the role of forcing

The computational starting point is Turing's finite machine model [38]. Landauer's discussion of retained intermediate information [19, Section 3] and Bennett's explicit history recording [4, Eqs. (9)–(11)] explain how an irreversible computation can be embedded in an invertible mechanism. We use that principle, but prove the finite guarded tables and their complete inverses here. No complexity bound or history-erasure procedure from those works is asserted for these tables.

Moore's generalized shifts turn symbolic updates into affine operations on positional expansions [23, 24, 16, 36, 5]. Our gapped expansions serve the same purpose. Moore also realizes invertible generalized shifts by smooth planar diffeomorphisms and three-dimensional flows [24, Section 6, Theorem 12 and its corollary]. Cardona, Miranda, Peralta-Salas and Presas extend the affine maps on neighborhoods of separated planar blocks to an area-preserving disk diffeomorphism, using contraction, transport, expansion and Moser's area correction [26] [9, Proposition 5.1]; see also [17, 3] for volume-preserving isotopy extension by localized potentials. This is a geometric predecessor of the routing problem here. Our local curl formulas implement positive diagonal determinant-one motions in three dimensions directly; the proof supplies effective support and neighborhood bounds and controls observation at intermediate times.

Computational universality has been established for stationary Euler flows with adapted geometry [9, 6, 10, 8]. Cardona, Miranda and Peralta-Salas also construct universal Euclidean Beltrami flows [13], with infinite energy, and robust simulations of tape-bounded machines by unforced Navier–Stokes flows on a flat torus [7, arXiv version 3, Theorem 1 and Remark 29]. Dyhr, González-Prieto, Miranda and Peralta-Salas obtain stationary unforced Navier–Stokes universality using an adapted metric and the Hodge Laplacian [12, Theorem A]; for averaged Navier–Stokes circuit blowup see [37].

Here the ambient metric is fixed and flat, the viscosity is prescribed, and the initial velocity is zero. The machine is placed in an external body force. Given an explicit solenoidal velocity U, the identity $f = U_t + (U \cdot \nabla)U - \nu\Delta U$ supplies that force. The mathematical work is the effective construction of U and its all-time observation, followed by uniqueness in the stated comparison class. We do not assert a regularity theorem for arbitrary fluid data.

The companion [28] gives an entry construction using solenoidal shears, and [29] treats prefix instructions and other planar or normal-compensated realizations. They explain adjacent geometric choices; the proofs of the present full-box mechanisms are given here. The final input-offset application explicitly imports the planar processor and spatial clock of [30], with the exact parameters and conclusions stated in Section 15.2. The companion [31] detects computation from an integral of

the velocity field on $\mathbb{R}^2 \times \mathbb{T}$ via an advection–diffusion estimate. The end of Section 8 compares that event with the path of one material particle studied here.

1.2 Equation, effectiveness and reading path

Throughout $\nu > 0$ is fixed, $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and the fluid equation on $\mathbb{R}^3$ or the flat unit torus is

$$u_t + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \qquad \operatorname{div} u = 0, \qquad u(0) = 0. \tag{1}$$

A material label a has path $\dot{\Phi}_t(a) = u(t, \Phi_t(a))$, $\Phi_0(a) = a$. We also write $X(t; a) = \Phi_t(a)$. The smooth constructed velocities have global material flows. On the torus pressure is normalized to have mean zero. Whole-space pressure assumptions are specified in Section 2 and in the individual statements. Unless explicitly projected, the force is a general body force and the constructed pressure is zero.

An effective field is a finite program that evaluates every requested mixed derivative at computably supplied arguments to any prescribed rational error and computes the bounds used in the proof. The finite machine and its finite input determine the coefficients, cutoffs and routing schedule. Evaluating the field cannot first compute the selected machine run. All algorithmic statements concern computable ν; the same formulas are valid for arbitrary positive ν, with evaluation relative to that parameter [32, 40]. The encoded trajectories are exact real trajectories. No uniform precision tolerance for an unbounded computation is asserted.

Section 2 states the analytic comparison tools. Section 3 proves the box motions, and Section 4 gives the balanced-stack application. Sections 5 and 6 establish reusable recorders and full closed-rectangle coding; Section 7 supplies the common interpolation formulas. The subsequent constructions can be read according to the requirement being changed:

Choice	Where it is developed
Initialization and time periodicity	Section 8 compares a separate loader, direct placement of the initial code, and a loading branch repeated in every period. An untargeted start square permits the last choice.
History domains	Sections 10 and 11 distinguish persistent marks, fresh history and neighboring-cell guards. Their inverses on arbitrary allowed tapes determine the separated image boxes.
Observer geometry	Section 12 treats bounded boxes, slabs and half-spaces. Sections 13 and 14 develop storage, detours and fixed torus charts with the corresponding intermediate-time bounds.
Decay and stationarity	The logarithmic clock in Section 9 produces separate decaying forces. Section 15 treats further profile choices, including the imported input-offset processor.

The five whole-space comparison classes are stated in Lemma 2.2. Each alternative fixes its own label, observer, support, pressure class and time dependence; these are separate constructions rather than simultaneous guarantees for one force.

2 Analytic classes and changes of physical coordinates

The geometric constructions prescribe a smooth velocity first. This section records exactly which other solutions are excluded by the energy argument. Pressure assumptions are kept separate:

continuity in a pressure norm, a norm at each individual time, and an integrable pressure gradient are different conditions.

Write $\mathcal{F}_\nu[U] = U_t + (U\cdot\nabla)U - \nu\Delta U$. All noncompact results below use $\mathbb{R}^3$; their velocities and forces have one fixed compact spatial support unless the individual statement says otherwise. This property concerns the constructed solution, not a competing solution. On $[0,T]$, write $CH^k = C([0,T];H^k(\mathbb{R}^3))$ and $C^1L^2 = C^1([0,T];L^2(\mathbb{R}^3))$. Spatial constants in pressure may depend on time.

Lemma 2.1 (The pressure term without a pressure norm)**.** *If $w, g \in L^2(\mathbb{R}^3;\mathbb{R}^3)$ satisfy $\operatorname{div} w = 0$ and $\operatorname{curl} g = 0$ in distributions, then $\langle w, g\rangle_{L^2} = 0$. In particular, any distributional gradient $g = \nabla p \in L^2$ has zero pairing with w, without an assumption that $p \in L^2$.*

Proof. An L^2 field is a tempered distribution. The differential identities, initially tested on compactly supported smooth functions, extend to Schwartz tests by multiplying by cutoffs tending to one and using L^2 convergence of the tests and their first derivatives. Fourier transformation gives $\xi\cdot\widehat{w} = 0$ and $\xi_j\widehat{g}_k - \xi_k\widehat{g}_j = 0$. These are identities of locally integrable functions, so they hold almost everywhere. Away from $\xi = 0$, $\widehat{g}$ is parallel to ξ and $\widehat{w}$ is perpendicular to it. Their Hermitian product is zero. The exceptional point has measure zero; Plancherel proves the assertion; compare the classical Helmholtz–Weyl orthogonal decomposition [39, 41, 11, 14]. □

Lemma 2.2 (Residual realization and separate comparison classes)**.** *Let $\nu > 0$ and let U be a prescribed smooth divergence-free velocity on $[0,\infty)\times\mathbb{R}^3$, with $U(0) = 0$. On each finite interval, assume $U \in CH^2 \cap C^1L^2$ and $U, \nabla U$ are bounded. Put $f = \mathcal{F}_\nu[U]$. Then $(U, 0)$ is a solution. Its velocity is unique in each of the following separately stated comparison classes, with the same force and initial data:*

- *(i) Smooth classical $v \in CH^2 \cap C^1L^2$, with bounded v, and a pressure representative $p \in CH^1$.*
- *(ii) Smooth classical $v \in CH^2 \cap C^1L^2$, with bounded $v, \nabla v$, and an H^1 pressure representative at each time, with no required time continuity in that pressure norm.*
- *(iii) Strong solutions with $v \in CH^4 \cap C^1H^2$ and distributional $\nabla p \in CL^2$.*
- *(iv) Smooth classical $v \in CH^2 \cap C^1L^2$, with bounded $v, \nabla v$, and a pressure representative $p \in CL^2$.*
- *(v) Smooth classical $v \in CH^2 \cap C^1L^2$, with bounded $v, \nabla v$, without an additional integrability or normalization condition on pressure.*

The pressure gradient is unique. An L^2 pressure representative, when specified, is zero; otherwise pressure is determined up to a function of time. The reference material flow is a smooth diffeomorphism at every finite time and exists for every finite forward time.

On a flat torus $\mathbb{T}^3_L$, the same conclusion holds in the classical class where v, v_t, spatial derivatives through order two, and $p, \nabla p$ are continuous on finite closed cylinders, with periodic mean-zero pressure. No Sobolev conditions at infinity are involved in that assertion.

Proof. Substitution proves existence of the prescribed solution. Set $w = v - U$. The difference equation is

$$w_t + (v\cdot\nabla)w + (w\cdot\nabla)U = \nu\Delta w - \nabla p, \qquad \operatorname{div} w = 0.$$

The stated velocity regularity implies $w \in C^1L^2 \cap CH^2$, so $\frac{d}{dt}\|w\|_2^2 = 2\langle w, w_t\rangle$ at each time. In case (iii), the stronger assumptions imply these inclusions. They also imply bounded velocity and spatial derivatives through order two: for $j \le 2$, Cauchy–Schwarz in Fourier space uses the finite integral $\int_{\mathbb{R}^3} |\xi|^{2j}(1+|\xi|^2)^{-4}\,d\xi$.

For the transport term, insert a cutoff χ_R that equals one on the ball of radius R, is supported in the ball of radius $2R$, and satisfies $|\nabla\chi_R| \le C/R$. Its boundary error is at most $CR^{-1}\|v\|_\infty\|w\|_2^2$, which tends to zero. Diffusion is integrated by parts in H^2; alternatively its cutoff error is bounded by $CR^{-1}\|w\|_2\|\nabla w\|_2$. The deformation term is bounded by $\|\nabla U\|_{\infty,\mathrm{op}}\|w\|_2^2$.

We justify the pressure pairing according to the stated alternatives. For (i) and (ii), at each fixed time approximate the chosen H^1 pressure by compactly supported smooth functions; distributional incompressibility and $w \in L^2$ give $\langle \nabla p, w\rangle = 0$. Thus no continuity of p in time is used in (ii). For (iv), integration by parts on the cutoff gives a pressure boundary term bounded by $CR^{-1}\|p\|_2\|w\|_2$, also tending to zero. For (iii), apply Lemma 2.1 directly. Finally, in (v) the equation itself gives at each time

$$\nabla p = f - v_t - (v\cdot\nabla)v + \nu\Delta v \in L^2, \qquad \|(v\cdot\nabla)v\|_2 \le \|v\|_\infty\|\nabla v\|_2.$$

Here $f \in L^2$ follows from the identical estimate for U and its stated time and space regularity. The smooth pressure gradient is curl free, so Lemma 2.1 again cancels the term. This does not choose an L^2 representative of the pressure.

In every case the resulting pointwise energy inequality is

$$\frac{1}{2}\frac{d}{dt}\|w\|_2^2 + \nu\|\nabla w\|_2^2 \le \|\nabla U\|_{\infty,\mathrm{op}}\|w\|_2^2.$$

The coefficient is bounded on each finite interval and $w(0) = 0$, so an integrating factor gives $w = 0$. The pressure has already disappeared before time integration; the proof imposes no unlisted temporal pressure norm. Substitution then gives $\nabla p = 0$. A spatially constant L^2 function on $\mathbb{R}^3$ is zero, which proves the normalization assertion.

On the torus every integration by parts is periodic and has no boundary term. The specified classical regularity justifies norm differentiation, diffusion, and pressure cancellation. The same energy inequality gives uniqueness and the mean-zero normalization fixes pressure. In either domain, the smooth reference velocity and its bounded spatial derivative give unique material trajectories; bounded speed prevents finite-time escape. Solving backwards on a finite interval gives the inverse flow, and smooth ODE dependence gives its smoothness. □

Cases (i)–(v) identify the precise hypotheses used by the applications; their presence in a common lemma does not replace a theorem's declared comparison class by a different one. In particular, a pressure norm is not silently imposed in a gradient-only application. The proof is the classical energy comparison and integrating-factor uniqueness argument for prescribed smooth solutions [20, Section 18] [15, 33, 34, 35, 18], with its noncompact pressure pairings made explicit.

For the flow Φ_t, differentiation of its trajectory equation gives

$$U(t) = (\partial_t\Phi_t)\circ\Phi_t^{-1}, \qquad \partial_{tt}\Phi_t(a) = (f - \nabla p + \nu\Delta U)(t,\Phi_t(a)).$$

These identities describe the material form of the same constructed solution; they do not add an independent existence claim.

Lemma 2.3 (Physical scaling)**.** *For $a, b > 0$, $x = x_0 + ay$, and $s = bt$, let $U(t,x) = abV(bt, (x - x_0)/a)$. Then trajectories are transformed by this change of variables, incompressibility is preserved, and at the prescribed physical viscosity ν,*

$$\mathcal{F}_\nu[U](t,x) = ab^2\left[V_s + (V\cdot\nabla_y)V - \frac{\nu}{a^2 b}\Delta_y V\right](bt, (x - x_0)/a).$$

Proof. The trajectory equation gives $\dot{x} = abV$; differentiating in space gives $\operatorname{div}_x U = b\operatorname{div}_y V$. The three residual terms scale by ab^2, ab^2, and b/a, respectively, proving the formula. □

Thus a side-10 torus is kept as a physical side-10 torus. If a chart is rescaled, the residual is recomputed with the fixed ν; the equation is not identified with a differently normalized viscosity.

Proposition 2.4 (Projection on a flat torus)**.** *Let g be a smooth force on $[0,\infty)\times\mathbb{T}^3_L$ with zero spatial mean at each time. There is a unique smooth mean-zero potential ϕ satisfying $\Delta\phi=\operatorname{div} g$. Put $Qg=\nabla\phi$ and $Pg=g-Qg$. Then Pg is solenoidal and mean zero, and replacing (g,p) by $(Pg,p-\phi)$ preserves the velocity and the pressure normalization [11, 14].*

Use normalized spatial measure, frequencies $\xi_k=2\pi k/L$, and norms

$$\|G\|_{H^s}^2=\sum_{k\in\mathbb{Z}^3}(1+|\xi_k|^2)^s|\widehat{G}(k)|^2,\qquad \|G\|_{C^r}=\max_{|\beta|\le r}\|\partial_x^\beta G\|_\infty.$$

For all integers $h,r\ge 0$, at each time,

$$\begin{aligned}\max\{\|\partial_t^h Pg\|_{C^r},\|\partial_t^h Qg\|_{C^r}\}&\le S_L\|\partial_t^h g\|_{H^{r+2}},\\ \|\partial_t^h\phi\|_{C^{r+1}}&\le S_L d_L\|\partial_t^h g\|_{H^{r+2}},\end{aligned}\tag{2}$$

where $S_L=(1+52(L/(2\pi))^4)^{1/2}$ and $d_L=(1+(L/(2\pi))^2)^{1/2}$.

Uniform bounds, temporal L^2 bounds on spatial suprema, and qualitative uniform decay of all mixed derivatives pass from g to Pg,Qg,ϕ. A quantitative envelope for time order h and output spatial order r passes to Pg,Qg through order r and to ϕ through order $r+1$ when the input derivatives through spatial order $r+2$ satisfy that envelope. The constants may depend on the derivative order. Periodicity and stationarity, including either property on a time tail, are preserved on their respective intervals.

For computable $L>0$, effective mixed-derivative evaluation and effective derivative bounds for g on finite time slabs give effective evaluation and such bounds for Pg,Qg,ϕ. Corresponding supplied global input bounds give effective global bounds in each assertion above.

Proof. For $k\neq 0$ define

$$\widehat{\phi}(t,k)=-\frac{i\xi_k\cdot\widehat{g}(t,k)}{|\xi_k|^2},\qquad \widehat{\phi}(t,0)=0.$$

The nonzero-frequency symbols of Q and P are complementary orthogonal projections. Their H^s operator norms are at most one, while the multiplier for ϕ has norm at most d_L from H^s to H^{s+1}. For $|\beta|\le r$, Cauchy–Schwarz bounds the absolute sum of the Fourier series for $\partial_x^\beta G$ by

$$\left(\sum_{k\in\mathbb{Z}^3}(1+|\xi_k|^2)^{-2}\right)^{1/2}\|G\|_{H^{r+2}}.$$

The shell $|k|_\infty=n\ge 1$ contains $24n^2+2\le 26n^2$ points. Since $\sum_{n\ge1}n^{-2}\le 2$, the factor is at most S_L. For the tail $|k|_\infty>N\ge1$, it is at most

$$\Big(26(L/(2\pi))^4/N\Big)^{1/2}.\tag{3}$$

Applying these estimates to the multipliers proves (2), including the estimate for ϕ with one additional spatial derivative. On compact time slabs these tails converge uniformly for every mixed derivative. Thus the series define smooth fields and may be differentiated term by term. The Fourier identities give the Poisson equation, uniqueness of its mean-zero solution, and $\operatorname{div} Pg=0$. Also $-\nabla(p-\phi)+Pg=-\nabla p+g$.

For every integer $n \geq 0$, the multinomial expansion and Parseval give

$$\|G\|_{H^n}^2 \leq \sum_{|\beta| \leq n} \frac{n!}{(n-|\beta|)!\,\beta!} \|\partial_x^\beta G\|_\infty^2. \tag{4}$$

Use $n = r + 2$ and $G = \partial_t^h g$. This finite sum proves the uniform bounds and qualitative decay assertions. If its input suprema are bounded by $C_{h,\beta} a_h(t)$, it gives the same envelope a_h for the output, with the constants specified by (2)–(4). Integrating the squared estimates proves the temporal L^2 assertion from the corresponding input norms. The operators act at each time, so they preserve the stated temporal symmetries.

For effective evaluation, the supplied derivative bounds and (3) give a computable Fourier truncation on each finite time slab. The finitely many coefficient integrals are computed by Riemann sums. After scaling the torus to $[0,1]^3$, the integrand for $\widehat{G}(k)$ has Lipschitz bound

$$L\sqrt{3} \max_j \|\partial_{x_j} G\|_\infty + 2\pi|k| \|G\|_\infty.$$

This gives an effective quadrature error and hence evaluation of every mixed derivative of all three fields. The displayed finite sums also compute output bounds from the supplied input bounds. □

The estimates apply to the pressure shift ϕ; in the constructions with original pressure zero, the projected pressure is $-\phi$. Projection generally changes pressure. It supplies no compact-support conclusion on Euclidean space and is distinct from the direct zero-pressure solenoidal shear construction.

Effective flat profiles. For later cutoff formulas put

$$\rho(s) = \begin{cases} e^{-1/s}, & s > 0, \\ 0, & s \leq 0, \end{cases} \qquad \sigma(s) = \frac{\rho(s)}{\rho(s) + \rho(1-s)}. \tag{5}$$

Positive-side derivatives of ρ are polynomials in $1/s$ times $e^{-1/s}$ and extend flatly at zero. The estimate $v^m e^{-v} \leq (m+k)! v^{-k}$ gives effective error bounds near the seam, and $\rho(s) + \rho(1-s) \geq e^{-2}$. Products, translations, rescalings, and derivatives therefore yield effective cutoffs with all requested derivative bounds. Periodization uses zero collars; at approximate real arguments a finite superset of possible translates is evaluated. No real equality test at a seam or execution of the encoded computation is used.

3 Transporting complete boxes

We allow an instruction to change all three side lengths while preserving their product. The geometric question is concrete: given finitely many disjoint closed boxes and prescribed determinant-one affine maps onto another disjoint family, can one carry out every map by a smooth incompressible motion? The two families may overlap each other. Consequently a destination cannot be occupied until every source that obstructs it has been removed.

We first extend an explicitly separated motion to a velocity field. We then give two ways to produce that motion. Separate horizontal projections permit a simple lift and transfer. For general solid boxes we use parking and detours around the other occupied boxes. Figure 1 contrasts the height separation with the evacuation order that makes overlapping source and target families harmless.

3.1 An affine motion and its curl extension

Throughout, a smooth switch is obtained by translating and rescaling σ in (5), with its transition strictly inside its allotted interval. Thus successive motions are stationary on neighborhoods of their joining times.

Lemma 3.1 (A velocity on a neighborhood of each moving box)**.** *Let $K_i(t) = c_i(t) + D_i(t)K_i^0$, $1 \le i \le N$, where K_i^0 is an axis-parallel box, possibly with zero side lengths, and $D_i(t)$ is positive diagonal with determinant one. Suppose the centers and scales are smooth and constant near the ends of a compact time interval. For the effective conclusion, assume computable endpoints for the initial boxes K_i^0, effective evaluation of the centers' and scales' derivatives, bounds for those derivatives, and positive lower bounds for every scale on that interval. If the coordinatewise 2η enlargements of the moving boxes are pairwise disjoint and bounded, for a known $\eta > 0$, a compactly supported smooth divergence-free field realizes these motions on open neighborhoods of the initial boxes. It vanishes near the time endpoints. When all enlarged boxes lie in an interior torus chart, the field extends to a mean-zero field on that torus. The field is effective under the effective-data assumptions above; smooth existence itself does not require computable data.*

Proof. Choose a smooth cutoff χ_i which is one on the η enlargement of $K_i(t)$ and supported in its 2η enlargement, using products of the effective switches. Put $y = x - c_i$ and $L_i = \dot{D}_i D_i^{-1}$; then $\operatorname{tr} L_i = 0$. Take the curl of

$$\chi_i\{\tfrac{1}{2}\dot{c}_i \times y + \tfrac{1}{3}(L_i y) \times y\}. \tag{6}$$

The identity $\nabla \times (G \times y) = 2G - y \operatorname{div} G + (y \cdot \nabla)G$ shows that this curl equals $\dot{c}_i + L_i y$ on the plateau; compare [17, 3] for extending local volume-preserving isotopies via cut-off potentials. Disjoint supports permit summation over i. The path $c_i(t) + D_i(t)D_i(t_0)^{-1}(x_0 - c_i(t_0))$ has exactly that derivative and remains in its prescribed box. ODE uniqueness proves the formula. On the compact time interval the normalized matrix norm has a finite bound M, computable under the effective-data assumptions; an initial perturbation less than $\eta/(2M)$ remains on the plateau. This proves the neighborhood claim, also for zero-width initial boxes. The potential is zero near time endpoints because the centers and scales are constant there. Its support is bounded. Within a torus chart extend the potentials by zero; periodic curls have zero integral. □

Proposition 3.2 (Fluid realization and a pressure refinement)**.** *Fix $\nu > 0$. A finite collection of the preceding motions, repeated with period one, or preceded by one finite loading interval, yields a smooth prescribed velocity U with $U(0) = 0$. Suppose its spatial support is contained in one compact set in $\mathbb{R}^3$, or its potentials are contained in interior torus charts. Then*

$$f = U_t + (U \cdot \nabla)U - \nu\Delta U \tag{7}$$

has the same support and time periodicity and bounded mixed derivatives of every order. The solution is $(U, 0)$, with the uniqueness and material-flow conclusions of Lemma 2.2. In $\mathbb{R}^3$ its uniqueness statement also holds when the competing pressure has an H^1 representative at each time, without requiring continuity into H^1, while the other velocity hypotheses of Section 2 are retained. On the torus U and f have zero mean. Under the effective-data assumptions of Lemma 3.1, the formula is $f_0 + \nu f_1$ with effective coefficients; evaluation is effective for computable ν, and relative to ν otherwise.

Proof. The residual identity and Lemma 2.2 give existence, uniqueness and the material flow. In the whole-space pressure refinement use case (ii): the reference field has compact support and smooth bounded mixed derivatives, hence belongs to $CH^2 \cap C^1L^2$ with bounded velocity and gradient;

the competing velocity and pointwise H^1 pressure are exactly that case's hypotheses. No pressure continuity in time is added. Integrating the residual on a torus uses $\int U = 0$, $\int \Delta U = 0$ and $\int (U \cdot \nabla)U = \int \operatorname{div}(U \otimes U) = 0$.

Each derivative of $e^{-1/s}$ for $s > 0$ is a polynomial in $1/s$ times $e^{-1/s}$. The estimate $v^m e^{-v} \leq (m+k)! v^{-k}$ supplies effective endpoint error bounds, and the denominator defining σ is at least e^{-2}. Under those effective-data assumptions all positive scales have effective positive lower bounds. Finite compositions, differentiation, and the logarithms of positive computable scales are therefore effective. At an approximate time, a coarse bound gives a finite superset of possible translates of the zero-extended unit template. This evaluates the periodic field without deciding whether time is an integer. None of these operations iterates the machine execution: the program contains the entire finite instruction list. □

3.2 Three geometric schedules

For later use we describe the exact hypotheses of the schedules. First suppose $P_i, Q_i \subset \mathbb{R}^2$ are finite families of closed rectangles, possibly with zero side lengths, separately pairwise disjoint, with affine maps

$$F_i(x) = q_i + \operatorname{diag}(\lambda_i, \lambda_i^{-1})(x - p_i), \qquad \lambda_i > 0,$$

where p_i, q_i are their centers, and require $F_i(P_i) = Q_i$. Their side lengths have a common finite bound. Source and target rectangles belonging to different families need not be disjoint. The smooth existence statements allow arbitrary real rectangle coordinates and scales. For the effective conclusions, assume rational rectangle endpoints and rational scales, together with computable chosen heights and vertical half-widths; then a rational side-length bound and positive gap margins can be computed.

Lemma 3.3 (Separated projections and parking)**.** *These reciprocal maps have the following smooth compactly supported unit-time realizations, stationary on time collars. They are effective under the rational-data assumptions just stated.*

1. *For the products $P_i \times [z_0 - h, z_0 + h]$, with common $h \geq 0$ and $z_0 \in \mathbb{R}$, lift all boxes with their source projections fixed, transfer them at sufficiently separated private heights, and lower them with their target projections fixed. The middle motion may use $g_i(s) = \lambda_i^s$ or $g_i(s) = 1 + s(\lambda_i - 1)$ and matrix $\operatorname{diag}(g_i, g_i^{-1}, 1)$, with the center interpolated linearly in s. Each horizontal width stays between its endpoint widths. In the linear case every point satisfies*

$$X_1(s) = (1 - s)X_1(0) + sX_1(1). \tag{8}$$

2. *Alternatively, choose disjoint parking rectangles away from both families. Move every source to its parking site before delivering any target, using an upward, horizontal, and downward translation. Reshape at the parking sites, then deliver by the same three translations. This uses $7N$ motions. The same schedule applies to assigned onto positive diagonal determinant-one maps between rational solid boxes whose centers lie in a common horizontal plane and whose source and target horizontal projections are separately disjoint. Require both horizontal widths during reshaping to stay inside their reserved parking rectangles and the travel height above the common plane to exceed twice the largest vertical half-width.*

Both constructions act on neighborhoods of the whole source boxes.

Proof. In the first construction choose padding smaller than a quarter of the within-family horizontal gaps and choose height differences larger than twice the padded vertical half-width. During lift and descent, horizontal separation suffices. During transfer, vertical separation suffices. The two choices of g_i are positive and monotone between 1 and λ_i; their reciprocals are monotone as well. Thus the width claim holds. The affine path is $c_i(s) + \operatorname{diag}(g_i, g_i^{-1}, 1)(X_0 - c_i(0))$. Its first coordinate gives (8) exactly for the linear choice. Lemma 3.1 now applies.

For the second construction choose padding smaller than the gaps in each union of source projections with reserved parking rectangles, and of target projections with those rectangles. In the evacuation pass, vertical motion is separated from uncollected sources and parked boxes. During delivery it is separated from already filled targets and boxes still parked. At travel height, the moving box is above every resting box. A shape change stays within its own reserved projection. These are the occupied sets at each actual time; source–target intersection causes no interference because all sources have left before delivery starts. Lemma 3.1 applies to each motion with every other box fixed. Finite concatenation gives the required neighborhood map. □

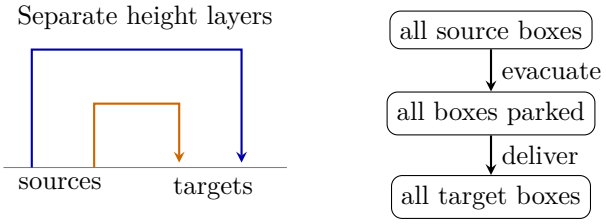


Figure 1: The two separation mechanisms in Lemma 3.3. Left: horizontal projections separate the lift and descent, and distinct heights separate the transfer; the second horizontal coordinate is suppressed. Right: a temporal schedule, not a spatial drawing. Every source is removed before any target is occupied, so overlap between the two families is harmless.

There is also a useful stream-function implementation of the first schedule. Suppose a_i, a_i' are the lower left corners of P_i, Q_i. At height z_i, put $d_i = a_i' - a_i$, $b_i = \log \lambda_i$ and $g_i(s) = a_i + s d_i$. On a neighborhood of the swept rectangle use

$$\xi_i(s) = g_i(s) + \operatorname{diag}(\lambda_i^s, \lambda_i^{-s})(\xi - a_i). \tag{9}$$

The Hamiltonian

$$H_i = d_{i1} y - d_{i2} x + b_i (x - g_{i1})(y - g_{i2})$$

generates this motion. Put $\widehat{H}_i = \chi_i H_i$, where χ_i is one near the whole sweep and supported in its reserved layer. Use

$$\dot{s}(\partial_y \widehat{H}_i, -\partial_x \widehat{H}_i, 0).$$

Disjoint height layers separate these cutoffs. Vertical phases use $\nabla \times (0, x\chi_i, 0)$, equal to $(0, 0, 1)$ on their column plateaus. If $M = \max_i(1, \lambda_i, \lambda_i^{-1})$ and the column plateau margins are δ, δ', an initial thickening smaller than $\min(\delta/(4M), \delta'/(4M), 1/(8M))$ stays on every plateau. Thus the affine formula also holds on open neighborhoods. For points of the original rectangle it has the additional bound

$$X_1(s) \geq \min(a_{i1}, a_{i1}'). \tag{10}$$

The bound follows because the first relative coordinate in (9) is nonnegative.

3.3 Solid boxes with overlapping projections

The preceding lift requires separate horizontal projections. We now remove that restriction; this is needed when all three coordinates encode independent stacks.

Theorem 3.4 (Expansion, obstacle detours, and delivery)**.** *Let $Q_i = a_i + \prod_{j=1}^{3}[-h_{ij}, h_{ij}]$ and $Q_i' = b_i + \prod_{j=1}^{3}[-k_{ij}, k_{ij}]$ be rational solid boxes, with positive half-widths, each family pairwise disjoint. Suppose $s_{ij} = k_{ij}/h_{ij}$ and $\prod_j s_{ij} = 1$. An effective compactly supported smooth solenoidal field, zero outside the time interval $[1/4, 3/4]$, has time-one map*

$$a_i + y \longmapsto b_i + \operatorname{diag}(s_{i1}, s_{i2}, s_{i3})y$$

on a neighborhood of each Q_i.

Proof. For an empty family take the zero field. Otherwise let $R = \max(1, h_{ij}, k_{ij})$. Choose a rational $\Lambda > 10$ so that centers in each of the lists (Λa_i) and (Λb_i) are separated by more than $20R$ in maximum norm. This is possible because distinct disjoint boxes have distinct centers. Let

$$K = \max(4\Lambda, |\Lambda a_{i1}|, |\Lambda b_{i1}|), \qquad d_i = (K + 30Ri, 0, 0).$$

The d_i are parking centers, separated from each other and both scaled families by at least $30R$ in their first coordinate.

First multiply all source centers by a common factor increasing from 1 to Λ, while keeping shapes fixed. Next change all shapes at those separated centers by factors s_{ij}^{θ}, $0 \le \theta \le 1$. Each half-width is the geometric interpolation $h_{ij}^{1-\theta} k_{ij}^{\theta} \le R$. Move the boxes one at a time from Λa_i to d_i, then, after all have been parked, from d_i to Λb_i. Finally shrink all target centers by a common factor from Λ to 1, with target shapes fixed.

Here is a finite construction of every single-box path. Around each stationary center place the closed cube of maximum-norm radius $4R$. These cubes are disjoint since stationary centers are separated by more than $20R$, and neither endpoint is in a cube. Start with the straight segment between the endpoints. Its intersections with the cubes are rational intervals, found by linear inequalities. Replace each portion inside a cube by a polygonal path on its boundary: join the entry point to a vertex of a containing face, use cube edges to a face containing the exit point, and join to that exit point. The cube is convex face by face, so these segments remain on its boundary; no other cube meets them. A tangency needs no replacement. The resulting rational polygonal path has distance at least $4R$ from every stationary center. Traverse its segments with flat switches, stopping to all orders at vertices. Its geometric image is unchanged, so smoothing the time parameter does not reduce clearance.

For a uniform localization margin, let g_a be the minimum over $i < j$ of

$$\max_{\ell}(|a_{i\ell} - a_{j\ell}| - h_{i\ell} - h_{j\ell}),$$

and define g_b similarly. These are positive; for a singleton list set both to R. During outward expansion a separating coordinate gap can only increase; during final contraction the target gap is its minimum. During deformation centers are more than $20R$ apart. During a single-box path the center distance is at least $4R$, leaving a coordinate gap at least $2R$. Hence padding smaller than $\frac{1}{8}\min(R, g_a, g_b)$ satisfies Lemma 3.1 throughout. Put all stages, and all straight segments, in consecutive subintervals of $[1/4, 3/4]$. Their flat switches give a smooth finite motion, with positive diagonal determinant-one linear parts. That lemma provides the velocity and the neighborhood assertion. All choices use rational inequalities and effective elementary functions, which also supply bounds for every derivative. □

4 A complete balanced three-stack realization

In a positional code with base B, replacing a prefix of length r by one of length s gives an affine map with slope B^{r-s}. Thus a three-stack instruction has determinant one when it preserves the sum of the three prefix lengths. A third stack supplies reversible history under this constraint: we first move a fresh blank from beyond a work-tape boundary onto the history stack, then replace that blank by the record.

Theorem 4.1 (Balanced three-stack realization)**.** *For every machine and finite input, there are effective coefficients f_0, f_1 such that $f = f_0 + \nu f_1$ on $\mathbb{R}^3$ has an explicit zero-data solution $(U, 0)$. Force and velocity have one compact spatial support and bounded mixed derivatives of every order, and are one-periodic after $t = 1$. At the fixed label $x_* = (4, 0, 0)$,*

$$\exists t \geq 0 : \quad X(t; x_*) \in (-1, 2)^3 \quad \Longleftrightarrow \quad \textit{the machine halts.}$$

For a fixed machine its repeated field is independent of the input. The comparison class is $u \in CH^2 \cap C^1L^2$, bounded $u, \nabla u$ on finite intervals, and pressure modulo constants in CH^1. The formula is valid for every fixed real $\nu > 0$, with effective evaluation when ν is computable and relative evaluation otherwise.

Proof. Normalize the finite machine to one halting state h, directing missing nonhalting instructions to it by unchanged-letter stay moves. This preserves initial halting as well. Let Γ be the work alphabet and $b \in \Gamma$ its blank. Use three infinite top-first stacks L, R, J [21, 22]. A rule replaces three finite prefixes, leaving their tails untouched. Introduce boundary $\star$, marked work letters $\bar{a}$, and controls A_q, S_q, T_q, D_q; only A_h is terminal. For each $q \neq h$, the preparatory rules are

$$(A_q; \varepsilon, a, \varepsilon) \to (S_q; \bar{a}, \varepsilon, \varepsilon), \tag{11}$$
$$(S_q; \varepsilon, c, \varepsilon) \to (S_q; c, \varepsilon, \varepsilon), \tag{12}$$
$$(S_q; \varepsilon, \star b, \varepsilon) \to (T_q; \varepsilon, \star, b), \tag{13}$$
$$(T_q; c, \varepsilon, \varepsilon) \to (T_q; \varepsilon, c, \varepsilon), \tag{14}$$
$$(T_q; \bar{a}, \varepsilon, \varepsilon) \to (D_q; \varepsilon, a, \varepsilon). \tag{15}$$

Here $a, c \in \Gamma$. For $\delta(q, a) = (q', d, \sigma)$ add

σ	source	target
0	$(D_q; \varepsilon, a, b)$	$(A_{q'}; \varepsilon, d, \kappa)$
-1	$(D_q; c, a, b)$	$(A_{q'}; \varepsilon, cd, \kappa) \quad (c \in \Gamma)$
1	$(D_q; \varepsilon, ac, b)$	$(A_{q'}; d, c, \kappa) \quad (c \in \Gamma)$
1	$(D_q; \varepsilon, a \star b, b)$	$(A_{q'}; d, b\star, \kappa)$.

(16)

Every individual row occurrence, including its choice of c, receives a distinct record symbol κ.

Full-domain separation and initialized behavior. Every rule preserves total prefix length: it is one in the transfer rows, two in the blank-supply row, and respectively two, three, three, four in the final rows. Source branches at A_q test the right top; at S_q the right top separates work letters from the boundary; at T_q the left top separates work and marked letters. At D_q the right top fixes the work instruction, with the next right symbol or left top separating its remaining branches. Target branches at S_q have distinct marked or ordinary left tops; at T_q distinct boundary or ordinary right tops; at D_q distinct restored a; and at an A control distinct history tops κ. This proves separation for arbitrary independent tails.

Initialize

$$(A_{q_0}; b^\infty, v \star b^\infty, b^\infty), \qquad v = \begin{cases} \omega, & \omega \neq \varepsilon, \\ b, & \omega = \varepsilon. \end{cases}$$

At a checkpoint, L lists cells strictly left of the head and $R = a_1 \cdots a_m \star b^\infty$, $m \geq 1$, lists the current cell and a finite right block; beyond it the tape is blank. J contains completed records, newest first. The first rule marks a_1 while moving it left; $m-1$ forward transfers expose $\star$. The blank row removes one filler blank beyond that boundary and pushes it on J, without changing the infinite blank filler. The reverse transfers and restoration of a_1 restore exactly the original two work stacks. The final rule replaces the fresh history blank by its record and performs the work instruction. A left move pops the next cell from L; a right move with $m \geq 2$ pops the current cell from R; when $m = 1$ the last row exposes a blank and retains a nonempty right block. Thus the invariant holds and one work step costs $1 + (m-1) + 1 + (m-1) + 1 + 1 = 2m + 2$ rules. An initially halted input needs none. The records recover the head displacements as well.

From balanced words to solid boxes. Give the enlarged alphabet Σ the odd digits $1, 3, \ldots, 2|\Sigma| - 1$ in base $B = 2|\Sigma| + 1$. For a finite word v put $e(v) = \sum_{j=1}^{|v|} d(v_j) B^{-j}$; for an infinite stack ξ put $e(\xi) = \sum_{j \geq 1} d(\xi_j) B^{-j}$. The prefix interval is $I(v) = e(v) + [0, B^{-|v|}]$, with $I(\varepsilon) = [0, 1]$. Every infinite tail satisfies

$$\frac{1}{B-1} \leq e(\xi) \leq \frac{2|\Sigma| - 1}{B-1} < 1,$$

so its code lies strictly inside each applicable prefix interval. Distinct incompatible prefixes have positive gaps, since digits differ by at least two while the remaining interval has width at most one unit in the differing place. Place control s at $o_s = (4i_s, 0, 0)$, with $i_{A_h} = 0$ and distinct positive integers for the others. The code of $(s; L, R, J)$ is $o_s + (e(L), e(R), e(J))$. A rule with input prefixes $\alpha_1, \alpha_2, \alpha_3$ and output prefixes $\alpha'_1, \alpha'_2, \alpha'_3$ gives boxes

$$Q_i = o_s + \prod_j I(\alpha_j), \qquad Q'_i = o_{s'} + \prod_j I(\alpha'_j).$$

Both families are separately disjoint by the preceding full-prefix separation. In control-relative coordinates their exact maps are

$$y_j = e(\alpha'_j) + B^{|\alpha_j| - |\alpha'_j|}(x_j - e(\alpha_j)). \tag{17}$$

The identity $e(\alpha\xi) = e(\alpha) + B^{-|\alpha|} e(\xi)$ proves their action on every code. Initialized stacks are eventually blank, with rational codes $e(vb^\infty) = e(v) + B^{-|v|} d(b)/(B-1)$. The affine determinant equals $B^{\sum |\alpha_j| - \sum |\alpha'_j|} = 1$, exactly by balance. Apply Theorem 3.4 to these full boxes.

Every intermediate time. All half-widths are at most $1/2$, so that lemma uses $R = 1$. Sources have first coordinate at least four, since no rule leaves A_h. Targets with nonhalt control have the same property. For such a branch the initial expansion and final contraction preserve first coordinate at least four. During deformation its center has first coordinate at least 4Λ. Every scaled halt target center has first coordinate at most Λ, whereas the endpoints of each nonhalt parking path have first coordinate at least 4Λ. Since $\Lambda + 4 < 4\Lambda$, the original segment cannot meet an obstacle cube around a halt target. Any encountered obstacle has first coordinate at least 4Λ, so its boundary detour remains at least $4\Lambda - 4$. Subtracting a half-width at most one still leaves first coordinate greater than two. Hence

$$\Phi_\tau(Q_i) \subset \{x_1 > 2\} \quad (0 \leq \tau \leq 1) \quad \text{for every branch with nonhalt target.} \tag{18}$$

This includes delivery after halt targets have already been filled.

Finally load $x_* = (4, 0, 0)$ along the straight segment to the rational initial code using the translation part of Lemma 3.1. It avoids $(-1, 2)^3$ for a nonhalt initial control. Repeat the box field from time one. Induction at integer times applies the exact full-box maps; the finite compiler cycles reach a code in $(0, 1)^3$ precisely upon halting. Equation (18) excludes every intermediate false visit. The residual proposition gives the force and uniqueness; compact support gives all finite-slab Sobolev conditions and uniformly bounded kinetic energy. Every choice was finite and effective, including the obstacle detours. The input enters only the first loading path, completing the proof. □

5 History records and complete symbolic domains

The balanced-stack theorem is complete. The remaining realizations use alternative recorders with tape data in two coordinates. A motion of boxes is invertible. To use it for an ordinary machine, we must retain whatever information an instruction erases. We use finite history records in the sense of reversible computation, but prove the local inverses below: reversibility of the initialized execution would not by itself separate the full boxes.

A work instruction is $r = (q, a) \mapsto (p_r, b_r, d_r)$, with $d_r \in \{-1, 0, 1\}$. The work tape is indexed by $\mathbb{Z}$, starts with head zero, and has only finitely many prescribed nonblank cells. Halting states have no outgoing work instruction. Missing instructions can be replaced by write-preserving stationary steps to a fresh halt state. Unless a construction explicitly adds a dummy initial step, an initially halting state is already a halt checkpoint. When a construction uses a single halting state h, merge the original halting states and redirect their incoming transitions; this preserves initial halting. In the recorder tables Q is the work-state set and Γ the work alphabet. Radix bases are chosen from the full cell alphabet, including all history and mark tracks.

5.1 A one-cell code and its exact inverse test

A partial one-cell table sends a control and a scanned letter to a new control, a written letter, and a displacement. Its two incoming conditions are: the new control determines the displacement, and the new control together with the written letter determines the entire old control and read letter [2, 25].

Lemma 5.1 (From one-cell tables to closed rectangles)**.** *Give an alphabet of size m digits separated by at least two, between 0 and $B - 2$. Put*

$$E(A) = \sum_{j \ge 1} d(A_j) B^{-j}, \qquad I(v) = \sum_{j=1}^{|v|} d(v_j) B^{-j} + [0, B^{-|v|}].$$

For a head-relative tape $(a_j)_{j \in \mathbb{Z}}$ use the coordinates

$$(x, y) = \big(E(a_{-1} a_{-2} \cdots), E(a_0 a_1 \cdots)\big).$$

After writing at cell zero, a displacement e makes the old cell e the new cell zero. A deterministic partial one-cell table satisfying the incoming conditions acts, after separate state translations, by a finite list of reciprocal affine maps on full closed rectangles. Each source family and each target family is separately disjoint. For a rule reading a, writing b, and moving by e, the branches are

e	*source*	*target*	(x', y')
1	$[0,1] \times I(a)$	$I(b) \times [0,1]$	$((d(b) + x)/B, By - d(a))$
0	$[0,1] \times I(a)$	$[0,1] \times I(b)$	$(x, y + (d(b) - d(a))/B)$
-1	$I(c) \times I(a)$	$[0,1] \times (d(c) + I(b))/B$	$(Bx - d(c), [d(c) + (d(b) + By - d(a))/B]/B)$.

(19)

The last row is split over every letter c. Each linear part is $\mathrm{diag}(B^{-e}, B^{e})$. *With odd digits* $1, 3, \ldots, 2m-1$ *and* $B \in \{2m+1, 2m+2\}$, *every code lies in the interior of its applicable rectangle, and unscaled within-family gaps are at least* B^{-2}.

Proof. The identity $E(aA) = (d(a) + E(A))/B$ gives all three formulas and their action on the independent tails. At the first differing digit, incompatible prefix intervals have a positive gap: the digit difference is at least two while the child interval has length one in that scale. Thus the source state, read letter, and any left split separate sources. At one target state the displacement is fixed. For a right or stationary move the written letter separates the relevant target coordinate; for a left move the pair (c, b) separates the two-letter prefix. The incoming inverse condition rules out duplicates. These arguments concern the complete intervals, including their endpoints. For odd digits the infinite tail lies between $1/(B-1)$ and $(2m-1)/(B-1) < 1$, proving interiority. The longest target prefix has length two, giving the asserted gap. Uniformly scaling state squares by s scales gaps by s and leaves the linear parts unchanged. □

5.2 Moving the work head before recording

We first place a frontier to the right of every possible work head. A temporary mark lets the history excursion return to the correct cell even though its length grows with the computation.

Lemma 5.2 (Move-first recorder)**.** *There is an effective one-cell table satisfying the incoming conditions whose checkpoint configurations simulate the work machine. After* n *work steps its frontier is at* $n+2$, *its records occupy* $2, \ldots, n+1$, *and its temporary marks are all zero. Each nonhalting checkpoint has a finite positive continuation to the next one, with no intervening checkpoint.*

Proof. Use letters (c, ℓ, j) with work letter c, history $\ell \in \{e, F\} \sqcup \{r\}$ and mark $j \in \{0, 1\}$. The controls are C_q, J_r, S_r, P_p, T_p. The complete partial table is

$$
\begin{array}{lll}
C_q : (a, \ell, 0) & \mapsto J_r : (b_r, \ell, 0), d_r & \ell \neq F, \\
J_r : (c, \ell, 0) & \mapsto S_r : (c, \ell, 1), 1 & \ell \neq F, \\
S_r : (c, \ell, 0) & \mapsto S_r : (c, \ell, 0), 1 & \ell \neq F, \\
S_r : (c, F, 0) & \mapsto P_{p_r} : (c, r, 0), 1, & \\
P_p : (c, e, 0) & \mapsto T_p : (c, F, 0), -1, & \\
T_p : (c, \ell, 0) & \mapsto T_p : (c, \ell, 0), -1 & \ell \neq F, \\
T_p : (c, \ell, 1) & \mapsto C_p : (c, \ell, 0), 0 & \ell \neq F.
\end{array}
$$

Initialize the frontier at two, all other history empty, and all marks zero. At checkpoint n the head satisfies $|h_n| \leq n$. The first row updates the work tape and moves to $h' \leq n+1$, strictly left of the frontier $f = n+2$. The second row marks h'. The forward scan writes r at f, puts a new frontier at $f+1$, and the backward scan stops exactly at the mark, which it clears. Its length is $2(f - h') + 4$, so it is finite and positive. This proves the invariant inductively, including the exact work update and halt equivalence.

For the full-domain inverse, r recovers the old work pair into J_r; mark one distinguishes entry into S_r from its mark-zero loop; a written record determines entry into P_p; a written F distinguishes entry into T_p from its loop; and entry into C_p uniquely restores mark one. Every destination has the single incoming displacement displayed in the table. No invariant on remote cells was used. □

5.3 Recording before the work-head displacement

The next table leaves its mark at the old work head. Its final step performs the work displacement; storing that displacement in the checkpoint control keeps incoming moves unambiguous.

Lemma 5.3 (Direction-recording scan)**.** *Let $D = \{-1, 0, 1\}$ and let O be either a singleton or $\{0, 1\}$. An effective one-cell table on $\Gamma \times (\{\bot, F\} \sqcup \mathcal{R}) \times \{0, 1\} \times O$, where $\mathcal{R} = \{(q, e, a) : q$ nonhalting$, e \in D, a \in \Gamma\}$, satisfies the incoming conditions. It preserves the O track in its physical cells. Starting with frontier at one, its checkpoint at work step n has frontier $n + 1$, records at $1, \ldots, n$, zero marks, and the correct work configuration. A step from head i takes $5 + 2(n - i)$ local rules.*

Proof. Use main controls $M(q, e)$, forward controls $G(r)$, and $A(p, d), B(p, d)$ for every work state p and $d \in D$, including pairs not produced by a work instruction. Copy the O component in each row. In the first three rows use every $r = (q, e, a) \in \mathcal{R}$ with $\delta(q, a) = (p, c, d)$. The final three rows apply independently to every $(p, d) \in Q \times D$, including pairs that are not outputs of any work instruction. The variables $v \in \Gamma$ and $h \in \{\bot, F\} \sqcup \mathcal{R}$ range over all letters subject to the displayed restrictions. The whole table is

$$\begin{array}{lll}
M(q,e) : (a,h,0) & \mapsto G(r) : (c,h,1), 1 & h \neq F, \\
G(r) : (v,h,0) & \mapsto G(r) : (v,h,0), 1 & h \neq F, \\
G(r) : (v,F,0) & \mapsto A(p,d) : (v,r,0), 1, & \\
A(p,d) : (v,\bot,0) & \mapsto B(p,d) : (v,F,0), -1, & \\
B(p,d) : (v,h,0) & \mapsto B(p,d) : (v,h,0), -1 & h \neq F, \\
B(p,d) : (v,h,1) & \mapsto M(p,d) : (v,h,0), d & h \neq F.
\end{array}$$

Starting at $M(q_0, 0)$, the first row writes and marks the old head. The scan reaches frontier $j = n+1$, stores its record, creates the next frontier, returns to the unique mark and makes the work move. The count is $2(j - i) + 3 = 5 + 2(n - i)$. Since $i \leq n < j$, every test is satisfied and both scans are finite.

Into $G(r)$ the written mark distinguishes entry and loop; r recovers the old state and work symbol. Into $A(p, d)$ the record recovers the old $G(r)$. Into $B(p, d)$ a written frontier distinguishes arrival and loop. Into $M(p, d)$ only clearing a mark is possible. All remaining components were copied. This proves the inverse on arbitrary tapes and the fixed incoming displacement. Deleting the passive O coordinate is an exact specialization of every rule, not merely an equivalence of initialized runs. When $O = \{0, 1\}$, a unique initial one at physical cell zero records the absolute origin. □

5.4 A frontier on the left and post-halt continuation

Lemma 5.4 (Two left-frontier tables)**.** *There are two effective incoming-separated tables: a stopping table, and a table which continues with idle work steps after halting. At checkpoint n their frontier is $f_n = -2 - n$ and their marks vanish. If the next work move ends at i', the next checkpoint is reached after $4 + 2(i' - f_n)$ local rules.*

Proof. For the stopping version let $J = (Q \setminus \{h\}) \times \Gamma$. For the continuing version take $J = Q \times \Gamma$ and add $\delta(h, a) = (h, a, 0)$. Use letters $(c, \eta, j) \in \Gamma \times (\{\bot, F\} \sqcup J) \times \{0, 1\}$, controls C_q, D_q, T_q, P_r, S_r, and precisely

$$\begin{array}{lll}
C_q : (a,\eta,0) \mapsto P_r : (b_r,\eta,0), d_r, & & \\
P_r : (c,\eta,0) \mapsto S_r : (c,\eta,1), -1 & \eta \neq F, & \\
S_r : (c,\eta,0) \mapsto S_r : (c,\eta,0), -1 & \eta \neq F, & \\
S_r : (c,F,0) \mapsto D_{p_r} : (c,r,0), -1, & & \\
D_q : (c,\bot,0) \mapsto T_q : (c,F,0), 1, & & \\
T_q : (c,\eta,0) \mapsto T_q : (c,\eta,0), 1 & \eta \neq F, & \\
T_q : (c,\eta,1) \mapsto C_q : (c,\eta,0), 0 & \eta \neq F. &
\end{array} \tag{20}$$

In the first row only, the stopping version requires $\eta \neq F$; the continuing version allows every η. These domains remain distinct even though initialized runs never use that difference.

Initialize F at -2, empty history elsewhere and zero marks. At checkpoint n, $i \geq -n = f_n + 2$. After the work update, $i' \geq f_n + 1$. Marking i' and scanning left therefore reaches f_n, records r, puts the next frontier at $f_n - 1$, and returns right to the mark without crossing the new frontier. Clearing it completes the next work checkpoint. The two finite traversals and four endpoint actions give the stated count. The continuing version has a next rule even after every halt checkpoint.

Into P_r the record index recovers the work pair. Into S_r mark one distinguishes entry from loop; into D_q the written record recovers the scan; into T_q a written F distinguishes entry from loop; into C_q there is only stationary unmarking. This proves the incoming conditions for both full partial domains. It also gives the configuration inverse: reverse the displacement, inspect the written cell, and invert its uniquely determined rule. □

5.5 A guarded update on two neighboring history cells

One may advance the frontier and write its record in a single local instruction. The resulting return loop needs a neighboring-cell guard. Without that guard, its image would overlap the logging image.

Lemma 5.5 (Three-cell guarded window)**.** *There is a finite partial injective map on relative tapes with controls* $\mathrm{Ready}(q, e)$, $\mathrm{Go}(r)$ *and* $\mathrm{Back}(q, d)$. *Here* $r = (q, e, a)$ *records an instruction, with* $\delta(q, a) = (p(r), b(r), d(r))$. *Use work, history and mark tracks* A, H, J. *At cursor position* i, *the Ready row has* $a = A_i$ *and* $r = (q, e, a)$. *The entire table is*

control	*test*	*action*
$\mathrm{Ready}(q, e)$	q *nonhalting*, $J_i = 0$	$A_i \leftarrow b(r),\ J_i \leftarrow 1,\ \mathrm{Go}(r), +1$
$\mathrm{Go}(r)$	$J_i = 0, H_i \neq P$	$\mathrm{Go}(r), +1$
$\mathrm{Go}(r)$	$J_i = 0, H_i = P, H_{i+1} = E$	$H_i \leftarrow r, H_{i+1} \leftarrow P,\ \mathrm{Back}(p(r), d(r)), -1$
$\mathrm{Back}(q, d)$	$J_i = 0, H_{i+1} \neq P$	$\mathrm{Back}(q, d), -1$
$\mathrm{Back}(q, d)$	$J_i = 1$	$J_i \leftarrow 0,\ \mathrm{Ready}(q, d), d.$

Unmentioned cells and tracks are copied. Initialized with frontier at $r_0 \in \{1, 2\}$, *other history* E, *zero marks and* $\mathrm{Ready}(q_0, 0)$, *the checkpoint at step* n *has frontier* $p_n = n + r_0$ *and correct head* h_n. *Its next work step uses exactly* $2(p_n - h_n) + 1 \leq 4n + 2r_0 + 1$ *local rules.*

Proof. The first rule writes and marks h_n. Since $p_n > h_n$, the forward scan reaches the frontier, moves it one cell right, and the return scan reaches the mark. Its neighboring-cell guard holds because the new frontier is $p_n + 1$. The last rule clears the mark and performs the work displacement. Counting the two traversals and endpoint actions gives $2(p_n - h_n) + 1$; induction gives $|h_n| \leq n$.

For injectivity on all relative tapes, into Go the old cursor is output offset -1: a mark there distinguishes entry from the loop, and r restores the overwritten work letter and old Ready control. Into Back the old cursor is offset $+1$. An output pointer at $+2$ identifies a log insertion, with record at $+1$ recovering r and old pair (P, E). A nonpointer at $+2$ identifies the guarded loop. Into Ready the corresponding Back control uniquely restores the cleared mark. These are all possibilities, proving the full inverse. □

Lemma 5.6 (Closed rectangles for a finite guarded window)**.** *Suppose a finite partial injective local tape map reads and writes only indices* $[-a, b - 1]$, *then shifts the cursor by* e, *where* $-a \leq e \leq b$. *Split its domain by every complete word in that window. With gapped radix coding the input prefix lengths are* (a, b) *and output lengths* $(a + e, b - e)$; *the affine branch has linear part* $\mathrm{diag}(B^{-e}, B^{e})$. *Both full rectangle families are separately disjoint.*

Proof. After the writes the new left prefix consists of old indices $e-1, e-2, \ldots, -a$, and the right prefix of $e, e+1, \ldots, b-1$. The two remaining tails are free and unchanged, so the image is the whole output cylinder. Distinct source cylinders have disjoint images by injectivity. If both pairs of output prefixes were compatible, a common extension would belong to both images; hence at least one coordinate has incompatible prefixes. The gapped interval argument then separates the complete closed rectangles. The prefix lengths give the two reciprocal scales directly. □

6 History conventions and branch subdivision

The general box motions now let us choose a recorder and an observer separately. We begin with the two frontier recorders in Section 5, which retain history in the sense of Bennett [4]. Later we retain the distinct neighboring-cell guards and persistent-mark tables needed by other realizations. A full-domain inverse, not merely agreement along an initialized run, is the criterion for sharing a compiler.

For these applications let A be the work alphabet with distinguished blank, Q the states, q_0 the initial state, and $H \subseteq Q$ the halting states. Write h_n for the work head after n steps and

$$\delta(q,a) = (q^+(r), b(r), d(r)), \qquad r = (q,a) \in \mathcal{R} = (Q \setminus H) \times A, \quad d(r) \in \{-1,0,1\}.$$

The initial head is zero and the tape has finitely specified nonblank cells. Replace missing instructions by unchanged-letter stay moves to a halt. If one halt square is needed, merge halting states and redirect their incoming instructions; no outgoing rule is changed. This preserves initial halting as well. An unreachable halt state can be added when needed. These are the same normalizations as in Section 5.

6.1 Frontier recorders and their notation

We use the move-first recorder of Lemma 5.2 with its original notation and complete partial domain. The delayed-head recorder of Lemma 5.3 records the work displacement until its final return step. Its table can also be initialized with the frontier one cell farther to the right.

Lemma 6.1 (Six-rule history recorder)**.** *For $r_0 = 1$ or $r_0 = 2$, there is a partial table with the two incoming properties of Lemma 5.1, frontier $r_0 + n$ at checkpoint n, and checkpoint control recording both the work state and the preceding work displacement. Its next checkpoint takes $2(r_0 + n - h_n) + 3$ instructions. The work head moves only on the last instruction.*

Proof. Specialize Lemma 5.3 to a singleton passive track O and delete that coordinate. Rename its controls $(M(q,e), G(\tau), A(q,d), B(q,d))$ as $(M_{q,e}, R_\tau, F_{q,d}, L_{q,d})$, and its frontier F as P. The full alphabet is $A \times (\{\bot, P\} \sqcup \mathcal{T}) \times \{0,1\}$, where $\mathcal{T} = (Q \setminus H) \times \{-1,0,1\} \times A$. Every row and guard is the same under this bijection, so both incoming properties hold on the entire partial domain, including unused controls. Only $M_{q,e}$ with $q \in H$ are halting controls.

Initialize $M_{q_0,0}$ with all marks zero, frontier at r_0, and empty history elsewhere. The case $r_0 = 1$ is the initialization already proved in the shared lemma. For either allowed r_0, checkpoint n has frontier $g = r_0 + n > h_n$. Mark the old head, scan right to g, write the record and install the new frontier at $g+1$, then return to the mark. All intervening history is nonfrontier, the new frontier cell is empty, and there is only one mark. Hence the same rows apply when $r_0 = 2$. There are $g - h_n - 1$ continuing right moves, $g - h_n$ continuing left moves, and four other instructions, giving $2(g - h_n) + 3$. The last row clears the mark and makes the work movement; the invariant is restored with frontier $g+1$. This proves finite completion, indefinite legality for a nonhalting run, and the stated halt equivalence, including initial halting. □

6.2 From symbolic cylinders to closed rectangles

Radix encodings turn a push or pop of a tape letter into an affine map; this is the generalized-shift viewpoint of Moore [23, 24]; for odd-digit gapped base-B stack encodings see also [16, 36]. The following calculation adds the separation of full closed rectangles needed for smooth localization.

Lemma 6.2 (Separated rectangle codes)**.** *Let a deterministic partial tape table on an alphabet Γ of size m have the two incoming properties of Lemma 5.1. Assign distinct odd digits $e(a) \in \{1, 3, \dots, 2m-1\}$ and choose an integer $B \geq 2m+1$. Its local rules admit effective positive diagonal affine maps of determinant one between finite families of closed planar rectangles. Sources are pairwise separated, as are targets. Both the minimal subdivision and the subdivision by both adjacent letters given below realize every tape in their full symbolic cylinders.*

Proof. Use Lemma 5.1 with digits $d(a) = e(a)$. Its digit hypotheses hold for every $B \geq 2m+1$. In the notation

$$C(a_1 a_2 \cdots) = \sum_{j \geq 1} e(a_j) B^{-j}, \quad h_a(v) = \frac{e(a) + v}{B}, \quad I_{a_1 \cdots a_l} = h_{a_1} \cdots h_{a_l}([0,1]),$$

the left stream starts immediately left of the head and the right stream starts at the head. Place state s in $k_s + a[0,1]^2$ with a common rational $a > 0$. The minimal branch maps are exactly (19); scaling both charts by a leaves $\operatorname{diag}(B^{-d}, B^d)$ unchanged.

For the additional full subdivision, restrict the right- and stay-move sources to $x \in I_\gamma$, for every full letter γ. Their targets become $I_{\beta\gamma} \times [0,1]$ and $I_\gamma \times I_\beta$, respectively, for written letter β. The already split left-move targets remain $[0,1] \times I_{\gamma\beta}$. At a fixed target state the incoming direction is fixed, and the pair (β, γ) distinguishes the branches in every case. Thus the added subdivision preserves separate target separation, while source separation follows by restriction of the shared source rectangles. All these statements concern the filled closed rectangles and arbitrary symbolic tails.

For completeness the quantitative gaps remain valid for the entire range $B \geq 2m+1$: distinct first-digit intervals have gap at least B^{-1} and distinct second-digit intervals at least B^{-2}. The source and target families therefore have within-state gaps at least aB^{-2}; different state squares have their own chosen gaps. Tail values lie between $1/(B-1)$ and $(2m-1)/(B-1) < 1$. An initial constant tail after l digits contributes $B^{-l} e(a_{\text{tail}})/(B-1)$, so initialized codes are rational. At checkpoints the history records, or the known absolute origin of the record block, also recover the absolute work-head position. □

6.3 Conventions for the remaining geometric realizations

Unless a different table is specified, subsequent routes use the move-first recorder of Lemma 5.2, with initial frontier two. The controls $(C_q, J_r, S_r, P_q, T_q)$ there are written $(W_q, P_r, B_r, C_q, D_q)$ here, and its empty/frontier/record symbols (e, F, r) are written (E, F, R_r). Permuting work, history and mark coordinates in every row preserves every guard. This identifies the full partial table, its inverse and the exact checkpoint count $2(2 + n - h') + 4$, including arbitrary allowed tapes. A nonterminal branch means one with both endpoint controls nonterminal; a final branch into a halt can enter the observer.

The full branches are those of Lemma 5.1. For an alphabet of size m we use one of the following explicitly indicated conventions:

$$(B, \{d_a\}) = (2m+2, \{1, 3, \dots, 2m-1\}), \quad (2m+1, \{1, 3, \dots, 2m-1\}), \quad (2m, \{0, 2, \dots, 2m-2\}). \tag{21}$$

All satisfy its digit bound and separation hypotheses. In the last convention the full blank has digit zero; boundary codes remain in the closed-rectangle and neighborhood conclusions. Write $C(a_1a_2\cdots)=\sum_{j\ge1} d_{a_j}B^{-j}$ and $I_v=\sum_{j=1}^{|v|} d_{v_j}B^{-j}+[0,B^{-|v|}]$. A constant tail after k positions contributes $d_{\mathrm{tail}}B^{-k}/(B-1)$, so the initial codes are rational. Guarded three-cell rules use Lemma 5.6 with $(a,b)=(1,2)$ after their own full-domain inverse has been proved.

6.4 Analytic conventions for the geometric variants

The whole-space baseline in these variants means smooth $u\in CH^2\cap C^1L^2$, bounded $u,\nabla u$ on finite intervals, and a pressure representative in CH^1. When the competitor-gradient condition is omitted explicitly, the other conditions remain unchanged. Both cases fall under Lemma 2.2(i); separately stated pressure classes are retained. The torus baseline is its classical torus clause. Unless a projection is specified, the force is

$$f_U=U_t+(U\cdot\nabla)U-\nu\Delta U,\qquad (u,p)=(U,0). \tag{22}$$

Torus projection is the separate choice in Proposition 2.4, with changed pressure. Localized affine motion is supplied by Lemma 3.1 on a compact time interval, using effective derivative bounds, positive lower scale bounds and the positive cutoff margins specified in each route. The potential is (6); the resulting flow fixes every region disjoint from its support. In particular exponential diagonal scales and linear reciprocal scales have respectively generators $\theta'\log D$ and $\mathrm{diag}(l'/l,-l'/l,0)$.

Lemma 6.3 (Finite forcing mechanisms)**.** *A finite family of the preceding pulses, joined with flat time collars, can be used once for initialization and repeated with period one thereafter. It gives a smooth effective velocity and residual force with all mixed derivatives uniformly bounded. In $\mathbb{R}^3$ both have one fixed compact spatial support, and the velocity has uniformly bounded kinetic energy. In a torus chart, periodized curl or planar Hamiltonian pulses have zero spatial mean, as does their residual force. Initial velocity and pressure are zero. Uniqueness holds in the comparison classes of Section 2; in the whole-space class the competitor's gradient bound can be omitted if the remaining conditions are retained.*

Proof. Each pulse is a finite expression in rational data, effective smooth switches, their derivatives, positive diagonal scales and logarithms. Finite positive separation margins bound every cutoff derivative. A finite maximum bounds a period and the loading interval, and repetitions preserve that bound. To evaluate near a join, a coarse enclosure of time gives a finite superset of possibly active pulses. Summing them avoids an exact comparison of a computable time with an integer. The flat collars give smoothness. Compact support gives every required spatial Sobolev bound. Lemma 2.2 gives the asserted solution and comparison; on a torus the integral of each curl or Hamiltonian derivative vanishes, and the convection term is a divergence. Its flow satisfies $\det D\Phi_t=1$, because differentiating this determinant gives $(\operatorname{div}U)(t,\Phi_t)\det D\Phi_t=0$ and its initial value is one. The velocity and acceleration identities in that lemma consequently apply to these volume-preserving diffeomorphisms. □

For later use, spatial chart changes are always made at the velocity level. If $x=o+ay$, set $\widetilde{U}(t,x)=aU(t,(x-o)/a)$ and then compute (22) at the prescribed physical ν. The time and convection terms scale by a, while the Laplacian term scales by a^{-1}. Thus chart rescaling does not silently change the fixed physical viscosity.

7 Affine motion on neighborhoods of boxes

Rectangle separation controls the beginning and end of a transfer; distinct heights separate its middle. We now specify the paths and cutoff margins used by the applications. Interpolating one planar scale and its reciprocal preserves thickness and, for a linear first scale, gives a convex formula for each first coordinate. Interpolating both planar scales independently gives that formula in both coordinates, at the cost of a compensating normal strain during the motion.

7.1 Reciprocal paths and their localization

Use the flat step σ of (5) and put

$$\theta(s) = \sigma(2s - 1/2), \qquad w_j(s) = \theta(3s - j + 1) \quad (j = 1, 2, 3).$$

The switches w_1, w_2, w_3 perform the lift, horizontal transfer and descent in separate thirds of the unit interval. All are constant near their phase endpoints. The following version of Lemma 3.3(i) records a common vertical half-width of one and explicit neighborhoods; the same formulas allow other heights and thicknesses as described immediately afterward.

Lemma 7.1 (Two explicit interpolations of separated boxes)**.** *Let $S_i, T_i \subset \mathbb{R}^2$, $1 \le i \le N$, be closed rational rectangles of positive side lengths. Suppose each list is pairwise disjoint and*

$$F_i(y) = c_i^T + \operatorname{diag}(\lambda_i, \lambda_i^{-1})(y - c_i^S), \qquad \lambda_i \in \mathbb{Q}_{>0},$$

maps S_i onto T_i, with c_i^S, c_i^T their centers. There is a smooth nondecreasing switch $\eta : [0,1] \to [0,1]$, zero near 0 and one near 1, for which either scale path $r_i = \lambda_i^\eta$ or $r_i = 1 - \eta + \eta\lambda_i$ yields an effective smooth compactly supported solenoidal unit-time field, zero on time collars, realizing $(y, z) \mapsto (F_i(y), z)$ on the whole box $S_i \times [-1, 1]$ and a neighborhood of it. The paths are affine; each horizontal half-width remains between its endpoint values, and the horizontal sweep of each original box lies in the coordinate bounding rectangle of S_i and T_i. With the linear choice, a particle's first coordinate is the convex interpolation of its endpoint first coordinates.

Proof. For $N = 0$ take the zero field. For $N \ge 1$ apply Lemma 3.3(i) with vertical half-width one. Use the shared phase switches with $\eta = w_2$ and set

$$C_i(s) = \big((1 - \eta)c_i^S + \eta c_i^T,\, 6i(w_1 - w_3)\big), \qquad D_i(s) = \operatorname{diag}(r_i, r_i^{-1}, 1).$$

The affine path is $G_i(s, y) = C_i(s) + D_i(s)(y - (c_i^S, 0))$. Choose ϵ to be one quarter of the minimum of 1 and all pairwise max-norm distances in the two planar lists, omitting empty pair lists. The ϵ-padded moving boxes are disjoint: source and target gaps are at least 4ϵ during the vertical phases, and the middle-phase height spacing is $6 > 2(1 + \epsilon)$.

For later loading formulas, record the centered cutoff explicitly:

$$\chi(C, L, \epsilon; x) = \prod_{j=1}^{3} \sigma\!\left(\frac{L_j + \epsilon + x_j - C_j}{\epsilon/2}\right) \sigma\!\left(\frac{L_j + \epsilon - x_j + C_j}{\epsilon/2}\right).$$

Here L is the vector of half-widths; the plateau has padding $\epsilon/2$ and the support has padding ϵ. If $L_i(s)$ are the moving half-widths, $Y = x - C_i(s)$ and $A_i = D_i' D_i^{-1}$, the field supplied by (6) is

$$V(s, x) = \sum_i \nabla \times \Big\{ \chi(C_i, L_i, \epsilon; x) \Big[\tfrac{1}{2} C_i' \times Y + \tfrac{1}{3}(A_i Y) \times Y \Big] \Big\}. \tag{23}$$

Indeed $\det D_i = 1$, and the preceding separation verifies Lemma 3.1 with localization parameter $\epsilon/2$. Its neighborhood conclusion applies with $L_* = \max_i \max(1, \lambda_i, \lambda_i^{-1})$ because $\|D_i(s)\|_\infty \le L_*$ and $D_i(0) = I$. It therefore gives the stated affine motion on the full boxes and on their initial max-norm neighborhoods of radius $\epsilon/(4L_*)$. The paths are stationary for $s \le 1/12$ and $s \ge 11/12$. Their finite center ranges and endpoint width bounds give common compact support. The effective positive lower bound $r_i \ge \min(1, \lambda_i)$ verifies the scale requirement. For the linear scale, the first-coordinate identity is

$$(G_i(s, y))_1 = (1 - \eta)y_1 + \eta(F_i(y_1, y_2))_1.$$

The second coordinate need not interpolate its own endpoint values. Instead, if b_{i2} is the source's second half-width, convexity of the reciprocal function gives

$$\frac{b_{i2}}{(1 - \eta) + \eta\lambda_i} \le (1 - \eta)b_{i2} + \eta\frac{b_{i2}}{\lambda_i}. \tag{24}$$

The first half-width is exactly its endpoint interpolation. For the exponential scale, convexity of the exponential gives the corresponding upper bounds in both coordinates, since $\lambda_i^{\pm\eta} \le (1 - \eta) + \eta\lambda_i^{\pm 1}$. In either case the center is interpolated linearly and each half-width is at most the interpolation of its endpoint half-widths. Therefore each lower edge is at least the interpolation of the endpoint lower edges, and each upper edge is at most the corresponding interpolation of upper edges. This proves the claimed bounding rectangle for the entire sweep. The individual endpoint-width bounds also follow directly from monotonicity of the positive scales and their reciprocals. □

The same construction applies to $P_i \times [-h, h]$ for any common $h \ge 0$, including plates and rectangles with zero side lengths, under the hypotheses of Lemma 3.3(i). Replace $6i$ by heights H_i whose pairwise gaps exceed $2h + 2\varepsilon$, and choose ε below one third of every within-family planar gap. The formula for C_i uses the source and target centers in place of c_i^S, c_i^T; its support padding is ε and plateau padding is $\varepsilon/2$. Source projections, distinct heights and target projections separate the three phases in that order. Empty pair lists impose no gap restriction, and the empty family gives the zero field. Arbitrary real data give smooth existence; the effective-data assumptions of the shared lemma give computable scales, derivative bounds and margins. The same perturbation argument gives an open neighborhood even when a side length is zero.

7.2 Transporting an initial cutoff

The transported-initial-cutoff convention is also useful. Write the affine path as $G_i(t)x = J_i(t)x + g_i(t)$ and localize by $\psi_i(G_i(t)^{-1}x)$, where ψ_i has support padding ε_0 around the initial box and equals one on a smaller positive padding. If $\lambda_i \in [B^{-1}, B]$, its initial padding ε_0 expands horizontally by at most B. Choose $2B\varepsilon_0$ below every target gap, $2\varepsilon_0$ below every source gap, and $\varepsilon_0 \le \varepsilon$. These transported supports stay separated in the same three phases. The affine generator is $v_i + A_i x$, with $A_i = \dot{J}_i J_i^{-1}$ and $v_i = \dot{g}_i - A_i g_i$. Its potential is

$$\tfrac{1}{2}v_i \times x + \tfrac{1}{3}(A_i x) \times x.$$

The identity proved in Lemma 3.1 gives the exact generator on the transported plateau. Its neighborhood argument then proves the same full-box endpoint map. The factor B in the margin choice is essential: transported padding does not stay fixed under stretching.

7.3 Independent planar interpolation

For an observer that needs both planar coordinates to follow their own endpoint interpolations, the reciprocal path is insufficient. The next construction interpolates the two factors independently and compensates the temporary planar area change in the third direction. Here the named targets may be disjoint containers of the actual affine images; only the image centers enter the motion.

Lemma 7.2 (Independent planar scales and normal compensation)**.** *Let $P_i, Q_i \subset \mathbb{R}^2$ be finite families of closed rectangles, separately pairwise disjoint, and let assigned positive diagonal affine maps F_i satisfy $F_i(P_i) \subseteq Q_i$. Write their diagonal factors as $\lambda_{i1}^*, \lambda_{i2}^* > 0$ with product one. There is a smooth compactly supported solenoidal unit-time field, zero on time collars, whose endpoint map is $(y, z) \mapsto (F_i(y), z)$ on a neighborhood of each entire plate $P_i \times \{0\}$. During the horizontal phase both planar factors interpolate linearly, while the normal factor compensates their product. Each planar coordinate of a point on the plate is its convex endpoint interpolation.*

Proof. Use the height phases in Section 7.1. For source center p_i, end its center path at its actual image $F_i(p_i)$; in the onto case this is the named target center. Choose separated heights H_i as above and set

$$c_i = \big((1 - w_2)p_i + w_2 F_i(p_i),\ H_i(w_1 - w_3)\big),$$
$$\lambda_{ij} = 1 - w_2 + w_2\lambda_{ij}^*, \qquad D_i = \operatorname{diag}(\lambda_{i1}, \lambda_{i2}, (\lambda_{i1}\lambda_{i2})^{-1}).$$

The volume-preserving path has logarithmic derivative

$$T_i = \dot{D}_i D_i^{-1} = \operatorname{diag}\left(\frac{\dot\lambda_{i1}}{\lambda_{i1}}, \frac{\dot\lambda_{i2}}{\lambda_{i2}}, -\frac{\dot\lambda_{i1}}{\lambda_{i1}} - \frac{\dot\lambda_{i2}}{\lambda_{i2}}\right).$$

This matrix has trace zero, so (6) localized around the moving plate gives the desired field. Its planar motion is $c_i + (\lambda_{i1}(y_1 - p_{i1}), \lambda_{i2}(y_2 - p_{i2}), 0)$. Normal displacement changes by the factor $(\lambda_{i1}\lambda_{i2})^{-1}$. The factors have positive lower bounds; choose the initial thickness less than their product's minimum times the normal plateau width. A small enlargement of the planar rectangle is allowed by the positive collars as well. Thus a whole thin neighborhood follows the volume-preserving affine motion. The normal factor returns to one at the endpoint. The final plate image is contained in $Q_i \times \{0\}$. Source and target-container gaps protect the vertical phases and distinct heights protect the middle phase. Linear interpolation of each planar factor gives both convex-coordinate identities on the plate. This proves smooth existence for arbitrary real data. Under the effective input assumptions of Lemma 3.1, the positive scale bounds and finite gap margins can be computed and the construction is effective. □

7.4 Forcing and effective evaluation

Lemma 7.3 (Residual forcing for the finite processors)**.** *Fix computable $\nu > 0$. Let W be smooth, solenoidal, initially zero, with support in one compact subset of $\mathbb{R}^3$ and every mixed derivative bounded. Then*

$$F_W = W_t + (W \cdot \nabla)W - \nu\Delta W \tag{25}$$

has the same support and derivative properties. The zero-data solution $(W, 0)$ is unique among smooth solutions with $u \in C_t H^2 \cap C_t^1 L^2$, bounded u on finite intervals, and $p \in C_t H^1$. The corresponding torus assertion holds in the classical comparison class of Section 2. A sum of curls in a torus chart gives both W and F_W mean zero. Finite formulas from the effective-data instances of the preceding constructions, finite loading, and repetition of one cycle give effective evaluation of every force derivative.

Proof. Fixed compact support and bounded mixed derivatives imply all reference velocity hypotheses of Lemma 2.2. Case (i) gives exactly the stated whole-space comparison class, without a competing gradient bound; its torus clause gives the stated classical comparison class. The same lemma provides the pressure normalization and global material flow. The product rule applied to (25) proves the force's support and derivative bounds. A chart curl has zero mean, and integration of the residual uses $\int \Delta W = 0$ and $\int (W \cdot \nabla) W = \int \operatorname{div}(W \otimes W) = 0$. For these effective-data instances, gate derivatives and seam evaluation are those of (5) and Proposition 3.2. The explicit affine scales and every reciprocal denominator have positive computable bounds. Thus differentiating the finite formulas, using a finite superset of possible period indices, evaluates every derivative without a machine execution or a real equality test at a seam. □

8 Choosing initialization and the observer

The preceding recorders supply complete instruction tables. We now specify how their boxes are positioned and how the fixed material label becomes the initial code. We compare one-time loading, a loading branch repeated in every cycle, and chart translations that place the initial code at the label itself. The latter two choices can preserve periodicity from time zero. Each construction also needs a bound between integer rule times; the endpoint coding alone does not give a material event.

Unless stated otherwise, the force in this section is the residual (7), with pressure zero. Its smoothness, all-order mixed derivative bounds, finite effective prescription and uniqueness are as in Proposition 3.2. Whole-space forces have one compact spatial support; torus forces have zero spatial mean. All statements use a fixed positive computable viscosity, with the relative interpretation of that proposition for arbitrary fixed positive viscosity.

8.1 Torus loading and direct chart placement

Proposition 8.1 (A move-first material test)**.** *The move-first recorder has a unit-torus realization, periodic after $t = 1$, whose fixed label $(1/4, 1/2, 1/4)$ enters $\{x_1 \in (1/2, 1) \pmod 1\}$ exactly when the work machine halts. For a fixed machine only the loading interval depends on the input.*

Proof. Use Lemma 5.2, let Σ be its full cell alphabet, and enumerate its N controls by $i = 1, \dots, N$. Set $s = 1/[16(N+1)]$ and give the ith control a square of side s centered at

$$m_i = (o_i + i/[4(N+1)], 1/2), \qquad o_i = \begin{cases} 5/8, & C_q \text{ with } q \text{ halting}, \\ 1/8, & \text{otherwise}. \end{cases}$$

Use odd digits and $B = 2|\Sigma| + 2$. The code at height $z_0 = 1/4$ is $(m_i + s(E(L) - 1/2, E(R) - 1/2), z_0)$. It is rational initially. Lemma 5.1 gives J branches with separately separated rectangles, all side lengths at most s and endpoints on the grid of mesh $[32(N+1)B^2]^{-1}$. For branch j choose height $z_j = 1/2 + j/[4(J+1)]$. Use the three-phase lift of Lemma 3.3, with exponential middle scale λ_j^θ and center interpolation. A padding parameter

$$\epsilon = \min\{1/128, [320(N+1)B^2]^{-1}, [40(J+1)]^{-1}\}$$

separates the enlarged sheets in the source, height, and target phases and keeps them inside the unit cube. Their thickness may then be chosen positive by Lemma 3.1. For $J = 0$ the repeated field is zero.

During $[0, 1]$ a localized curl translation takes the fixed label along the segment to the initial code. Both segment and a small padding lie in the chart. Repeat the branch pulse thereafter. At

each subsequent integer time the particle is the next table code. If both endpoint controls are nonhalting, the interpolated center and half-width bounds give $x_1 < 3/8 + s < 1/2$ throughout the motion. The loader also stays below $1/2$ in a nonhalting run. A halting checkpoint lies in the upper half, reached in finitely many positive local steps; an initial halt is reached at the end of loading. This proves the event for every real time. □

Proposition 8.2 (A guarded-window material test)**.** *The table of Lemma 5.5 has a unit-torus realization, periodic after $t = 1$, with fixed label $(1/4, 1/4, 1/4)$ and event $x_1 \in (1/2, 7/8) \pmod 1$, occurring exactly when the work machine halts. At work step n its next checkpoint occurs after exactly $2(p_n - h_n) + 1 \le 4n + 5$ unit periods.*

Proof. Choose $r_0 = 2$ in Lemma 5.5: initially the frontier is at cell two, all other history cells are empty, and all marks are zero. Thus $p_n = n + 2$, with records in cells $2, \dots, n + 1$. Split the full three-cell windows $[-1, 1]$ and apply Lemma 5.6 with $(a, b) = (1, 2)$. Let $\mathcal{A}$ be the full cell alphabet of this window table and use odd digits in base $B = 2|\mathcal{A}| + 1$. For S controls let $\epsilon = 1/[16(S + 1)]$ and place their squares at $o_s + \epsilon[0, 1]^2$, where

$$o_s = (\xi_s, 1/4 + 2\kappa(s)\epsilon), \quad \kappa(s) \in \{1, \dots, S\}, \quad \xi_s = \begin{cases} 5/8, & s \text{ halting Ready}, \\ 1/4, & \text{otherwise}. \end{cases}$$

Write J for the number of branches and e_j for branch j's cursor displacement. The full source and target rectangle gaps are at least ϵB^{-3}. Put their common height at $1/4$, assign branch j height $1/2 + j/[4(J + 1)]$, and use the exponential middle scale $\operatorname{diag}(B^{-e_j\theta}, B^{e_j\theta}, 1)$. Take one tenth of $\min(1/32, \epsilon B^{-3}, 1/[4(J + 1)])$ as padding. Horizontal source and target gaps and the private heights separate the three phases, while all widths remain at most ϵ. The strain can equivalently be generated by $(0, 0, -e_j\dot{\theta} \log B\,(x_1 - c_{j1})(x_2 - c_{j2}))$, retaining the explicit diagonal-strain implementation.

A first unit interval loads the label along the segment to the rational initial code. Thereafter the exact rectangle action and the compiler lemma give checkpoint times $t_0 = 1$ and $t_{n+1} = t_n + 2(p_n - h_n) + 1$. A nonhalting center stays in $[1/4, 1/4 + \epsilon]$, with half-width at most $\epsilon/2$; thus every intermediate first coordinate remains below $1/2$. The loader has the same exclusion. Halt codes are in $[5/8, 5/8 + \epsilon]$, inside the observer, including an initial halt. The history block starts at the known absolute cell two, so the head-relative code also recovers absolute indexing at checkpoints. □

Proposition 8.3 (Two direction-recording placements)**.** *The direction recorder gives both of the following realizations.*

1. *On $\mathbb{R}^3$, with its passive origin bit, there is a force periodic from time zero and a fixed-origin event $X_1(t; 0) < -5$. Input dependence can be confined to two translations of the state charts. On every implemented full box the first coordinate is a convex combination of its endpoint first coordinates.*

2. *On the unit torus, without the passive track, a force periodic after $t = 1$ has fixed label $(1/8, 1/8, 1/4)$ and event $x_1 \in (1/2, 7/8) \pmod 1$. For a fixed machine its repeated part is independent of the input.*

Both events are equivalent to halting.

Proof. For the first case introduce a fresh terminal state † and replace every missing instruction and every originally terminal state–symbol pair by a write-preserving stationary transition into

it. The original initial control is therefore nonterminal in this normalized table, even when the original machine is initially halted. Apply Lemma 5.3 with its passive origin track, and let $\mathcal{A}$ be the resulting full cell alphabet. Use odd digits in base $B = 2|\mathcal{A}| + 1$ and initial fractions x_*, y_*. Index the initial main state by zero, the main † states by distinct negative integers, and the rest by distinct positive integers. State s has placement

$$(x, y) \longmapsto (10k(s) + x - x_*, y - y_*).$$

The initialized point is zero; nonhalt codes have first coordinate > -1, and halt codes < -5. Thicken the rectangles of Lemma 5.1 by $[-1, 1]$, give branch i height $10i$, and use the linear scale $g_i = 1 + \theta(\lambda_i - 1)$ in the lift schedule. A padding smaller than one tenth of the minimum of one and all within-family planar gaps suffices. The middle strain has potential $(0, 0, (\dot{g}_i/g_i)(x_1 - c_{i1})(x_2 - c_{i2}))$. Equation (8) excludes every nonhalting visit to $x_1 < -5$. The table and routing are fixed for the machine; only x_*, y_* change with the input. The passive origin bit remains at physical cell zero, so no absolute tape information is discarded.

For the second case first merge all halting work states into a single state q_h, redirecting incoming transitions and retaining initial halting. Delete the passive track from the direction recorder, and now let $\mathcal{A}$ denote that full alphabet. Use odd digits with $B = 2|\mathcal{A}| + 2$ and $\lambda = 1/[16(J + 1)]$, where J counts the controls. Number them from zero and set

$$S_s(x, y) = (A_s + \lambda x, 1/4 + \lambda y), \qquad A_s = 2\lambda \operatorname{index}(s) + \begin{cases} 5/8, & s = M(q_h, e), \\ 1/4, & \text{otherwise.} \end{cases}$$

The nonhalt squares lie in $[1/4, 3/8] \times [1/4, 1/4 + \lambda]$; halt squares lie in $[5/8, 3/4] \times [1/4, 1/4 + \lambda]$. Write m for the number of branches. Use coding height $1/4$, branch heights $1/2 + i/[4(m + 1)]$, and the same linear first scale. One hundredth of the minimum of $1/16$, $1/[4(m + 1)]$, and the source and target planar gaps is valid padding; absent pair gaps are omitted. All supports lie inside the cube, and all horizontal widths are at most λ. A localized translation with padding $1/1000$ loads the rational initial point from the fixed label in one unit interval.

In a nonhalting period the center has first coordinate at most $3/8$ and half-width at most $\lambda/2$, hence is below $1/2$; loading has the same bound. Halting codes lie in the observer. The exact sampling induction gives one table step per period, with checkpoint cycle $2(k + 1 - h_k) + 3$. The record block beginning at cell one recovers absolute indexing at checkpoints. An initial halt is observed at time one. □

8.2 A one-time torus loader

The next placement uses a separate first interval to load the code. Only the instruction cycle is then repeated, so periodicity begins at time one.

Theorem 8.4 (A separate one-unit loader on the torus)**.** *For every machine and finite input and every fixed computable $\nu > 0$, there is an effective smooth mean-zero force, one-periodic for $t \geq 1$ with bounded mixed derivatives, whose unique smooth zero-data solution with zero normalized pressure satisfies*

$$\exists t \geq 0 : \ \Phi_t(1/2, 1/2, 1/4) \in (1/16, 1/4) \times \mathbb{T}^2 \quad \Longleftrightarrow \quad \textit{halting.}$$

Uniqueness holds in the classical class of Section 2.

Proof. Use the single-halt move-first recorder of Lemma 5.2, without an entry state. Let Γ be its full alphabet and N its number of controls. Put $B = 2|\Gamma| + 1$, $a = 1/(16N)$, the halt square at

$(1/8, 1/4) + a[0,1]^2$, and the other squares at $(1/2 + 2ja, 1/4) + a[0,1]^2$, $0 \le j \le N-2$. Use the minimal subdivision, so right and stay moves are not split by the left letter. Its gaps are at least a/B^2. For M branches choose

$$h = \frac{1}{64(M+1)}, \quad Z_i = \frac{1}{2} + \frac{i}{4(M+1)}, \quad \varepsilon = \min\{1/128, h/4, a/(4B^2)\}.$$

Apply linear-first-scale full-box routing from height $z_0 = 1/4$. Middle heights differ by $16h$; source and target gaps handle the vertical phases. Centers lie in $[1/8, 5/8] \times [1/4, 5/16]$, widths are at most a, and all padded boxes lie in $(1/16, 15/16)^3$. This is a construction on the specified unit torus, with no change of physical viscosity.

In the first unit interval translate a box of side $1/64$ from the fixed label to the rational initial code, using Lemma 3.1. The segment and a small padding lie inside the same chart. Repeat only the instruction cycle afterward. Code induction proves the checkpoint identities. A halting code has first coordinate in $[1/8, 3/16]$, including an initial halt reached by loading. For a nonhalting run both centers of every used branch have first coordinate at least $1/2$, with half-width at most $a/2$; loading has the same lower center bound. Hence the particle always stays to the right of $1/4$. Residual realization proves every analytic assertion. The separate loading field accounts for the qualification $t \ge 1$ in the periodicity statement. □

8.3 Repeated torus loading branches

A loader can instead be one branch of every period. To keep its target separate from all computational targets, place the initial code in a state square with no incoming instruction. This condition concerns the entire target family, not only the distinguished execution. We implement the same startup principle first with full boxes, then with fixed columns.

Theorem 8.5 (A loader repeated as part of the processor)**.** *For every finite machine and input, and every fixed computable $\nu > 0$, there is an effective smooth mean-zero force on the unit flat torus, one-periodic from time zero, with all mixed derivatives bounded. Its zero-data solution has pressure zero, is unique in the classical class of Section 2, and satisfies*

$$\exists t \ge 0 : \ \Phi_t(1/8, 1/8, 1/4) \in \{5/8 < X_1 < 7/8\} \quad \Longleftrightarrow \quad \textit{the machine halts.}$$

The one repeated cycle implements all instruction branches and its loading branch.

Proof. Normalize to one halt state and prepend a new nonhalting state q_I whose rule preserves the scanned letter, stays, and enters the original initial state. No machine rule targets q_I. Apply the move-first recorder of Lemma 5.2 and omit the unused controls P_{q_I}, T_{q_I}. Then the entry control C_{q_I} has no incoming local instruction. Write Γ for this recorder's full alphabet. Put $B = 2|\Gamma| + 1$, let N be the number of controls, and set $a = 1/(16(N+1))$. Use the halt square $(3/4, 1/4) + a[0,1]^2$ and nonhalting squares $(1/4 + 2al, 1/4) + a[0,1]^2$, $0 \le l \le N-2$. The full subdivision of Lemma 6.2 gives source half-widths $a/(2B)$ in both coordinates.

Add a square with those half-widths centered at $(1/8, 1/8)$ and a translation to the square centered at the initial entry code. Each initial tail is at distance at least $1/(B-1) > 1/(2B)$ from the endpoints of $[0,1]$, so the new target lies strictly inside the entry square. There is no other target in that square. Both enlarged branch lists therefore remain separately separated, with gaps at least a/B^2. The extra source is separated from all state squares.

Number the branches, including loading, by $1 \le i \le M$. Set

$$z_0 = \tfrac{1}{4}, \qquad h = \frac{1}{16(M+1)}, \qquad Z_i = \tfrac{1}{2} + \frac{i}{4(M+1)}, \qquad \varepsilon = \frac{1}{10}\min(h, a/B^2).$$

Thicken each planar source by $[z_0 - h, z_0 + h]$. Lift the resulting full box to center height Z_i, move it horizontally, and lower it to z_0. For an instruction with displacement d_i, use $D_i = \operatorname{diag}(B^{-d_i w_2}, B^{d_i w_2}, 1)$; loading has $d_i = 0$. Its center is the planar endpoint interpolation plus $(Z_i - z_0)(w_1 - w_3)e_3$. Adjacent middle heights are $4h$ apart, while the padded half-thickness is $h + \varepsilon < 2h$. Source and target gaps handle the other phases. Horizontal half-widths never exceed $a/2$, horizontal centers lie between $1/8$ and $3/4 + a$, and vertical centers lie between $1/4$ and $3/4$. The padded motions are thus strictly inside the unit cube. Lemma 3.1 gives a chart field, extended periodically in space and time. Its zero time collars give zero initial velocity.

The first cycle loads the fixed particle. Each subsequent cycle maps its exact code to the next local code until the first halt, or forever in a nonhalting run. Lemma 5.2 identifies the finite checkpoint times with machine steps. The added entry state handles an original initial halt. A halt code is in the target slab. For a nonhalting execution every used source and target center, including the loader, has first coordinate at most $3/8$. Exponential interpolation keeps the center below that bound and the half-width below $a/2$; the particle cannot reach $5/8$ at any intermediate time. The final transition into a halt is not subject to this nonhalting-endpoint bound. Lemma 7.3 gives the force, its mean, its effective derivatives, and uniqueness. Repeating the loading field is harmless to the initialized trajectory and makes the force periodic from zero. □

A fixed-column implementation. The same untargeted-entry condition also permits a repeated initializer using fixed source and target columns. The next construction retains its explicit column potentials and its fixed label; its middle motion uses linear interpolation of the first planar scale.

Theorem 8.6 (A periodic initializer on the unit torus)**.** *There is an effective halting realization on $\mathbb{T}^3$ with fixed label $a_* = (3/8, 1/4, 1/2)$ and fixed strip $5/8 < x_1 < 7/8$, using a smooth mean-zero general force of period one from time zero. Its initial velocity and prescribed pressure are zero, every mixed derivative is bounded, and its kinetic energy is uniformly bounded. The solution is unique in the torus comparison class.*

Proof. Normalize to a single halt and add a new nonhalting start q_s, with no incoming original transition, which performs a dummy unchanged stay move to the old initial state. This preserves even initial halting. Use the move-first recorder of Lemma 5.2, with the control names of Section 6.3, and retain its frontier-writing and return controls C_q, D_q only for q in the set Q^+ of targets of the normalized work table, including the new dummy transition. In particular its destination, the old initial state, belongs to Q^+, whereas $q_s \notin Q^+$. No recorder rule therefore targets the ready control W_{q_s}. Its initial frontier is still two and the simulation proof is unchanged.

Use odd digits in base $B = 2K + 2$, where K is the recorder alphabet size. If there are n nonterminal controls, put $\tau = 1/[16(n + 1)]$. Give nonterminal states squares of side τ at lower corners $(1/8 + 2\tau j, 1/4)$, $0 \le j < n$, and put the terminal square at $(3/4, 1/4)$. All nonterminal first coordinates are below $1/4$. Refine every rule, including right and stationary rules, by the left letter ℓ. The full source is $I_\ell \times I_\gamma$, and the three image rectangles are respectively

$$I_{\beta\ell} \times [0, 1], \qquad I_\ell \times I_\beta, \qquad [0, 1] \times I_{\ell\beta}. \tag{26}$$

They follow from the same affine formulas as Lemma 5.1. The refined sources are separately disjoint; the incoming displacement and written symbol, together with the extra left prefix where applicable, separate the images. Thus full geometric separation still holds.

Let $w_* = (3/8, 1/4)$ and let w_{in} be the rational input code in the untargeted start square. Add the translation from $w_* + [-\delta, \delta]^2$ to $w_{\mathrm{in}} + [-\delta, \delta]^2$, where $\delta = \tau/(10B^2)$. The source lies away from every state square. The target lies strictly inside the start square: odd-digit codes have distance at

least $1/(B-1)$ from zero and at least $2/(B-1)$ from one, both larger than $1/(10B^2)$. It meets no computational image because none targets that state. This extended list still has separately disjoint source and target families, even though a target can overlap a source.

We give the torus transport explicitly to keep the startup claim separate from the whole-space column construction. All rectangles initially lie in $z_0 = 1/2$. For m branches assign heights $z_j = 1/2 + j/[4(m+1)]$. Choose planar source and target cutoffs equal to one on neighborhoods of their rectangles, with padding less than one tenth of every relevant pairwise gap and chart-face margin; a missing pairwise minimum is omitted. Let $\eta(z)$ equal one near $[1/2, 3/4]$ and have padding $1/16$. The source column field

$$Z_j = \nabla \times (0, x\chi_j(x,y)\eta(z), 0)$$

equals e_3 along its source column; the target analogue Z'_j does so along the target column. These fields have compact support in the open unit cube and zero spatial mean after periodization.

For the middle phase use the center path $c_j = (1-\mu)p_j + \mu q_j$ and scale $l_j = 1 - \mu + \mu a_j$, where a_j is the branch's horizontal scale. Choose ξ_j equal to one near the planar coordinate bounding rectangle of the endpoints, and disjoint height cutoffs ρ_j at z_j with padding $1/[16(m+1)]$. For three successive flat ramps $\theta_1, \theta_2, \theta_3$, put $c_j = c_j(\theta_2)$, $l_j = l_j(\theta_2)$ and

$$\Psi_j = \xi_j \rho_j \left[\dot{c}_{j,1}(y - c_{j,2}) - \dot{c}_{j,2}(x - c_{j,1}) + \frac{\dot{l}_j}{l_j}(x - c_{j,1})(y - c_{j,2}) \right].$$

The full velocity is

$$V = \sum_j (z_j - z_0)\theta_1' Z_j + \sum_j (\partial_y \Psi_j, -\partial_x \Psi_j, 0) + \sum_j (z_0 - z_j)\theta_3' Z'_j. \tag{27}$$

It lifts all full rectangles, performs the reciprocal affine motions in separate layers, and lowers them. In the middle phase the exact planar path is $c_j + \operatorname{diag}(l_j, l_j^{-1})(\zeta - p_j)$. The swept-rectangle bound in Lemma 7.1 keeps it inside the coordinate bounding rectangle of the endpoints. The first coordinate is an endpoint convex combination. The cutoffs are therefore one along every claimed path. Differentiation and uniqueness prove the assigned full maps. The entire construction is supported in the open unit cube and has zero collars in phase.

Extend it with period one in phase and periodically in space. It is zero at time zero, so its residual has the required period from time zero. During the first period the distinguished particle follows the initializer; its first coordinate stays below $1/2$. Thereafter it follows the recorder. Every nonterminal transition remains below $1/4$, whereas a terminal code lies inside $5/8 < x_1 < 7/8$. It cannot return to the initializer source, whose square lies at $x_1 = 3/8$. The dummy instruction handles initial halting without changing the observation rule. Lemma 6.3 now proves all analytic assertions, including uniform energy on the compact torus. The force formula includes every branch, the initializer included, in each period; it uses no precomputed execution. □

8.4 A repeated loading instruction

The next construction incorporates loading in a different symbolic table: its instructions replace finite prefixes of two stacks. A replacement can change planar area, so the third coordinate compensates for the product of the two planar scales. Evacuation and delivery then realize the resulting volume-preserving boxes in every period.

Theorem 8.7 (Periodic prefix-box realization)**.** *For every deterministic machine and finite input, a compactly supported smooth force on $\mathbb{R}^3$, periodic with period one from time zero, realizes the fixed-origin halting event*

$$\exists t \geq 0 : \quad X(t;0) \in (-4,-1) \times (-1,2) \times (-1,1) \quad \Longleftrightarrow \quad \textit{halting.}$$

The force has bounded mixed derivatives and the unique zero-data solution in the class of Proposition 3.2. Its finite program uses full volume-preserving boxes; no solenoidal condition is imposed on the force itself.

Proof. Normalize to one halting work state q_h, redirecting incoming transitions to it and replacing missing nonhalting rules by unchanged-letter stay moves into it. This preserves initial halting. Write Γ for the work alphabet, b for its blank, q_0 for the initial control, and ω for the finite input word. For top-first stacks L, R, a rule $(s, \alpha, \beta) \rightsquigarrow (r, \gamma, \eta)$ replaces the two displayed prefixes and leaves their independent infinite tails unchanged. Introduce fresh state s_0, boundary letter S and mark letter M. The preliminary rules are

$$(s_0, \varepsilon, \varepsilon) \rightsquigarrow (q_0, S, \omega)$$

and, for each work transition $(q, a) \mapsto (r, c, d)$,

d	preliminary rules
1	$(q, \varepsilon, a) \rightsquigarrow (r, c, \varepsilon)$
0	$(q, \varepsilon, a) \rightsquigarrow (r, \varepsilon, c)$
-1	$(q, v, a) \rightsquigarrow (r, \varepsilon, vc) \quad (v \in \Gamma)$,
	$(q, S, a) \rightsquigarrow (r, S, bc)$.

The final row exposes the implicit blank beyond the left boundary. Index each preliminary occurrence by j, with target $r(j)$, and introduce a private control g_j and record letter ℓ_j. Replace that occurrence by $(s, \alpha, \beta) \to (g_j, \gamma, M\eta)$ and add

$$\begin{aligned} (g_j, v, \varepsilon) &\to (g_j, \varepsilon, v) && v \in \Gamma, \\ (g_j, S, \varepsilon) &\to (k_{r(j)}, S\ell_j, \varepsilon), \\ (k_q, \varepsilon, v) &\to (k_q, v, \varepsilon) && v \in \Gamma, \\ (k_q, \varepsilon, M) &\to (q, \varepsilon, \varepsilon). \end{aligned} \tag{28}$$

Sources are separated by control and tested top symbols. Into g_j, entry has right top M, whereas a loop has a work letter there. Into k_q, entry has left prefix $S\ell_j$, whereas a loop has a work letter at left top; distinct entries have distinct ℓ_j. Into a work state only the last rule enters. Thus both source and target cylinders are disjoint for arbitrary tails.

After a preliminary replacement the stacks have form wSB, MY, with $w \in \Gamma^*$. The forward loop transfers w, leaving $SB, w^{\mathrm{rev}}MY$. The boundary rule inserts ℓ_j behind S, the reverse loop restores w, and the final rule removes M. After $2|w| + 2$ further instructions the stacks are $wS\ell_j B, Y$ at control $r(j)$. At a work control, w lists represented cells left of the head, nearest first; R lists the current cell and right tail, and cells beyond S on the left are implicit blanks. The preliminary rules perform exactly the work write and move. Starting at $(s_0, b^\infty, b^\infty)$, the first work visit is at rule count three, and consecutive work visits differ by $3 + 2|w_{\mathrm{new}}|$. This proves finite continuation and the halting equivalence, including an initially halted input. The retained records also recover all head displacements.

Enumerate the enlarged alphabet, blank first, by even digits $0, 2, \dots, 2m-2$ in base $K = 2m$. Use prefix intervals $I(v)$ and codes E as in Lemma 5.1. Set offsets $o_{q_h} = -3$, $o_{s_0} = 0$, and $3, 6, 9, \dots$ for the other controls. Encode (s, L, R) by $(o_s + E(L), E(R), 0)$; its initial value is zero. For a rule $(s, \alpha, \beta) \to (s', \gamma, \eta)$ put

$$a_i = K^{|\alpha|-|\gamma|}, \quad b_i = K^{|\beta|-|\eta|}, \quad c_i = (a_i b_i)^{-1}.$$

The full source box is $(o_s + I(\alpha)) \times I(\beta) \times [-1, 1]$; the target is $(o_{s'} + I(\gamma)) \times I(\eta) \times [-c_i, c_i]$. The center-to-center map has matrix $\operatorname{diag}(a_i, b_i, c_i)$ and determinant one. It implements the prefixes on all codes. Disjoint cylinders give separated full planar projections in each family, including blank-tail endpoints.

Park box i at $(4i, 4, 0)$, reserving the horizontal rectangle $[4i - 1/2, 4i + 1/2] \times [7/2, 9/2]$. Let $R = \max(1, c_1, \dots, c_N)$ and travel at height $Z = 2R + 3$. Evacuate all sources to their parking sites, reshape there by $\operatorname{diag}(a_i^s, b_i^s, c_i^s)$, then deliver all targets. The two horizontal widths remain at most one, and vertical half-widths at most R. Let g be the minimum of one and the within-family gaps in the source-plus-parking and target-plus-parking projections. Padding smaller than $g/8$ works during vertical motion and strain; at travel height the moving bottom is at least $Z - R > R + 2$. Lemma 3.3 gives the full $7N$-motion schedule.

Repeat this unit pulse from zero. Nonhalting endpoint boxes and all parking boxes have nonnegative first coordinate. Translations preserve that bound, and strain takes place in positive parking envelopes. Therefore a nonhalting path never enters the negative observation box. A halting code lies in $[-3, -2] \times [0, 1] \times \{0\}$ inside it. The residual proposition finishes the proof. Loading was a repeated instruction, so no exceptional first-time force is required. □

Remark 8.8 (A different observation regime). For comparison, the distinct velocity-field detector of [31, Theorem 1.1(2) and Section 5] uses $\mathbb{R}^2 \times \mathbb{T}$ and the event $\int_{\{X_2>0\}\times\mathbb{T}} u_3 > 1/2$. Its force has globally bounded mixed derivatives and horizontal support compact on each finite time interval. The nonnegative scalar velocity has diffusion tails [27, 1]; after loading its total mass is one. Its cutoff estimate gives loss $\delta < 1/24$, nonhalting mass at most δ in the observed half-space, and halting mass above $3/4$. Those conclusions follow from the cited advection–diffusion proof and have no uniform spatial support assertion. They are a separate force and observation regime from the material test proved here.

9 Compact support and onto slow clocks

A whole-space processor can use arbitrarily separated finite lanes. We shall construct a periodic cycle with fixed compact support and also give a separate decaying force with the same material event. We first establish the time change used for the latter choice. Its clock must tend to infinity: traversing only a finite amount of machine time would not preserve the halting event.

9.1 The logarithmic clock at every derivative order

Lemma 9.1 (Logarithmic clock with full derivative bounds)**.** *Let $W(s, x)$ be a smooth solenoidal field for $s \geq 0$, initially zero, with every mixed derivative bounded and support in one compact set K. For fixed computable $\nu > 0$, set*

$$s(t) = \log(1 + t), \quad \rho(t) = (1 + t)^{-1}, \quad U(t, x) = \rho(t) W(s(t), x), \quad F = U_t + (U \cdot \nabla)U - \nu \Delta U.$$

Every derivative $\partial_t^k \partial_x^\alpha U$ and $\partial_t^k \partial_x^\alpha F$ is uniformly $O((1+t)^{-1-k})$ and belongs to space-time L^2. Both supports lie in K. If Ξ is the material flow of W, that of U is $\Xi(s(t), a)$, so every fixed spatial

reachability event is unchanged. Effective finite formulas for W give effective formulas for U, F and all derivatives.

Proof. Since $s' = \rho$ and $\rho' = -\rho^2$,

$$\partial_t\{\rho^j Y(s(t), x)\} = \rho^{j+1}(\partial_s - j)Y(s(t), x). \tag{29}$$

For $P_{k,j}(D) = \prod_{r=0}^{k-1}(D - j - r)$, with $P_{0,j} = 1$, induction yields

$$\partial_t^k\{\rho^j Y(s(t), x)\} = \rho^{j+k} P_{k,j}(\partial_s)Y(s(t), x). \tag{30}$$

Spatial differentiation commutes with this identity. Taking $j = 1$ proves the velocity bound at every order. The residual is exactly

$$F = \rho^2\{W_s - W + (W \cdot \nabla)W\}(s(t), x) - \nu\rho\Delta W(s(t), x). \tag{31}$$

Every mixed derivative of the braces is bounded by the finite product rule and the hypotheses. Apply (30) with $j = 2$ and $j = 1$ to obtain orders ρ^{2+k} and ρ^{1+k}, respectively. This gives the force bound, including every spatial multi-index. In particular no temporal derivative of the convection term has been omitted.

Supports do not increase under differentiation or multiplication. For either field Z, the squared integral of an indicated derivative is at most

$$|K|C_{k,\alpha}^2 \int_0^\infty (1 + t)^{-2-2k}\, dt = \frac{|K|C_{k,\alpha}^2}{1 + 2k}.$$

The zero datum remains zero, so residual realization applies. The chain rule shows that $\Xi(s(t), a)$ solves the U trajectory equation with initial label a; uniqueness identifies it with the actual trajectory. The clock is increasing onto $[0, \infty)$ with inverse $e^s - 1$, so it loses no finite computational time. Finally logarithm on positive arguments, the rational powers of $1 + t$, and the finite-index evaluation of the template and its derivatives are effective. This proves all assertions. □

9.2 Loaded Euclidean processors

Theorem 9.2 (Two full-box processors on Euclidean space)**.** *For every finite machine and input, and every fixed computable $\nu > 0$, either the move-first recorder of Lemma 5.2 or the six-rule recorder of Lemma 6.1 yields effective smooth forcing on $\mathbb{R}^3$ with one compact spatial support, all mixed derivatives bounded, and period one for $t \geq 1$, such that*

$$\exists t \geq 0 :\ (\Phi_t(0))_1 < -1 \quad \Longleftrightarrow \quad \textit{halting.}$$

The zero-data solution is unique in Lemma 7.3's class, has pressure zero and uniformly bounded energy. Separately, the six-rule processor admits a force square-integrable on all space-time with the same support and event.

Proof. Write Γ for the full alphabet of the recorder being used. For the six-rule table use initial frontier 2 and base $B = 2|\Gamma| + 2$. Give nonhalting controls lanes $3i + [0, 1]$ and halting controls lanes $-3i + [0, 1]$, enumerating each group from 1; the second coordinate is $[0, 1]$. The minimal subdivision has gap B^{-2}. Use full thickness $[-1, 1]$, heights $10i$, and the transported-cutoff construction in Section 7.2, with initial padding $1/(10B^3)$. Target horizontal padding is at most $1/(10B^2)$, below half the gap; vertical middle intervals are also separated.

For the move-first table use a single halt lane $-3+[0,1]$, other lanes $2+2j+[0,1]$, base $B=2|\Gamma|+1$, the minimal subdivision, thickness $[-1,1]$, and heights $4i$. Use current-box padding $1/(10B^2)$. The middle-phase separation is $2(1+1/(10B^2))<4$; the planar gaps handle the other phases. This localization has a fixed current padding, as distinct from the transported initial padding in the six-rule version.

The rational initial code P_* is reached from zero along $\theta(t)P_*$ by a localized translation during $[0,1]$. Repeating the chosen cycle produces a template W with one compact support and bounded mixed derivatives. The code identity at times $1+n$ follows by induction through the local rules; the respective recorder invariant supplies every next checkpoint in finite positive time. A halt lies below -1 in the first coordinate. In a nonhalting run all branch endpoints have positive first coordinate, and (8) keeps it positive throughout every cycle. Loading is nonnegative. Lemma 7.3 completes the PDE argument. Uniform support and boundedness give a uniform energy bound. The separate square-integrable choice follows from Lemma 9.1, without claiming periodicity in physical time for that choice. □

9.3 Two realizations of a left frontier

Proposition 9.3 (Separate columns or solid-box parking)**.** *Both left-frontier tables have compact whole-space realizations, periodic after $t=1$, with fixed label zero and event $X_1(t;0)<-1/2$. The stopping version can use distinct-height stream functions and has uniqueness with pointwise-in-time H^1 comparison pressure. The continuing version transports full solid boxes by evacuation, reshaping and delivery, and represents every local step, including post-halt idle computation. It has in particular uniqueness with comparison pressure in $C([0,T];H^1)$.*

Proof. Let Σ be the full work–history–mark alphabet of the chosen left-frontier table, including its record symbols and any halting idle records in the continuing version. Use odd digits in base $B=2|\Sigma|+2$, offset -2 for C_h and $2,4,6,\dots$ for other controls. The two head-relative fractions and height zero give a rational initial code Z_*. The rightmost nonempty history cell is the original cell -2, an absolute landmark.

For the stopping table, apply the stream-function construction (9) at heights $z_i=3i$. Choose source and target column paddings no greater than $1/2$ and one third of their respective family gaps. A middle layer padding $1/2$ keeps distinct layers disjoint. The explicit thickening bound following that construction proves the neighborhood action. Write $Z_*=(x_*,y_*,0)$. With a flat switch θ from zero to one, load the origin along $\theta(t)Z_*$ using the velocity $\nabla\times(0,0,H)$, where $H(t,x,y,z)=\theta'(t)\chi(x,y,z)(x_*y-y_*x)$. Choose χ compactly supported in $\mathbb{R}^3$ and equal to one near the entire segment. The gate θ' vanishes on the time collars. All nonhalt source and target lower-left first coordinates are at least two. Inequality (10) therefore excludes the observer during each period; loading has nonnegative first coordinate. Halt codes lie in $(-2,-1)$.

For the continuing table retain its unrestricted first-row history symbol and its idle work rules. Thicken every rectangle by $[-1,1]$. Choose L ten larger than the maximum of zero and every source or target right endpoint. Reserve parking projections $(L+4i,0)+[-1,1]^2$ and let δ be the minimum of one and the gaps in each union of one rectangle family with these reservations. Use padding $\delta/10$, travel height five, and $7N$ slots: evacuate all sources by three translations each, reshape all parked boxes by $\operatorname{diag}(1+\theta(\lambda_i-1),[1+\theta(\lambda_i-1)]^{-1},1)$, and deliver by three translations each. Horizontal widths stay at most one, vertical half-width is one, and the reservations miss both original families. Lemma 3.3 proves exact action on the whole boxes and their neighborhoods even with source–target overlap. A first localized translation loads Z_* from the origin. The sampling identity is now valid for every $k\ge 0$, not just until halting: $\Phi_{1+k}(0)=Z(C_k)$ for the continued table.

For a nonhalting run all used endpoint centers have first coordinate at least two, parking centers are farther right, and half-widths are at most $1/2$. Translation, parking strain, and waiting therefore keep first coordinate positive. A halt checkpoint lies in $(-2,-1)$. Proposition 3.2 proves the analytic assertions for both constructions. Their compact support also gives uniformly bounded kinetic energy. □

10 Persistent marks and guarded history routes

The preceding recorders clear the work-head mark at each checkpoint. We now retain that mark, or replace the moving history pointer by a guarded boundary between occupied and unused cells. These changes give different partial maps on arbitrary tapes, even though their initialized executions simulate the same machine. We prove each inverse on its declared domain before using it to separate the geometric branch images. The use of retained history follows the reversible-computation idea of Bennett [4]; the local guards and geometric interfaces are established here.

In this section an input word occupies cells $0, 1, \ldots$, the other work cells are blank, and the initial head is at zero. All constructions include initial halting. When one halt state h is used, identify the original halting states with h and redirect incoming instructions; there are no outgoing halting instructions to preserve. If necessary adjoin an unreachable halt state. We write $X(t;a) = \Phi_t(a)$ for the trajectory from a.

10.1 A persistent mark and input-dependent placement

A force can be periodic from time zero if no separate loading interval is needed. We achieve this by translating the initial control's coding square so that the prescribed particle already represents the input. The translation depends effectively on the finite input. First we give the recorder that this periodic processor implements.

Lemma 10.1 (A recorder with a persistent work-head mark)**.** *Let a machine have work alphabet Γ, halt state h, and instructions $\gamma = (q,a) \mapsto (q_\gamma^+, a_\gamma^+, d_\gamma)$, where $q \neq h$ and $d_\gamma \in \{-1,0,1\}$. There is an effective finite partial one-head table with these properties. At the checkpoint after n work steps, it has the correct work tape, state and head, a unique mark at that head, and a history pointer at $p_n = n+2$. Each target control determines the incoming displacement; the target control and written full cell symbol determine the source control and read full symbol on the entire partial domain. If the next work-head position is $w' = w_n + d_\gamma$, the next checkpoint is reached after exactly $2(p_n - w') + 4$ local steps.*

Proof. A full symbol is $(a,e,s) \in \Gamma \times \{0,1\} \times \mathcal{H}$, where $\mathcal{H} = \{E,P\} \sqcup \{r_\gamma\}$ records empty history, a pointer, or an instruction. All displayed control families are disjoint. The following is the complete table; free cell indices range over their finite alphabets:

control	read and guard	target	write	d
A_q	$(a,1,s),\ s \neq P$	B_γ	$(a_\gamma^+, 0, s)$	d_γ
B_γ	$(c,0,s),\ s \neq P$	R_γ	$(c,1,s)$	1
R_γ	$(c,0,s),\ s \neq P$	R_γ	$(c,0,s)$	1
R_γ	$(c,0,P)$	$C_{q_\gamma^+}$	$(c,0,r_\gamma)$	1
C_q	$(c,0,E)$	L_q	$(c,0,P)$	-1
L_q	$(c,0,s),\ s \neq P$	L_q	$(c,0,s)$	-1
L_q	$(c,1,s),\ s \neq P$	A_q	$(c,1,s)$	0

The first row has $q \neq h$ and $\gamma = (q, a)$; there is no rule from A_h. Initially the actual head and the unique mark are at zero, the work tape is the input, and history is E except for P at cell 2.

At checkpoint n, $|w_n| \leq n$, so after the first row the head is at $w' \leq n+1 < p_n$. That row clears the old mark, and the second installs the new one at w'. The right scan reaches the pointer without meeting another mark, writes r_γ, and visits the empty cell $p_n + 1$. The next row installs the new pointer there. The left scan returns through the newly written record to the unique mark and enters $A_{q_\gamma^+}$ without clearing it. Hence every invoked rule is defined, the work instruction is correct, and the checkpoint invariant is restored. The two continuing scans take $p_n - w' - 1$ and $p_n - w'$ steps; the other five steps give the stated total. This also handles $d_\gamma = 0$. Induction gives a defined execution until and including the first halting checkpoint, and indefinitely in the nonhalting case.

We check the inverse independently of this initialization. Targets $B_\gamma, R_\gamma, C_q, L_q, A_q$ have incoming displacements $d_\gamma, 1, 1, -1, 0$, respectively. Undo that displacement to locate the written cell. At B_γ, the index γ recovers the old control and overwritten work symbol, and the old mark was one. At R_γ, the written mark is one for entry and zero for the scan, so it identifies the source and the old mark. At C_q, the written record identifies γ and its former value P. At L_q, a written pointer identifies entry, whereas every loop has history different from P. At A_q, the unique source control is L_q and the full symbol is unchanged. All unspecified tracks are retained. Thus every possible output has at most one preimage on the full domain. □

Theorem 10.2 (An input-dependent entry chart)**.** *For each machine and finite input, and each fixed positive computable viscosity ν, there is an effective smooth force on $\mathbb{R}^3$ with one compact spatial support, bounded mixed derivatives, and period one from time zero, whose zero-data solution satisfies*

$$\exists t \geq 0 : \; X_1(t; (2, 0, 0)) < 0 \quad \Longleftrightarrow \quad \textit{the machine halts.}$$

The solution is smooth with pressure zero and is unique among smooth solutions for which, on every finite interval, $u \in C_t H^2 \cap C_t^1 L^2$, u and ∇u are bounded, and $p \in C_t H^1$. As a separate force choice, the same event is realized with every mixed derivative of velocity and force in space-time L^2 and bounded by a derivative-dependent multiple of $(1+t)^{-1}$.

Proof. Prepend a nonhalting state q_{in} which, on every work symbol, preserves it, stays put, and enters the original initial state. This includes original initial halting while keeping the new initial state nonhalting. Apply Lemma 10.1. For its full alphabet of size m use odd digits $e(\sigma) \in \{1, 3, \ldots, 2m-1\}$ in base $B = 2m + 2$, and write

$$c(\omega) = \sum_{j \geq 0} e(\omega_j) B^{-j-1}, \qquad (\xi, \eta) = \big(c(\sigma_{-1}\sigma_{-2}\cdots), c(\sigma_0\sigma_1\cdots)\big).$$

The initial values ξ_0, η_0 are rational, because the initialized streams have finite prefixes and known constant tails. Set

$$O(A_{q_{\text{in}}}) = (2 - \xi_0, -\eta_0), \qquad O(A_h) = (-2, 0), \qquad O(P) = (4 + 2j, 0)$$

for the remaining controls in a finite ordering. All squares $O(P) + [0,1]^2$ are separated. Every nonhalting square lies strictly to the right of $x_1 = 1$, and the halting square is $[-2, -1] \times [0, 1]$. The initial code at height zero is exactly $(2, 0, 0)$.

Apply Lemma 5.1 with digits $d(\sigma) = e(\sigma)$, base $B = 2m + 2$, and the state translations just specified. Its incoming hypotheses are exactly those of Lemma 10.1. With $I_\sigma = (e(\sigma) + [0,1])/B$, use the minimal left-split branches of (19), including every left letter for a left move. The state translations conjugate these maps without changing their reciprocal diagonal linear parts. The

shared lemma therefore gives the exact updates and separate source and target separation on full closed rectangles. If an absolute head position is wanted at a checkpoint, the number of contiguous records gives n, and the pointer's relative position locates the head relative to its known absolute position $n+2$. The gapped digits permit each finite prefix needed for this recovery to be read with increasing precision.

Apply the exponential interpolation of Lemma 7.1, with heights $10i$ and the following separated phase intervals: lifting on $[1/8, 2/8]$, horizontal motion on $[3/8, 5/8]$, and descent on $[6/8, 7/8]$. On each interval use a rescaled flat step σ from (5). Let δ be one quarter of the minimum of 1 and the positive planar source and target gaps. Use plateau padding $\delta/3$ and support padding $2\delta/3$. The same gap and height estimates in the lemma make these collars disjoint. Here the local potential can be written

$$\tfrac{1}{2}C_i' \times (x - C_i) + (0, 0, \eta' \log \lambda_i \, (x_1 - C_{i1})(x_2 - C_{i2})).$$

Its curl is the required translation plus $\eta' \log \lambda_i \, (x_1 - C_{i1}, -(x_2 - C_{i2}), 0)$. It realizes the whole cuboid and a neighborhood by the plateau argument.

Repeat this unit-time field as U, and put $f = U_t + (U \cdot \nabla)U - \nu \Delta U$ at the prescribed physical viscosity. The zero collars give $U(0) = 0$ and smooth period-one gluing. Lemma 7.3 gives the solution, uniqueness, support and effective mixed-derivative bounds. Its hypotheses hold because this is a finite curl formula on a compact spatial set and one compact time period. The formula includes every branch and never requires the run on the selected input.

At integer times, induction gives the exact local configuration through the first halt, or at every integer time for a nonhalting run. A halt therefore reaches the negative square. During a nonhalting run, both endpoint squares of every used branch are nonhalting. Every moving center has first coordinate greater than one and every first half-width is at most $1/2$, because the endpoint widths are at most one and the exponential scale is monotone between them. The actual particle stays positive during all phases. This proves the equivalence for all real times, including the added entry step.

For the separate decaying choice apply Lemma 9.1: $\widehat{U}(t, x) = (1+t)^{-1} U(\log(1+t), x)$, with its own residual at the same ν. That lemma gives the stronger temporal-order-k bound $O((1+t)^{-1-k})$ for every spatial derivative, hence the stated bound and space-time L^2 conclusion on the common compact support. The clock is onto $[0, \infty)$, and the chain rule identifies the new trajectory with $X(\log(1+t); (2, 0, 0))$. The fixed event is unchanged. □

10.2 Using a fresh half-line instead of a frontier symbol

The next rule reads a neighboring history cell. Its inverse therefore uses a full three-cell cylinder, rather than only the written cell.

Lemma 10.3 (Three-cell fresh-tail recorder)**.** *There is an effective partial rule on cells at offsets $-1, 0, 1$ that simulates the machine, is injective on its entire domain of relative tapes, and has incoming displacement determined by the target state. Its allowed cylinders admit separately separated closed-rectangle maps with linear parts* $\operatorname{diag}(B^{-d}, B^{d})$.

Proof. Use symbols $c_i = (a_i, \ell_i, \mu_i)$ in $C = A \times (\{\star, \perp\} \sqcup \mathcal{R}) \times \{0, 1\}$ and controls N_q, B_q, J_r, S_r. Only N_q with $q \in H$ halt. After writing, a move d reindexes a relative tape by $c_j' = \bar{c}_{j+d}$. The following rules give the complete domain:

1. At N_q, $q \notin H$, require $\mu_0 = 0$ and $\ell_0 \neq \perp$. With $r = (q, a_0)$ write $b(r)$, move $d(r)$, and enter J_r.

2. At J_r, require $\mu_0 = 0$ and $\ell_0 \neq \perp$. Mark cell zero, move right, and enter S_r.

3. At S_r, require $\mu_0 = 0$. If $\ell_0 \neq \perp$, scan right. If $\ell_0 = \ell_1 = \perp$, write record r at zero, move left, and enter $B_{q^+(r)}$.

4. At B_q, require $\ell_0, \ell_1 \neq \perp$. Scan left if $\mu_0 = 0$; if $\mu_0 = 1$, clear the mark, stay, and enter N_q.

Initialize history as $\star$ below absolute cell 2 and $\perp$ from 2 onward, all marks zero, the prescribed work tape, and control N_{q_0}. At checkpoint n the first fresh cell is $F = n + 2$, records occupy $2, \dots, F - 1$, and the work head satisfies $h \leq F - 2$. The work move reaches $k \leq F - 1$. Mark k, scan to F, write the record, and return from $F - 1$ to k. The return's two-history guard holds even at its first cell because cell F has just been filled. Clearing the mark restores the invariant with frontier $F + 1$. There are $F - k - 1$ instructions in each continuing scan and four others, so the duration is $2(F - k) + 2$. This proves finite completion and the checkpoint halt equivalence.

For the full-domain inverse, targets J_r, S_r, B_q, N_q determine moves $d(r), 1, -1, 0$, respectively. Undo the indicated shift. At J_r, r recovers the overwritten letter and old control. At S_r, the written mark distinguishes entry from the zero-mark scan. At B_q, target offset $+2$ is the former offset $+1$; its history is fresh for a recording entry and nonfresh for a return scan. In the entry case target offset $+1$ is the former scanned cell; its record identifies r. At N_q, restore the mark to one. All other tracks are preserved. The neighbor guard is what makes this inverse valid on arbitrary allowed tapes.

Use odd digits in base $B = 2|C| + 1$ and state offsets e_s whose unit intervals have gaps at least two. A permitted tuple (s, c_{-1}, c_0, c_1) has source $(e_s + I_{c_{-1}}) \times I_{c_0 c_1}$. The full inverse just proved checks the injectivity hypothesis of Lemma 5.6 with $(a, b) = (1, 2)$ and $d \in \{-1, 0, 1\}$. Only cell zero is written, which is within that window. The lemma therefore gives full target cylinders with prefix lengths $(1 + d, 2 - d)$, exact affine factors (B^{-d}, B^d), and separate closed-rectangle images. For one target state the incoming d is fixed, so its prefix lengths agree across incoming branches. An incompatible digit among at most three positions yields the additional quantitative target gap B^{-3}. This is separation of the filled rectangles, including their boundaries. □

Theorem 10.4 (Compact realization of fresh-tail cylinders)**.** *The fresh-tail recorder has an effective compactly supported smooth realization on $\mathbb{R}^3$, with zero initial velocity and force periodic after one unit of loading, whose event $\exists t \geq 0 : (\Phi_t(0))_1 < -1$ is equivalent to halting. A separate choice, with the same event and one compact support, satisfies for every $k \geq 0$ and spatial multi-index α*

$$\sup_x (|\partial_t^k \partial_x^\alpha u| + |\partial_t^k \partial_x^\alpha F|) \leq C_{k,\alpha} (1 + t)^{-1-k}. \tag{32}$$

Every displayed derivative is space-time square-integrable. Both forces have the unique zero-pressure solution of Lemma 7.3 at the fixed computable viscosity.

Proof. Use the full alphabet C of Lemma 10.3 and $B = 2|C| + 1$. Place halting controls at $e_s = -3j_s$ and all others at $3j_s$, with distinct positive integer labels. Use the three-cell rectangles as plates at height zero. Branch i has middle height $4i$ and padding $B^{-3}/10$. Interpolate both planar factors independently and apply Lemma 7.2, including its reciprocal normal strain. Planar gaps separate the vertical phases; middle heights separate the padded plates, whose collar half-thickness is less than one. Thus the specified motion holds on full neighborhoods. It retains independent planar interpolation throughout the cycle.

The initial code P_* is rational. A point-centered curl translation, with center $\theta(t)P_*$ and padding one, loads zero in one unit. Repeat the plate cycle thereafter. The finite-step invariant and code maps identify the trajectory at all times $1 + n$ until halt and for all n otherwise. A halt code has first coordinate below -1. For a nonhalting run, loading is nonnegative and each later pair of coded

endpoints is positive; the two-coordinate convex interpolation of Lemma 7.2 proves the all-time exclusion. Use residual realization for the periodic choice and Lemma 9.1 for the separate decay choice. □

10.3 Guarded history cylinders in a torus chart

An occupied prefix can replace the explicit pointer. The resulting inverse must distinguish arrival from the recording cell from an ordinary return-scan step. Testing the next history cell supplies that distinction. Unlike the fresh-tail recorder, the marked return below does not test its right neighbor; we retain this larger partial domain.

Lemma 10.5 (An occupied-prefix history recorder)**.** *Let a machine have state set Q, halting set H, alphabet Γ, and instructions $\tau = (q, a) \mapsto (q_\tau^+, w_\tau, d_\tau)$. There is an effective partial injective map on head-relative tapes whose domain depends only on the control and full cells at offsets $-1, 0, 1$. With the initialization below, occupied history at checkpoint n consists exactly of the absolute indices less than $F_n = n + 2$. A work step ending at head position j takes exactly $2(F_n - j) + 2$ local steps. Halting at a checkpoint, including the initial one, is equivalent to halting of the original machine.*

Proof. Full symbols are (a, h, m), with work letter a, mark $m \in \{0, 1\}$, and history $h \in \{\#, E\} \sqcup \mathcal{R}$, where $\mathcal{R} = (Q \setminus H) \times \Gamma$. History is called occupied when $h \neq E$. A move d reindexes the post-write relative tape as $\xi_k^{\text{new}} = \widetilde{\xi}_{k+d}$. The complete table is

control	guard	write at 0	target	d
Run_q	$m_0 = 0,\ h_0 \neq E$	$a_0 \leftarrow w_\tau$	Mark_τ	d_τ
Mark_τ	$m_0 = 0,\ h_0 \neq E$	$m_0 \leftarrow 1$	Seek_τ	1
Seek_τ	$m_0 = 0,\ h_0 \neq E$	unchanged	Seek_τ	1
Seek_τ	$m_0 = 0,\ h_0 = h_1 = E$	$h_0 \leftarrow \tau$	$\mathsf{Back}_{q_\tau^+}$	-1
Back_q	$m_0 = 0,\ h_0 \neq E,\ h_1 \neq E$	unchanged	Back_q	-1
Back_q	$m_0 = 1,\ h_0 \neq E$	$m_0 \leftarrow 0$	Run_q	0.

The first row requires $q \notin H$ and $\tau = (q, a_0)$; all unmentioned tracks are unchanged. Initialize the original work tape with head zero, all marks zero, history # at every index below 2, and E elsewhere.

At checkpoint n the original head has $|j_n| \leq n$. The work step ends at $j \leq n + 1 < F_n$, on occupied history. Mark that cell and scan right across occupied unmarked cells to F_n. Both F_n and $F_n + 1$ are unused, so the recording row applies and moves left. Every unmarked return-loop cell has occupied history at itself and its right neighbor, including the new record at F_n. The return reaches j, clears the mark, and enters the new work state. The continuing scans each take $F_n - 1 - j$ steps, with four other steps. This proves finite completion, the duration, and the new occupied boundary $F_n + 1$. Induction gives the stated computation and halt equivalence.

For the full-domain inverse, target controls determine displacement. The moves are d_τ into Mark_τ, one into Seek_τ, minus one into Back_q, and zero into Run_q. At Mark_τ, undoing the move and restoring the read work symbol encoded by τ recovers the predecessor. At Seek_τ, target offset -1 is the written cell; its mark distinguishes the marking entry from the scan. At Back_q, target offset 2 is the former offset 1. Its history is E for a recording entry and is occupied for a return loop. In the recording case, target offset 1 contains τ, identifying the old control and restoring its history to E. Finally, the only entry into Run_q clears a mark without moving, so the mark is restored to one. These inverses use only the table's guards and retained symbols, not the special shape of initialized history. □

Lemma 10.6 (The full cylinders of the guarded recorder)**.** *The recorder of Lemma 10.5 admits a finite effective list of positive-area closed source and target rectangles, separately pairwise disjoint. If $\mathcal{A}$ is its full alphabet and $B = 2|\mathcal{A}| + 1$, a branch of displacement d has an exact affine map with linear part* $\operatorname{diag}(B^{-d}, B^{d})$.

Proof. Apply Lemma 5.6 with window parameters $(a, b) = (1, 2)$ to the complete table of Lemma 10.5. That table reads only offsets $-1, 0, 1$, writes only offset zero, and has displacements $d \in \{-1, 0, 1\} \subset [-1, 2]$; its full-domain injectivity was proved above. Thus every hypothesis of the window lemma holds.

To fix the actual rectangles, choose odd digits $\gamma(a) \in \{1, 3, \dots, 2|\mathcal{A}| - 1\}$ and $B = 2|\mathcal{A}| + 1$. Write

$$e(P_1 \cdots P_k) = \sum_{j=1}^{k} \gamma(P_j) B^{-j}, \qquad I(P) = e(P) + [0, B^{-k}], \qquad I(\varnothing) = [0, 1],$$

using the infinite series for an infinite word. Choose separated state intervals $o_s + [0, 1]$ and use the code

$$r(s, \xi) = (o_s + e(\xi_{-1}\xi_{-2} \cdots),\ e(\xi_0\xi_1 \cdots), 0).$$

For each permitted triple (l, a, b) at offsets $-1, 0, 1$, let a' be its written full symbol. The source prefix pair is $((l), (a, b))$, and the target pair is

$$\begin{cases} ((a', l), (b)), & d = 1, \\ ((l), (a', b)), & d = 0, \\ (\varnothing, (l, a', b)), & d = -1. \end{cases}$$

Take the corresponding products of prefix intervals with the source and target state offsets. Both arbitrary tails are unchanged, so the window lemma gives the exact center-to-center affine map with factors B^{-d}, B^{d}, onto the whole specified target cylinder and filled rectangle. Its separation conclusion applies to these complete closed rectangles, including their boundaries. All endpoints and positive within-family gaps are effectively rational. The odd digits also put all infinite-word codes strictly in $(0, 1)$ in each normalized coordinate. □

Theorem 10.7 (A guarded processor in a fixed torus chart)**.** *For each machine and finite input, at any fixed positive computable viscosity there is an effective smooth mean-zero force on the unit flat three-torus, with bounded mixed derivatives and period one for $t \geq 1$, whose smooth zero-data solution has pressure zero and satisfies*

$$\exists t \geq 0 :\ X(t; [(-1/16, 0, 0)]) \in \{[x] : x_1 \bmod 1 \in (0, 1/2)\} \quad \Longleftrightarrow \quad \textit{the machine halts.}$$

The pressure is normalized to mean zero, and the solution is unique in the classical torus comparison class of Section 2.

Proof. Use the occupied-prefix recorder. Assign offsets $2j$, $j \geq 1$, to halting run controls and $-2j$ to all other controls, with distinct indices within each group. The initial code r_{in} is rational: the left stream is constant and the right stream is eventually constant. If N is the number of cylinders, set

$$O = 1 + \max_s |o_s|, \qquad K = O + 6\max(1, N) + 4, \qquad \sigma = (8K)^{-1}, \qquad b_* = (-K/2, 0, 0).$$

Use the linear version of Lemma 7.1 for the branch boxes of third interval $[-1, 1]$, obtaining V_1. Use the same lemma with scale one on the single pair of cubes of half-width one centered at b_* and r_{in},

obtaining a loader V_0. Extend both fields by zero outside their unit time intervals. Their supports lie in $(-K,K)^3$. For V_1, first-coordinate centers have magnitude at most O, second-coordinate centers lie in $[0,1]$, horizontal half-widths are at most $1/2$, and heights are at most $6N$. For V_0, the horizontal center is bounded by $\max(K/2,O)$, half-widths are one, and height is at most six. The padding is at most $1/4$ in either application. Each bound is strictly less than K.

Periodize the rescaled fields in space and the step field in time:

$$U(t,[x]) = \sigma \sum_{k\in\mathbb{Z}^3} \left[V_0(t,(x-k)/\sigma) + \sum_{n\geq 1} V_1(t-n,(x-k)/\sigma) \right].$$

Spatial support lies in chart boxes of radius $1/8$, so distinct spatial copies are disjoint. The time collars make the gluing smooth, initially zero and periodic after time one. Every mixed derivative is bounded. The rescaled blocks remain curls of compactly supported potentials, so U has zero spatial mean. Define the force at the physical viscosity by $f = U_t + (U\cdot\nabla)U - \nu\Delta U$. It has zero mean because the convection term is $\operatorname{div}(U\otimes U)$ and the Laplacian integrates to zero. Lemma 7.3 now gives the unique solution $(U,0)$. Recomputing the residual after the chart scaling is essential: it does not identify two different viscosities.

The prescription is effective without deciding a space or time seam. Given approximations $\widehat{x},\widehat{t}$ with error at most $1/4$, all possibly active spatial indices satisfy $\|\widehat{x}-k\|_\infty \leq 1$, and repeated time indices satisfy $|\widehat{t}-n|\leq 2$. Evaluate this finite superset using the flat cutoffs and their derivative bounds. The finite table, not an executed machine history, determines every coefficient.

Since $\sigma b_* = (-1/16,0,0)$, loading sends the fixed particle to $[\sigma r_{\mathrm{in}}]$. At time $1+j$ it is the code after j local steps, through the first halt or forever in the nonhalting case, by the exact whole-box maps. Halting, including initial halting, gives a positive first-coordinate representative below $1/8$, inside the target half-circle. For a nonhalting input both loading endpoints and both endpoints of every subsequent used branch have negative first coordinate. The linear-scale identity keeps the first coordinate negative at every intermediate time. All representatives stay in $(-1/8,0)$, disjoint from the target modulo one. This proves the all-time equivalence. □

11 Wider guards and alternative comparison classes

The transport lemmas also accept tables which differ on tapes never visited by the initialized computation. Those differences matter: they change the complete source and image cylinders on which the smooth extension is prescribed. We give the remaining tables with their actual guards and then isolate the extra pressure arguments.

11.1 A shuttle aligned at the origin

Theorem 11.1. *On the unit torus there is an effective smooth mean-zero force, periodic from time zero with period one and with bounded mixed derivatives, whose unique zero-data classical solution has*

$$\exists t\geq 0: \quad X_1(t;0) \pmod 1 \in (1/16,1/4) \quad\Longleftrightarrow\quad \textit{halting.}$$

The constructed pressure is zero. Alternatively the force may be projected to a solenoidal force with the same velocity and event, and with the correspondingly changed mean-zero pressure.

Proof. Add a fresh nonhalting initial state $*$, without incoming transitions, which writes back the scanned symbol and stays while entering the original initial state. This handles an originally

halted input. Use work, flag and history tracks $\Sigma = \Gamma \times \{0,1\} \times (\{\bot, E\} \sqcup \mathcal{R})$, and controls R_q, A_r, B_r, C_q, D_q. Its complete table is

$$\begin{aligned}
R_q : (a,0,g) &\mapsto A_r : (b_r,0,g), d_r, \\
A_r : (a,0,g) &\mapsto B_r : (a,1,g), 1, \\
B_r : (a,0,g) &\mapsto B_r : (a,0,g), 1 \qquad g \neq E, \\
B_r : (a,0,E) &\mapsto C_{p_r} : (a,0,r), 1, \\
C_q : (a,0,\bot) &\mapsto D_q : (a,0,E), -1, \\
D_q : (a,0,g) &\mapsto D_q : (a,0,g), -1 \qquad g \neq E, \\
D_q : (a,1,g) &\mapsto R_q : (a,0,g), 0 \qquad g \neq E.
\end{aligned} \tag{33}$$

In particular its first two rows do not exclude the frontier symbol; this is a wider domain than the move-first table. Into A_r the record identifies the work instruction; into B_r the output flag distinguishes entry and loop; into C_q the record determines the incoming scan; into D_q a written E distinguishes entry and loop; into R_q only clearing a flag is possible. The destination fixes the incoming displacement. These checks include every copied track and every permitted frontier case.

Initialize at R_*, with all flags zero and only E at cell two. At ready step n the frontier is $e_n = n+2$, the records are at $2, \dots, n+1$, and the head is h_n, $|h_n| \le n$. The work move reaches $h' = h_{n+1} < e_n$; the scan marks that cell, records at e_n, creates the next frontier, returns to the mark and clears it. Its count is $2(e_n - h') + 4$ and all its tests hold. Thus it has exactly the required initialized behavior despite its wider full domain.

Use even digits in base $b = 2|\Sigma|$ and initial fractions ℓ_*, r_*. Let S be the control set and m count the branches in Lemma 5.1, and put $L = 100(1 + |S| + m)$. Give R_* offset $-\ell_*$, other nonhalt states offsets $-4i$, and ready halt states offsets $L + 4i$. The code is $(o_s + \ell, r - r_*, 0)$, so the initial point is zero. The incoming conditions give separately disjoint full rectangle families. Thicken each by $[-1,1]$, lift to height $10i$, and use the linear reciprocal middle scale. Choose padding at most $1/10$ small enough for both family gaps. Vertical phases can use fixed column potentials $(0, \dot{Z}x\chi, 0)$, with columns covering $[-3, 10i+3]$; middle phases use horizontal Hamiltonians in disjoint height bands. All paths and supports are contained in $(-2L, 2L)^3$: their coordinate bounds are $[-4|S| - 3, L + 4|S| + 4]$, $[-4, 4]$, and $[-3, 10m + 3]$. This verifies the hypotheses of the lift schedule on neighborhoods of the whole boxes.

Scale by $\gamma = (8L)^{-1}$ into the torus patch $(-1/2, 1/2)^3$: $U(t,x) = \gamma W(t, x/\gamma)$. The support lies inside $(-1/4, 1/4)^3$, the potentials scale by γ^2, and the residual is recomputed at viscosity ν by Lemma 2.3. Periodicity starts at zero because the input is already encoded there and the pulse has idle endpoint collars. At integer times the particle is the correct code. Halt first coordinates before scaling lie between $L + 4$ and $2L$. For a nonhalt branch its endpoint first coordinates are at most one; the linear-scale identity keeps them at most one, and the weaker bound $3/2$ also follows from center and width estimates. After scaling every nonhalt path lies in $(-1/4, 1/16)$, disjoint modulo one from the event. The initialized dummy step makes the implication valid even for an originally halted machine.

For the optional projection solve $\Delta q = \operatorname{div} g$, $\int q = 0$, where $g = \mathcal{F}_\nu[U]$. Proposition 2.4 gives the effective force $f = g - \nabla q$ and pressure $p = -q$, with all mixed derivative bounds and periodicity retained. The pressure change is part of this alternative; the direct residual had pressure zero. □

11.2 A persistent marker and cursor shifts of two cells

Theorem 11.2. *There is a unit-torus realization, periodic for $t \ge 1$, with fixed label $a_* = (1/4, 1/4, 1/4)$. Its event*

$$\exists t \ge 0 : \quad X_1(t; a_*) \pmod 1 \in (2/3, 5/6)$$

occurs exactly when the machine halts. It uses an injective marked-head table whose cursor shifts may equal two. Its force is effective, smooth and mean zero, with bounded mixed derivatives and the zero-data uniqueness class of Proposition 3.2.

Proof. Normalize to one halt state and use work, permanent work-head marker, and history tracks $(a, m, g) \in \mathcal{A} := \Gamma \times \{0, 1\} \times (\{\bot, \#\} \sqcup \mathcal{R})$. Let S be the set of controls M_q, L_q, R_r. With the cursor at zero, make exactly the following tests and writes, then reindex by $T'_i = T_{i+e}$ for the displayed shift e.

1. In M_q, q nonhalting, require $m_0 = 1$ and, if $d_r \neq 0$, $m_{d_r} = 0$. Write b_r at zero, move the marker to d_r if necessary, and enter R_r with shift $e = d_r + 1$.

2. In R_r, a cell with $m_0 = 0, g_0 \neq \#$ gives shift 1 in the same control. If $m_0 = 0, g_0 = \#, g_1 = \bot$, write $g_0 = r, g_1 = \#$ and enter L_{p_r} with shift zero.

3. In L_q, if $m_0 = 0, g_0 \neq \#$, shift -1 in that control. If $m_0 = 1$, enter M_q with shift zero.

All unmentioned entries are retained. Into R_r, a main entry has output $m_{-1} = 1$, whereas a scan has $m_{-1} = 0$. The record recovers the overwritten work letter, displacement and marker changes. Into L_q, a log write has output $g_1 = \#$ and record r at zero, whereas a scan has $g_1 \neq \#$. Its tests recover the overwritten pair. Into M_q only the matching control change is possible. This is a full-domain inverse; uniqueness of a marker was not assumed on arbitrary tapes.

Initially the sole marker is at zero, the sole pointer at two, and all other history is blank. At work step n, head and marker are at h, pointer at $n + 2$, and records at $2, \dots, n + 1$. The work update puts the marker at $h' = h + d_r$ and the cursor at $h' + 1 \leq n + 2$. The forward scan reaches the pointer, logs, and returns to h'. All tests hold since $h' < n + 2$. The cycle length is

$$1 + [n + 2 - (h' + 1)] + 1 + [(n + 2) - h'] + 1.$$

This proves finite continuation, exact work simulation and halting.

Use even digits, blank zero, in base $B = 2|\mathcal{A}| + 1$. Enumerate every full word on $[-2, 2]$ with an applicable rule, giving J branches. Lemma 5.6 gives prefix lengths $(2, 3)$ and $(2 + e, 3 - e)$, with $e \in \{-1, 0, 1, 2\}$. Each output length is at most four, so gaps after chart scaling by $l = [100(|S|+1)]^{-1}$ are at least lB^{-4}. Place the halt square at $(3/4, 1/4)+l[0, 1]^2$ and other squares at $(1/4+2il, 1/4)+l[0, 1]^2$; the latter lie below $x_1 = 1/3$. At base height $1/4$, lift branch j to $1/2 + j/[4(J + 1)]$, use exponential scales $B^{-e_j\theta}, B^{e_j\theta}$, and lower. Padding $\min(1/200, 1/[32(J + 1)], lB^{-4}/3)$ separates supports. The middle Hamiltonian is

$$\dot{c}_x(y - c_y) - \dot{c}_y(x - c_x) - e_j\dot{\theta} \log B\,(x - c_x)(y - c_y).$$

It gives the intended derivative on the full moving rectangle. The plateau margins and finite scale bounds extend this to an open neighborhood, as in Lemma 3.1.

Load the fixed label along its segment to the rational initial code in time one, then repeat the unit field. Integer samples give the full local execution. In a nonhalt period the center is below $1/3$ and the width at most l, so the whole path remains below $1/2$; loading also avoids the event. Halt codes lie in $[3/4, 3/4 + l]$. Thus every real time is accounted for. The pointer site $n + 2$, record count n, and relative pointer offset recover the absolute head position at main checkpoints. The residual construction proves the remaining assertions. □

11.3 A split direction shuttle in the whole space

Theorem 11.3. *On $\mathbb{R}^3$ an effective compactly supported force with bounded mixed derivatives and period one after $t = 1$ realizes the halting event $X_1(t;(1,0,0)) < 0$. Uniqueness holds with bounded $u, \nabla u$, $u \in CH^2 \cap C^1L^2$ and pressure modulo constants in CH^1 on each finite interval. An alternative force is square-integrable in space-time, with the same label, event, support and derivative bounds.*

Proof. Use exactly the direction table of Lemma 5.3 without its passive track, renaming the controls by $M \mapsto S$, $G \mapsto R$, $A \mapsto I$, and $B \mapsto J$, and the frontier by $F \mapsto \#$. This is a bijection of its entire partial table, including the nonfrontier tests. Write Σ for the resulting three-track alphabet. Initialize frontier at one and zero flags. At work step j the frontier is $j+1$ and the cycle length is

$$2(j - h_j) + 5.$$

Use odd digits in base $B = 2|\Sigma| + 1$, offsets $3, 6, \dots$ for nonhalt controls and $-3, -6, \dots$ for halt main controls. In this realization split *every* one-cell rule by every left symbol c, including the stationary and right-moving rules. Its source is $I(c) \times I(a)$, and the three target rectangles, for moves $1, -1, 0$, are respectively

$$\varphi_b\varphi_c([0,1]) \times [0,1], \quad [0,1] \times \varphi_c\varphi_b([0,1]), \quad I(c) \times I(b), \qquad \varphi_a(t) = (d(a) + t)/B.$$

The maps themselves are the restrictions of (19). Source separation follows from (c, a), target separation from the fixed incoming move and (c, b). Their gaps are at least B^{-2}, and their linear parts are $\operatorname{diag}(B^{-d}, B^d)$.

Thicken by $[-1, 1]$, lift branch i to $4i$, use exponential middle scale and center interpolation, then descend. Padding smaller than $(4B^2)^{-1}$ is valid, by source gaps, height gaps, and target gaps in the three phases. A first localized translation takes $(1, 0, 0)$ to the rational initial code. Exact full-box transport then gives the local execution at every subsequent integer time. Nonhalt centers have first coordinate at least three and horizontal width at most one, so every intermediate point stays positive; the nonhalt loading segment does too. A halt sample has negative first coordinate, including initial halting after loading. Proposition 3.2 gives the force, pressure zero, and uniqueness. The decaying alternative follows from Proposition 15.1 below, applied to this same loaded template. □

11.4 Two-cell history and a gradient-only pressure condition

Theorem 11.4. *An effective smooth force on $\mathbb{R}^3$, periodic from time zero, with one compact spatial support and bounded mixed derivatives, realizes the fixed event*

$$\exists t \geq 0 : \quad X(t;(2,0,0)) \in (-2,-1) \times (0,1) \times (-1,1) \quad \Longleftrightarrow \quad \textit{halting}.$$

Velocity is unique in the class $u \in C([0,T];H^4) \cap C^1([0,T];H^2)$, $\nabla p \in C([0,T];L^2)$ for every finite T, allowing a distributional pressure gradient. Alternatively the force belongs to $L^2([0,\infty);H^j)$ for every integer $j \geq 0$, with all the same support, event and comparison-class conclusions.

Proof. Add a fresh initial state which writes back and stays, and replace all originally terminal state-symbol pairs by stay transitions into one fresh terminal state q_h. Normalize the other missing instructions similarly. Let $r = (q, a) \mapsto (p_r, b_r, d_r)$ denote the resulting nonterminal instructions, and let $\mathcal{A}$ be the work alphabet. The full cell alphabet is $\Gamma = \mathcal{A} \times \{0,1\} \times (\{e,p\} \sqcup \mathcal{R})$. Use its work, flag and log tracks W, F, L, and controls N_q, A_r, R_r, B_q. The complete rules are:

1. At N_q, $q \neq q_h$, write $W_0 = b_r$, enter A_r, and shift d_r, with no test on the other tracks.

2. At A_r, require $F_0 = 0$, set it to one, enter R_r and shift one.

3. At R_r, if $F_0 = 0, L_0 \neq p$, shift one in that control. If $F_0 = 0, L_0 = p, L_1 = e$, write $L_0 = r, L_1 = p$, enter B_{p_r} and shift minus one.

4. At B_q, if $F_0 = 0, L_1 \neq p$, shift minus one in that control. If $F_0 = 1$, clear it and enter N_q without moving.

The first output control records the overwritten work pair and shift. Into R_r, output flag one at -1 distinguishes entry from its loop. Into B_q, a log insertion has pointer at output index two and record at one, whereas a return loop has a nonpointer at two; the record and tests recover both overwritten log entries. Into N_q only unmarking is possible. Thus this map is injective on its whole domain, including the unrestricted first-row tracks.

Initialize zero flags and a sole pointer at cell three, all other log cells empty. At work step n, records occupy $3, \dots, n+2$ and pointer $g = n+3$. The work move reaches h' with $g - h' \geq 2$. The finite scan flags h', logs at g, advances the pointer, and returns to the flag. The return guard holds because the new pointer is at $g+1$. The cycle has $2(g - h' - 1) + 4$ steps. This verifies the work invariant and eventual terminal entry.

Use odd digits in radix $D = 2|\Gamma| + 1$ and write $\phi(\xi) = \sum_{j \geq 1} d(\xi_j) D^{-j}$ for the radix sum. Let L_*, R_* be the initialized left and right streams, ordered as in Lemma 5.1. Give the terminal state chart offset $(-2, 0)$, the initial chart offset $(2 - \phi(L_*), -\phi(R_*))$, and all other charts $(4 + 2j, 0)$. The initial code is $(2, 0, 0)$, every nonterminal interval lies in $x_1 > 1$, and terminal codes lie strictly in $(-2, -1) \times (0, 1)$. The infinite tails are in $[\delta, 1 - \delta]$, $\delta = 1/(D-1)$, so finite-prefix decoding of length b is possible from error less than $(\delta/2) D^{-b}$ without a boundary ambiguity.

Split full triples $[-1, 1]$. By Lemma 5.6, source prefix lengths $(1, 2)$ become $(1 + d, 2 - d)$, target cylinders are complete, and both rectangle families have gaps at least D^{-3}. Thicken by $[-1/4, 1/4]$, lift to height $4i$, use linear middle scale $\mu_i = 1 + (D^{-d_i} - 1)\theta$, and descend. Padding smaller than $D^{-3}/4$ separates all phases. Repeating the pulse, zero on $[0, 1/12]$ and $[11/12, 1]$, gives periodicity from zero without loading. At a nonhalting branch's lift and descent the first coordinate is greater than one; during the middle transfer the particle's third coordinate is exactly $4i$, outside $(-1, 1)$. Thus even a horizontal route passing over the observation box cannot cause a false visit. The integer samples and the compiler invariant prove halting equivalence. The record count and pointer site $n + 3$ recover absolute indexing at checkpoints.

For uniqueness apply Lemma 2.2(iii). The explicit compactly supported smooth field lies in $CH^4 \cap C^1 H^2$ on every finite interval and has pressure zero. The competing class in the theorem is precisely case (iii), including its distributional CL^2 gradient. Lemma 2.1 cancels that gradient against the solenoidal velocity difference without introducing a pressure L^2 norm. Thus the velocity and pressure gradient are unique, while pressure itself remains free up to a time-dependent spatial constant. Proposition 15.1 gives the stated decaying alternative. □

11.5 A flagged head and two-pass parking

Theorem 11.5. *For any machine and finitely specified initial tape, an effective smooth force on $\mathbb{R}^3$, periodic from time zero, with one compact spatial support and bounded mixed derivatives, realizes the event*

$$\exists t \geq 0 : \quad X(t; 0) \in (-6, -4) \times (-1, 1) \times (-1, 1) \quad \Longleftrightarrow \quad \textit{halting}.$$

The zero-data velocity is unique among classical smooth solutions with bounded $u, \nabla u$, $u \in CH^2 \cap C^1L^2$ and $p \in CL^2$ on every finite interval. The prescribed solution has uniformly bounded kinetic energy.

Proof. Merge terminal states into $q_\dagger$, send missing instructions there by a stay step, and add a fresh initial state q_*, with no incoming instruction, which writes back and stays while entering the old initial state. For each normalized nonterminal rule $r = (q, a) \mapsto (p_r, b_r, d_r)$ use cell letters

$$(w, h, j) \in \Gamma := \mathcal{A} \times (\{\circ, \star\} \sqcup \{[r] : r \in \mathcal{R}\}) \times \{0, 1\},$$

where $\mathcal{A}$ is the work alphabet and $\mathcal{R}$ is the finite set of normalized instruction labels. Thus Γ includes all three tracks. The complete table, with $h \neq \star$ wherever h occurs, is

$$\begin{aligned}
M_q : (a, h, 1) &\mapsto A_r : (b_r, h, 0), d_r, \\
A_r : (w, h, 0) &\mapsto R_r : (w, h, 1), 1, \\
R_r : (w, h, 0) &\mapsto R_r : (w, h, 0), 1, \\
R_r : (w, \star, 0) &\mapsto B_{p_r} : (w, [r], 0), 1, \\
B_q : (w, \circ, 0) &\mapsto L_q : (w, \star, 0), -1, \\
L_q : (w, h, 0) &\mapsto L_q : (w, h, 0), -1, \\
L_q : (w, h, 1) &\mapsto M_q : (w, h, 1), 0.
\end{aligned}$$

The destination fixes the incoming displacement. The record index recovers the first row's old work pair; a flag distinguishes the entries into R_r; a record distinguishes the entries into B_q; a written frontier distinguishes entry and loop into L_q; and entry into M_q is the unique cell-preserving last row. This proves the full one-cell inverse conditions.

Initialize the sole flag at zero and the sole frontier at three, with empty history elsewhere. At ready step k, only the work head i is flagged, records occupy $3, \dots, k+2$, and the frontier is $e_k = k + 3$. The work rule clears the flag and moves to $j = i + d_r \leq e_k - 2$, which is unflagged even for $d_r = 0$. It is flagged anew; the scan logs at e_k, advances the pointer, returns, and preserves the ready flag. The two loop lengths are $e_k - j - 1$ and $e_k - j$, with five other rules, for a total $2(e_k - j) + 4$. Thus all scans terminate and ready halting is exact.

Use odd digits in base $K = 2|\Gamma| + 2$, initial fractions x_*, y_*, and state offsets $a_{M_{q_*}} = 0$, $a_{M_{q_\dagger}} = -5$, and $4, 8, 12, \dots$ for the others. The code $(a_s - x_* + x, y - y_*, 0)$ starts at zero. Lemma 5.1 gives N full rectangle maps with scales $\lambda_i \in \{K^{-1}, 1, K\}$; thicken by $[-1/4, 1/4]$. The entire terminal block is inside the observation box because $0 < x_*, y_* < 1$. Every nonterminal block is in $x_1 > -1$. Eventually constant tails are rational even for an arbitrary finitely specified initial tape, and the pointer and record count recover its absolute head position at ready checkpoints.

For the required two-pass geometry, let g be the minimum of one and all positive separating coordinate gaps in either planar family, and set $\epsilon = g/20$. Choose

$$C = 20 + \max(\{0\} \cup \{\text{right endpoints of source and target boxes}\}), \qquad S_i = (C + 4i, 0, 0).$$

In a first pass give each source four consecutive slots: lift by ten, translate above S_i, descend, and reshape there by the linear reciprocal scale. In a second pass give each parked box three slots: lift by ten, translate above its target, and descend. The total is $7N$ slots, and every source is gone before a target is filled. Widths remain at most one. Source gaps protect the first lift, parking spacing protects strain and parking moves, target gaps protect delivery, and travel height separates every horizontal transfer from resting boxes. These inequalities remain strict under 2ϵ padding. Applying Lemma 3.1 in chronological order fixes every stationary box and gives the exact full-box map to the moving one. Thus overlapping source and target families cause no interference.

Repeat the flat-ended template from zero and take its residual. The integer sampling induction gives the table execution. In a nonhalting run every endpoint box lies in $x_1 > -1$; translations interpolate corresponding endpoint points, strain occurs far to the right, and waiting leaves the point fixed. The entire two-pass path therefore remains in $x_1 > -1$. A terminal sample lies inside the observer. This proves the event for all times.

For the exact pressure class use Lemma 2.2(iv). The prescribed compactly supported smooth velocity satisfies the reference $CH^2 \cap C^1L^2$ and bounded-gradient conditions. The competing velocity has precisely these bounds and a CL^2 pressure representative, as that case requires. Its cutoff argument uses the pressure itself, with error $CR^{-1}\|p\|_2\|u-U\|_2$, and assumes no extra integrability of its derivative. The lemma gives velocity uniqueness; the L^2 pressure representative is then zero. Compact support and bounded derivatives of the prescribed velocity give its uniform energy bound and its global material flow. □

11.6 A two-cell append and its guarded inverse

Use Lemma 5.5 with initial frontier $r_0 = 1$, renaming Ready, Go and Back as M, R, L and (E, P) as $(\perp, \star)$. The track order is work, mark, history. The Ready history coordinate and the marked Back row remain unrestricted; the unmarked Back row tests precisely the right-neighbor history. Thus the full table, not only its initialized execution, is identical. At checkpoint n, its frontier is $p_n = 1 + n > h_n$, where h_n is the work-head position, and the exact cycle length is $2(1 + n - h_n) + 1$. Lemma 5.6 applies with $(a, b) = (1, 2)$: the complete allowed triple fixes both history writes, and its full image has prefix lengths $(1 + e, 2 - e)$.

Theorem 11.6 (Guarded cylinders in a torus chart)**.** *On the unit torus, the fixed label* $(1/8, 0, 0)$ *and the strip* $\{x : x_1 \pmod 1 \in (-1/4, 0)\}$ *realize halting by a general mean-zero force with zero initial velocity and pressure. It is periodic after time one and all its mixed derivatives are bounded. The torus comparison class gives uniqueness.*

Proof. Let a be the augmented alphabet size and k the control count in Lemma 5.5. Use zero-blank base $B = 2a$, and enumerate nonterminal and terminal states with positive integer indices $j \geq 1$ and first-coordinate offsets respectively $L_0 + 3j$ and $-L_0 - 3j$, where $L_0 = 4k + ka^3 + 10$. Refinement by triples gives at most $N = ka^3$ branches. Their source prefix lengths are $(1, 2)$ and image lengths $(1 + e, 2 - e)$, so the smallest possible grid gap is B^{-3}. If P_i, Q_i are lower corners, the branch map is $Q_i + \operatorname{diag}(B^{-e_i}, B^{e_i})(y - P_i)$.

Choose three flat ramps α, β, γ, active respectively in $(1/8, 1/4)$, $(3/8, 5/8)$ and $(3/4, 7/8)$. Set

$$c_i(t) = ((1 - \beta)P_i + \beta Q_i,\ i(\alpha - \gamma)), \quad l_i = 1 - \beta + \beta B^{-e_i}.$$

The moving rectangle has side lengths $l_i B^{-1}$ and $l_i^{-1} B^{-2}$. Use a physical padding $\varepsilon = B^{-3}/4$ around each moving rectangle in three dimensions. During ascent and descent the planar endpoint gaps separate these neighborhoods; during the affine phase the unit height gaps separate them. Use Lemma 3.1 for the paths $c_i + \operatorname{diag}(l_i, l_i^{-1}, 1)(y - (P_i, 0))$ and sum the disjoint fields. The first coordinate is an endpoint convex combination even though the paths are based at lower corners rather than centers. All supports lie in $[-2L_0, 2L_0]^3$, by the definitions of L_0 and N.

Scale by $\rho = 1/(32L_0)$, so the repeated mechanism is supported in $[-1/16, 1/16]^3$. Load the fixed label $(1/8, 0, 0)$ to the scaled rational input code by a moving translation with cutoff radius $1/32$; its support stays in $(-1/4, 1/4)^3$. For a nonterminal initial code, the entire first coordinate during loading is positive, and subsequent nonterminal branches have first coordinate greater than $1/32$. Terminal codes have first coordinate less than $-1/32$. Hence the stated strip detects exactly

terminal entry, including initial halting, without any wraparound artifact. Periodize the compact curl fields and recompute the force at physical ν. Mean zero, bounded derivatives and comparison follow from Lemma 6.3. □

11.7 A filled half-line and mean-corrected vertical motion

The next recorder starts with an infinite constant history tail rather than a single frontier symbol. Its guard and its mean-zero vertical realization both have independent roles.

Lemma 11.7 (History at a filled half-line)**.** *A finite guarded table has a full-domain inverse and simulates the original machine from a tape with a separately constant tail in each direction. At checkpoint n its nonempty history cells are exactly the absolute indices below $2+n$.*

Proof. Normalize to one halt state h. For $i=(q,a)\in I=(Q\setminus\{h\})\times A$, write $\delta(q,a)=(q_i,b_i,e_i)$. History letters are $\mathcal{H}=\{\bot,\#\}\sqcup I$, with $\mathcal{H}_+=\{\#\}\sqcup I$. The tracks are work, mark and history; controls are $\mathsf{R}_q,\mathsf{B}_q,\mathsf{T}_i,\mathsf{S}_i$. Write ℓ_j for the history at relative cell j. Here is the complete table, with unmentioned cells unchanged:

input	output	move	condition
$\mathsf{R}_q(a,0,k)$	$\mathsf{T}_i(b_i,0,k)$	e_i	$i=(q,a),\ q\neq h$
$\mathsf{T}_i(c,0,k)$	$\mathsf{S}_i(c,1,k)$	1	$k\in\mathcal{H}_+$
$\mathsf{S}_i(c,0,k)$	$\mathsf{S}_i(c,0,k)$	1	$k\in\mathcal{H}_+$
$\mathsf{S}_i(c,0,\bot)$	$\mathsf{B}_{q_i}(c,0,i)$	0	$\ell_{+1}=\bot$
$\mathsf{B}_q(c,0,k)$	$\mathsf{B}_q(c,0,k)$	-1	$k\in\mathcal{H}_+$
$\mathsf{B}_q(c,1,k)$	$\mathsf{R}_q(c,0,k)$	0	$k\in\mathcal{H}_+$.

(34)

Initialize R_{q_0} at head zero with marks zero, history # at every index below two and $\bot$ elsewhere. At checkpoint n, the work head p_n has $|p_n|\le n$ and the boundary is $r_n=2+n$. After the work move, $p'=p_n+e_i\le n+1<r_n$. Thus the mark is placed on nonempty history, and the right scan reaches the first empty cell. Its right neighbor is also empty, so the append is allowed. A finite left scan returns to and erases the unique mark. The new boundary is r_n+1; no other work symbol changes. The filler boundary at absolute index two is permanent, so it also retains the tape's absolute alignment.

Into T_i the tag gives the old work instruction and shift. Into S_i, both shifts are rightward, and the mark at the old cell distinguishes entry from a loop. Into B_q, an output with empty history at offset $+1$ must be an append: its current record identifies i. An output of a leftward loop has nonempty history at that offset, since it is the old current cell. This is exactly why the append guard is necessary. Into R_q only mark erasure applies. Each case gives a unique predecessor on the full domain. In particular, entry into B_q may have displacement zero or minus one; the neighboring history, rather than the target control alone, recovers that displacement. Apply Lemma 5.6 with $(a,b)=(1,2)$, refining by the full cells at $-1,0,1$. Its full-domain injectivity hypothesis has just been proved, and every write and displacement lies in the allowed window. It therefore gives full image cylinders and separately disjoint closed rectangle families. □

Enumerate the resulting branches by $i=1,\dots,N$, with displacements e_i. Number the augmented alphabet by $0,\dots,k-1$, use base $b=3k+1$ and digits $1+3\ell$. Two different prefixes have gaps at least $2b^{-r}$ at their first differing place r. Codes lie in the interior of their intervals. If there are K controls, let $\sigma_0=1/[8(K+1)]$. Put the terminal state's lower corner at $(3/4,1/4)$ and the other lower corners at $(1/8+2\ell\sigma_0,1/4)$ for $0\le\ell<K-1$, giving every state square side σ_0. Their branch

rectangles have sides $\sigma_0 b^{-1}, \sigma_0 b^{-2}$ and image sides $\sigma_0 b^{-(1+e)}, \sigma_0 b^{-(2-e)}$. The families are separately disjoint by the full inverse. Every endpoint rectangle is contained in

$$F = [1/8, 7/8] \times [1/4, 3/8], \qquad z_* = 1/4. \tag{35}$$

All nonterminal state squares have first coordinate at most $3/8$.

Proposition 11.8 (Mean-corrected vertical transport)**.** *The preceding full rectangles at height z_* have an effective mean-zero solenoidal realization on the unit torus. Every tracked first coordinate is between its endpoint values throughout the transition.*

Proof. For a source rectangle D, choose a planar cutoff θ_D with padding $\varepsilon = \sigma_0 b^{-3}/4$, equal to one near D. Define

$$\kappa_D(x, y) = \theta_D(x, y) - \theta_D(x, y - 1/2)$$

periodically. It has integral zero. On its own source it equals one, and on all other sources it equals zero; the shifted copy lies away from F. Use the same construction for target rectangles E. Assign branch i the height $z_i = 1/2 + i/[4(N+1)]$ and choose disjoint height cutoffs ρ_i equal to one near z_i, with padding $1/[16(N+1)]$. The vertical fields with speeds

$$Z_D(x, y) = \sum_i (z_i - z_*)\kappa_{D_i}(x, y), \qquad Z_E(x, y) = \sum_i (z_i - z_*)\kappa_{E_i}(x, y)$$

are divergence free and mean zero, because they are independent of z.

Let χ equal one near F, with planar padding $1/32$. For branch i, let p_i, q_i be the source and target centers and put $d_i = q_i - p_i$, $c_i(\mu) = p_i + \mu d_i$, $l_i = 1 - \mu + \mu b^{-e_i}$ and

$$H_i = \chi \left[d_{i,1} y - d_{i,2} x + \frac{b^{-e_i} - 1}{l_i} (x - c_{i,1})(y - c_{i,2}) \right].$$

The middle field $W(\mu) = \sum_i \rho_i(z)(\partial_y H_i, -\partial_x H_i, 0)$ is divergence free and mean zero. The path

$$c_i(\mu) + \operatorname{diag}(l_i, l_i^{-1})(\zeta - p_i) \tag{36}$$

stays in F: the first side is interpolated linearly and the reciprocal inequality (24) bounds the second side by the interpolated endpoint sides. Thus $\chi = 1$ along the entire path, and differentiation gives its exact velocity. For three ordered flat ramps $\alpha_0, \alpha_1, \alpha_2$ on a unit interval, the complete field is

$$V = \alpha_0'(0, 0, Z_D) + \alpha_1' W(\alpha_1) - \alpha_2'(0, 0, Z_E). \tag{37}$$

It lifts each source, performs its affine map in its own height layer, and lowers it at its target. The first coordinate is unchanged during vertical motion and is an endpoint convex combination in the middle. All statements hold on the full rectangles by the cutoff plateaus and ODE uniqueness. The zero-mean correction acts away from the computational rectangles and therefore changes none of these paths. □

Theorem 11.9 (A fixed strip with half-line history)**.** *On $\mathbb{T}^3$, the fixed label $(1/8, 1/4, 1/4)$ and strip $2/3 < x_1 < 15/16$ realize halting by a general mean-zero force, with zero initial velocity and pressure, bounded mixed derivatives and period one after time one. For a fixed machine only the loading field depends on the input. The material maps are volume-preserving diffeomorphisms, and the velocity and material-acceleration identities of Lemma 2.2 hold.*

Proof. The two initial tape tails are separately constant, so their radix sums are computable rationals, even though the left history tail is filled. Load the fixed label to that code at height z_* by the planar Hamiltonian $\chi(d_x y - d_y x)$ with a flat scalar schedule. The segment stays in F. A nonterminal load and every nonterminal phase have first coordinate at most $3/8$; a terminal code lies in the stated strip. For an initially terminal machine the loading endpoint suffices. Repeat (37) after time one. Checkpoint simulation and the intermediate-time guard prove the event equivalence. The common analytic lemma proves force effectivity, uniqueness, boundedness and the asserted flow identities. In particular $U = (\partial_t \Phi_t) \circ \Phi_t^{-1}$ and $\partial_{tt} \Phi_t = (f - \nabla p + \nu \Delta U) \circ \Phi_t$; these are identities for the constructed flow, not additional unknown equations. □

12 Routing choices dictated by the observer

The preceding processors move every branch by a prescribed affine map. We now ask which parts of that motion must be controlled between the integer times. A bounded observation box allows horizontal work above the observer. A slab observer requires a bound on the horizontal coordinate throughout the motion. Finally, a half-space observer can be protected by moving all sources to storage before delivering any target. These three requirements lead to the constructions below.

12.1 Four motions above a bounded observer

We first separate ascent, scaling, translation and descent. The two planar motions occur at a fixed height, so their exclusion from the bounded observer follows without a horizontal coordinate estimate.

Let $D_i, E_i \subset \mathbb{R}^2$, $1 \le i \le N$, be closed rational axis-aligned rectangles with positive side lengths. Assume the D_i are pairwise disjoint and the E_i are pairwise disjoint, and let

$$A_i(b) = c_i' + \operatorname{diag}(\lambda_i, \lambda_i^{-1})(b - c_i), \qquad \lambda_i \in \mathbb{Q}_{>0}, \qquad A_i(D_i) = E_i,$$

where c_i, c_i' are their centers. No separation between a source and a target is assumed. All distances used to choose coordinate padding below are distances for the maximum norm.

Let $\vartheta = \sigma$ be the effective flat switch in (5). For a closed interval, possibly reduced to one point, define

$$\chi_{[a,b],\epsilon}(v) = \vartheta\left(\frac{2(v - a + \epsilon)}{\epsilon}\right) \vartheta\left(\frac{2(b + \epsilon - v)}{\epsilon}\right).$$

For a box K, let $\chi_{K,\epsilon}$ be the product over its three coordinates. It is one on the $\epsilon/2$ enlargement of K and zero outside its ϵ enlargement. The effective profile estimates following (5) apply to these products and rescalings.

Lemma 12.1 (Four-stage realization)**.** *The rectangle data determine effectively a smooth divergence-free field $W(s, x)$ supported in a compact subset of $(0, 1) \times \mathbb{R}^3$. Its time-one map is $(b, z) \mapsto (A_i(b), z)$ on a neighborhood of each $D_i \times \{0\}$. For a point of that plate, every planar motion occurs at height $4i$, while ascent and descent preserve its source and target planar coordinates, respectively. The scaling can use either linear interpolation of its positive first factor or linear interpolation of the logarithm of that factor.*

Proof. The empty family is realized by zero. Otherwise put $h_i = 4i$, $w_i = c_i' - c_i$, and

$$\Theta_k(s) = \vartheta(10s - (2k - 1)), \qquad 1 \le k \le 4.$$

Only the kth motion is active on the interval $[(2k-1)/10, 2k/10]$; the four intervals are disjoint. If a_{ij}, a'_{ij} are the source and target half-widths, respectively, take the four swept boxes

$$\begin{aligned}
K_{i1} &= D_i \times [0, h_i],\\
K_{i2} &= \{c_i + (u_1, u_2) : |u_j| \le \max(a_{ij}, a'_{ij})\} \times \{h_i\},\\
K_{i3} &= \mathrm{box}(E_i - w_i, E_i) \times \{h_i\},\\
K_{i4} &= E_i \times [0, h_i].
\end{aligned}$$

Here box denotes the smallest axis-aligned rectangle containing the two displayed rectangles. Within each of the four families these boxes are disjoint: source separation handles the first, target separation the fourth, and distinct heights the other two. Choose a positive rational $\epsilon < 1$ smaller than one quarter of every pairwise distance within these families. A family with fewer than two members contributes no restriction. Its ϵ enlargements are then still disjoint.

Choose either

$$\mu_i = 1 + (\lambda_i - 1)\Theta_2 \quad \text{or} \quad \mu_i = \lambda_i^{\Theta_2}, \qquad \gamma_i = \dot{\mu}_i/\mu_i.$$

The factors μ_i and μ_i^{-1} stay between their respective endpoint values. Use the coordinate potentials

$$\begin{aligned}
B_{i1} &= (0, h_i x_1, 0), & B_{i2} &= (0, 0, (x_1 - c_{i1})(x_2 - c_{i2})),\\
B_{i3} &= (0, 0, w_{i1} x_2 - w_{i2} x_1), & B_{i4} &= (0, -h_i x_1, 0).
\end{aligned}$$

For the logarithmic-scale construction we also retain the alternative translation potentials, with $e_3 = (0, 0, 1)$,

$$B_{i1} = \tfrac{1}{2}(h_i e_3) \times x, \qquad B_{i3} = \tfrac{1}{2}(w_{i1}, w_{i2}, 0) \times x, \qquad B_{i4} = \tfrac{1}{2}(-h_i e_3) \times x.$$

Their curls equal the same constant translation vectors. Their localized fields need not agree away from the plates. Either choice gives

$$W = \sum_i \left[\dot{\Theta}_1 \nabla \times (\chi_{K_{i1},\epsilon} B_{i1}) + \gamma_i \nabla \times (\chi_{K_{i2},\epsilon} B_{i2}) + \dot{\Theta}_3 \nabla \times (\chi_{K_{i3},\epsilon} B_{i3}) + \dot{\Theta}_4 \nabla \times (\chi_{K_{i4},\epsilon} B_{i4})\right].$$

This is a finite sum of spatial curls. Its spatial support is compact, its time support is contained in $[1/10, 8/10]$, and the lower bound $\mu_i \ge \min(1, \lambda_i) > 0$ proves smoothness and effectivity.

For $b \in D_i$, the successive paths are

motion	position
ascent	$(b, h_i\Theta_1)$
scaling	$(c_i + \mathrm{diag}(\mu_i, \mu_i^{-1})(b - c_i), h_i)$
translation	$(A_i(b) - w_i + \Theta_3 w_i, h_i)$
descent	$(A_i(b), h_i(1 - \Theta_4))$.

Each path stays in its swept box. On a neighborhood of that box the uncut curls are, in order,

$$h_i e_3, \qquad (x_1 - c_{i1}, -(x_2 - c_{i2}), 0), \qquad (w_{i1}, w_{i2}, 0), \qquad -h_i e_3.$$

After multiplication by the time coefficients, these are the prescribed affine velocities. The motion matrices are positive diagonal of determinant one, the plateau margin is $\epsilon/2$, and the support padding is ϵ. The disjoint swept boxes therefore permit the plateau argument of Lemma 3.1, with $\eta = \epsilon/2$. It identifies these particular localized fields with the displayed paths on the plateaux. The normalized diagonal factors are bounded by $L_i = \max(1, \lambda_i, \lambda_i^{-1})$, so that lemma gives the full initial neighborhood of radius $\epsilon/(4L_i)$. □

We use the following precise Euclidean comparison class in the applications: on each $[0,T]$,

$$\begin{aligned} u &\in C([0,T];H^2(\mathbb{R}^3)) \cap C^1([0,T];L^2(\mathbb{R}^3)),\\ u, \nabla u &\in L^\infty([0,T]\times\mathbb{R}^3), \qquad p \in C([0,T];H^1(\mathbb{R}^3)). \end{aligned} \tag{38}$$

The pressure representative is part of this condition; it is zero for our constructed solution. These comparison velocities and pressures satisfy case (i) of Lemma 2.2. We retain the stated bounded-gradient requirement here, although that case does not need it.

That lemma also gives a smooth material diffeomorphism on every finite interval. Differentiating its trajectory equation gives $\dot J = (D_x u)(t,\Phi_t(a))J$, $J(0)=I$, for $J = D_a\Phi_t$. Consequently

$$\frac{d}{dt}\det J = (\nabla\cdot u)(t,\Phi_t(a))\det J = 0, \qquad \det J = 1. \tag{39}$$

The same proof applies to periodic lifts on the torus. The residual formulas below are affine in viscosity, so they also define the asserted solutions for any real $\nu > 0$, with effectivity relative to ν.

Theorem 12.2 (Two bounded-box material tests)**.** *Fix a positive computable viscosity ν. A deterministic one-tape machine and finite input effectively determine each of the following general body forces on $[0,\infty)\times\mathbb{R}^3$. Both have zero initial velocity, a unique global smooth velocity in (38), and constructed pressure zero.*

1. *The force and velocity have one fixed compact spatial support, all mixed derivatives bounded, and period one for $t \ge 1$. The particle from 0 enters $(-5,-2)\times(-1,2)\times(-1,1)$ exactly when the machine halts.*

2. *The force and velocity have one fixed compact spatial support and, for all $k \ge 0$ and spatial multi-indices α, satisfy*

$$\|\partial_t^k\partial_x^\alpha u(t)\|_\infty + \|\partial_t^k\partial_x^\alpha f(t)\|_\infty \le C_{k,\alpha}(1+t)^{-1-k}. \tag{40}$$

   The particle from $(2,0,0)$ enters $(-1/2,3/2)\times(-1/2,3/2)\times(-1/2,1/2)$ exactly when the machine halts. In particular $f \in L^2([0,\infty);H^m(\mathbb{R}^3))$ for every integer $m \ge 0$.

The supports and constants may depend on the machine and input. The labels and observers do not. All force derivatives admit evaluation to any prescribed rational error by a finite program.

Proof. Normalize to a single halt state q_H and a complete nonhalting table, as in Section 6. Apply the move-first recorder of Lemma 5.2. Its history frontier starts at cell 2; all other history cells are empty and all mark bits are zero. At checkpoint n, the frontier is $g_n = n+2$, the work head satisfies $|h_n| \le n$, and the next work instruction ending at h' is completed in exactly $4 + 2(g_n - h')$ local steps. Thus every required checkpoint is reached after finitely many steps. The incoming-direction and written-symbol inverse properties hold on the whole partial table. The records recover the successive work instructions and hence the absolute head position, even though the geometric code is head-relative.

Let m be the full alphabet size. Use odd digits $\beta(a)$ in $\{1,3,\dots,2m-1\}$, with base $B = 2m+1$ in the first construction and $B = 2m+2$ in the second. Set

$$C(a_0a_1\cdots) = \sum_{r\ge0}\beta(a_r)B^{-r-1}, \quad P_a(v) = \frac{\beta(a)+v}{B}, \quad D_a(v) = Bv - \beta(a), \quad I_a = P_a([0,1]).$$

Let ξ encode the left string, nearest cell first, and η the string starting at the head. The complete right, stay and left branch maps are (19), with its digit $d(a)$ equal to $\beta(a)$ and its move e equal to d

here. The left move is split over every full symbol c. The shared lemma proves the full-rectangle endpoint formulas, arbitrary-tail action and linear part $\operatorname{diag}(B^{-d}, B^{d})$.

Translate each state square horizontally. In the first construction give the halt state offset -4 and the other states offsets $2 + 2r$, $r \geq 0$; in the second use halt offset 0 and other offsets $3r$, $r \geq 1$. Thus the code is $(o_s + \xi, \eta, 0)$. Lemma 5.1 applies with exactly these odd digits and either displayed base: its incoming conditions are those supplied by the move-first recorder, and (19) is its minimal left-split branch list. Thus both complete closed rectangle families are separately disjoint, with within-state gaps at least B^{-2}. This proves the hypotheses of Lemma 12.1; no separation between the source and target families is required.

Use its linear scale and coordinate potentials for the first construction; use its logarithmic scale and symmetric translation potentials for the second. In the latter case the scaling coefficient is exactly $(\log \lambda_i)\dot{\Theta}_2$ and the scale matrix is $\operatorname{diag}(\lambda_i^{\Theta_2}, \lambda_i^{-\Theta_2})$. Denote the resulting step field by W_{step}.

The initial code P_{in} is rational. A constant full-symbol tail of digit β_0 after M positions contributes $B^{-M}\beta_0/(B-1)$; all other contributions are finite. For the appropriate fixed label a_*, put $v_0 = P_{\mathrm{in}} - a_*$ and let J_0 be the smallest box containing $[a_*, P_{\mathrm{in}}]$. The loader

$$W_{\mathrm{in}}(s, x) = \frac{d}{ds}\vartheta(2s - 1/2)\,\nabla \times \left(\chi_{J_0,1}(x)\frac{v_0 \times x}{2}\right)$$

has the exact marked path $a_* + \vartheta(2s - 1/2)v_0$, since the uncut curl is v_0 and the entire segment lies in the plateau. Form

$$W(s, x) = W_{\mathrm{in}}(s, x) + \sum_{n \geq 1} W_{\mathrm{step}}(s - n, x).$$

The zero time collars make this smooth, initially zero, and periodic for $s \geq 1$, with common compact spatial support and bounded mixed derivatives.

At time $s = 1 + k$ the marked particle is exactly the code after k local instructions, through the first halt or indefinitely in a nonhalting run. This follows by induction from the full-rectangle maps. Each halt code lies in its stated observation box, including one reached by loading an initially halted machine. In a nonhalting run, the first loader has $x_1 \geq 0$ and later nonhalt codes have $x_1 \geq 2$; for the second these bounds are $x_1 \geq 2$ and $x_1 \geq 3$. Ascent and descent leave those planar coordinates unchanged. Every planar motion has height $4i \geq 4$, outside both observers. The intervening stationary intervals add no new positions. This proves exclusion at every real time.

For the first case take $u = W$ and its residual force. For the second set

$$s(t) = \log(1 + t), \quad \theta(t) = (1 + t)^{-1}, \quad u(t, x) = \theta(t)W(s(t), x),$$

whose force is

$$f = \theta^2\{-W + \partial_s W + (W \cdot \nabla)W\}(s(t), x) - \nu\theta\Delta W(s(t), x).$$

Lemma 9.1 applies to this fixed-support, bounded-derivative template. Its identity $\partial_t(\theta^j Y(s(t))) = \theta^{j+1}(\partial_s - j)Y(s(t))$ proves (40) for every k, α. Fixed support then gives the stated $L^2_t H^m_x$ conclusion. Its clock is onto, with inverse $t = e^s - 1$, so the material path is the old path composed with $s(t)$ and the event is unchanged.

In both cases smoothness, fixed support and bounded derivatives on finite time intervals put the reference velocity in $CH^2 \cap C^1L^2$ with bounded velocity and gradient. It is solenoidal and initially zero. Lemma 2.2(i) therefore gives the exact solution $(u, 0)$ and uniqueness in (38). The effective profiles following (5), applied to the finite table, rational initial code, finitely many boxes and repeated template, give all derivative evaluations. A coarse time approximation encloses finitely many possible active translates, avoiding equality tests at seams. The prescription never computes the machine's run. □

12.2 Small plates and a slab observer on the unit torus

A slab observer has no height restriction, so elevation alone cannot protect it. We now keep each nonhalting branch inside a narrow range of first coordinates throughout its motion. The delayed-head recorder also allows arbitrary finitely supported work data on both sides of the head.

Theorem 12.3 (Torus plates and square-integrable forcing)**.** *Fix a positive computable ν. Every deterministic one-tape machine with finitely specified tape, blank outside a finite set, effectively determines a smooth mean-zero general force on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, with bounded mixed derivatives and period one for $t \geq 1$. Its zero-data Navier–Stokes solution is globally smooth and unique among smooth solutions, with pressure normalized to mean zero and constructed pressure zero. For $P_* = (1/4, 1/2, 1/4)$ and $X(t) = \Phi_t(P_*)$, the event*

$$\exists t \geq 0 : \quad X_1(t) \pmod 1 \in (1/2, 7/8) \tag{41}$$

is equivalent to halting.

A separate effective force has the same zero data, event, uniqueness and mean-zero property, is supported in one compact sub-box of $(0,1)^3$, has all mixed derivatives bounded, and satisfies

$$\|\widehat{f}(t,\cdot)\|_\infty = O((1+t)^{-2/3}). \tag{42}$$

In particular $\widehat{f} \in L^2([0,\infty) \times \mathbb{T}^3)$.

Proof. Use Lemma 6.1 with the frontier initially at cell 1. Its main control $M_{q,e}$ records the last work displacement e, initially zero. In one work step it writes and marks the old head, records (q, e, a) at the frontier, returns to the mark, and only then makes the work displacement. All other initial history cells are blank and every mark is zero. At checkpoint n the frontier is $n+1$, $|h_n| \leq n$, and the next checkpoint takes exactly $2(n+1-h_n)+3$ local steps. These scans test history and marks, so the proof is unaffected by work symbols on either side of the head. Main controls with halting q have no outgoing rules. The full-domain incoming properties are those of the cited lemma.

Let S be the local state set, $N = |S|$, let S_h be its halting main states, and enumerate S by $n(s) = 0, \dots, N-1$. Put

$$\ell = \frac{1}{16(N+1)}, \quad z_0 = \frac{1}{4}, \quad x_s = \frac{1}{8} + 2\ell n(s) + \frac{1}{2}\mathbf{1}_{s \in S_h}, \quad T_s(\xi,\eta) = (x_s + \ell\xi, 1/4 + \ell\eta).$$

Nonhalting squares lie in $1/8 \leq x_1 < 1/4$ and halting squares in $5/8 \leq x_1 < 3/4$; distinct squares have gap at least ℓ. Give the full alphabet, of size m, odd digits in base $\beta = 2m+1$. Conjugating (19) by the source and target charts gives separately disjoint rectangle families R_j^0, R_j^1 with positive diagonal factors α_j, α_j^{-1}, $\alpha_j \in \{\beta^{-1}, 1, \beta\}$. Every chart has the same scale, so this remains the physical linear part. Lemma 5.1, with this uniform chart scaling, gives within-family coordinate gaps at least ℓ/β^2, including the two-letter target intervals for left moves.

For $1 \leq j \leq L$, write the centers as p_j^0, p_j^1, the source half-widths as a_j, b_j, and the target half-widths as $\alpha_j a_j, \alpha_j^{-1} b_j$. All endpoint half-widths are at most $\ell/2$. Choose

$$h_j = \frac{1}{2} + \frac{j}{4(L+1)}, \qquad \epsilon = \frac{1}{100}\min\left(1, \frac{\ell}{\beta^2}, \frac{1}{4(L+1)}\right).$$

With $\theta(s) = \vartheta(2s - 1/2)$ and $\Theta_k(\tau) = \theta(3\tau - k + 1)$ for $k = 1, 2, 3$, define

$$\begin{aligned} C_j(\tau) &= ((1-\Theta_2)p_j^0 + \Theta_2 p_j^1, z_0 + (\Theta_1 - \Theta_3)(h_j - z_0)), \\ \sigma_j &= 1 + (\alpha_j - 1)\Theta_2, \qquad \lambda_j = \dot\sigma_j/\sigma_j, \\ (w_{j1}, w_{j2}, w_{j3}) &= (a_j\sigma_j, b_j/\sigma_j, 0). \end{aligned}$$

These centers and half-widths describe moving plates. Their ϵ enlargements are disjoint: the first third uses source projections, the middle third distinct heights, and the last third target projections. The padding is at most one hundredth of every relevant gap. All plates and padding lie in a compact sub-box of $(0,1)^3$, since their centers interpolate between the stated chart centers, widths are at most ℓ, heights lie between $1/4$ and $3/4$, and $\ell \le 1/32$, $\epsilon \le 1/100$.

For $w \ge 0$ put

$$\kappa_\epsilon(w,v) = \vartheta\Big(\frac{2(w+\epsilon-v)}{\epsilon}\Big)\,\vartheta\Big(\frac{2(w+\epsilon+v)}{\epsilon}\Big), \qquad \chi_j = \prod_{a=1}^{3} \kappa_\epsilon(w_{ja}, x_a - C_{ja}).$$

Writing $y = x - C_j$, use

$$A_j = \tfrac{1}{2}\dot{C}_j \times y + (0,0,\lambda_j y_1 y_2), \qquad V(\tau,x) = \sum_j \nabla \times (\chi_j A_j).$$

Extend the potentials by zero outside the chart. The support remains away from its faces, so this gives a smooth torus field, zero near the time endpoints. On the whole plateau of plate j its velocity is $\dot{C}_j + (\lambda_j y_1, -\lambda_j y_2, 0)$; other summands vanish there. Hence the path from $(p_j^0 + (r,s), z_0)$ is exactly

$$C_j(\tau) + (\sigma_j(\tau) r, \sigma_j(\tau)^{-1} s, 0), \qquad |r| \le a_j, \quad |s| \le b_j. \tag{43}$$

It ends at the prescribed affine image. The plateau argument of Lemma 3.1 applies with $D_j = \operatorname{diag}(\sigma_j, \sigma_j^{-1}, 1)$, $\eta = \epsilon/2$, and diagonal bound $\max(1, \alpha_j, \alpha_j^{-1})$: the required enlarged-box separation was just proved. It gives the same motion on an initial neighborhood of radius $\epsilon/[4\max(1,\alpha_j,\alpha_j^{-1})]$, with constant normal displacement, and therefore includes boundary codes.

The initial code P_{in} is rational, since relative to constant blank tails only finitely many work and frontier digits differ. Load it from P_* along $C_* = (1-\theta)P_* + \theta P_{\mathrm{in}}$ using

$$V_0(t,x) = \nabla \times \left[\prod_{a=1}^{3} \kappa_{1/64}(0, x_a - C_{*,a}(t)) \frac{\dot{C}_*(t) \times (x - C_*(t))}{2} \right].$$

The segment and its collar stay within the chart: its first coordinate stays between $1/8$ and $3/4$, its second between $1/4$ and $1/2 + \ell$, and its third is $1/4$. Its plateau curl is $\dot{C}_*$. Take $U = V_0$ on $[0,1]$ and $U(t) = V(t-n)$ on $[n, n+1]$ for $n \ge 1$. Time collars make this smooth and initially zero. It has fixed compact chart support, bounded mixed derivatives, and period one after loading. Define $f = U_t + (U \cdot \nabla)U - \nu \Delta U$ using the stated physical viscosity. Both U and f have mean zero: U is a curl, the Laplacian integrates to zero, and $(U \cdot \nabla)U = \operatorname{div}(U \otimes U)$. The torus clause of Lemma 2.2 applies to this smooth, initially zero, solenoidal reference field and gives the unique smooth solution $(U, 0)$ with mean-zero pressure.

Equations (19) and (43) identify the particle at time $1 + k$ with the kth local configuration. The recorder invariant ensures that a nonhalting computation continues indefinitely. A halt, including an initial halt reached by loading, puts the particle in $5/8 \le X_1 < 3/4$, proving the event. For a nonhalting computation the loader has $X_1 \le 1/4$ and every later branch has both endpoint centers below $1/4$. Its center is their convex interpolation and its first half-width is at most $\ell/2$. Thus $0 < X_1 < 1/4 + \ell/2 < 1/2$ at every intermediate time. The path stays inside the chart, so reduction modulo one cannot create a false visit.

For the separate decaying force choose the onto clock

$$s(t) = (1+t)^{1/3} - 1, \qquad \widehat{U}(t,x) = s'(t) U(s(t), x).$$

Its residual is

$$\widehat{f} = s''U + (s')^2\{\partial_s U + (U\cdot\nabla)U\} - \nu s'\Delta U,$$

where the fields on the right are evaluated at $(s(t), x)$. Since $s' = (1/3)(1+t)^{-2/3}$ and $s'' = -(2/9)(1+t)^{-5/3}$, the bounded template derivatives prove (42). Every positive-order derivative of s is bounded; repeated product and chain rules therefore bound every mixed derivative of $\widehat{U}$ and $\widehat{f}$. The square of $(1+t)^{-2/3}$ is integrable. Support and mean zero persist, and $U(0) = 0$ retains the zero datum. The torus clause of Lemma 2.2 therefore applies to $\widehat{U}$ too. Finally the material path is the old one at $s(t)$, and s has inverse $t = (1+s)^3 - 1$, so the event is unchanged.

All choices are effective finite data followed by explicit smooth formulas. In particular $\sigma_j \geq \min(1, \alpha_j) > 0$; the cutoff estimates and zero collars allow derivative evaluation at approximate real arguments without tests for exact boundary equality. Only positive arguments enter the cube root. The program repeats the entire local table and never uses a computed execution. □

12.3 Storage and periodic forcing without a loader

The next construction absorbs initialization into the positions of the state rectangles. A zero digit for the full blank symbol places the initial left stack at zero; an input-dependent second-coordinate translation then places the entire initial code at one fixed particle. To protect a half-space observer during every period, we first evacuate all sources to separate storage rectangles and only afterwards fill the targets.

Lemma 12.4 (Zero-blank coding)**.** *Let an alphabet of size $m > 1$ have digits $0, \dots, m-1$, with full blank digit zero. Set $B = m+1$, $\kappa = (m-1)/m$, and*

$$\mathrm{val}(a_0 a_1 \cdots) = \sum_{r\geq 0} a_r B^{-r-1}, \qquad I_a = \frac{a + [0,\kappa]}{B}, \qquad I_{a,b} = \frac{a + I_b}{B}.$$

The code is injective, takes values in $[0,\kappa]$, and its distinct one-letter and two-letter cylinders have positive rational gaps. For a deterministic partial table with target-state-determined incoming direction and incoming rule determined by target state and written symbol, separated state translates of $[0,\kappa]^2$ yield separately disjoint full source and target rectangles by the following maps in state-relative coordinates:

d	*source*	(ξ', η')	*target*
1	$[0,\kappa] \times I_a$	$((b+\xi)/B, B\eta - a)$	$I_b \times [0,\kappa]$
0	$[0,\kappa] \times I_a$	$(\xi, \eta + (b-a)/B)$	$[0,\kappa] \times I_b$
-1	$I_c \times I_a$	$(B\xi - c, c/B + (b + B\eta - a)/B^2)$	$[0,\kappa] \times I_{c,b}$.

(44)

The last row is split over all c. Each map has positive diagonal linear part of determinant one and acts exactly on the full rectangle.

Proof. The maximum tail is $(m-1)/(B-1) = \kappa$, and $(m-1+\kappa)/B = \kappa$. Consecutive first-digit intervals have gap $(1-\kappa)/B > 0$. Within one first-digit interval the second-digit gaps are this quantity divided by B; different first digits already give a gap. The interval containing a code recovers its first digit, and $v \mapsto Bv - a$ then exposes the tail. Repetition proves injectivity, including endpoint codes.

Writing a to b changes the first right digit. A right move pushes b onto the left string and removes a from the right; a left move removes c from the left and prepends c, b to the old right tail. These identities give (44), including its images and reciprocal slopes. Determinism and digit gaps

separate sources by state, read digit, and c when needed. At one target state the common incoming direction determines where the written digit is visible: in I_b on the left or right, or in $I_{c,b}$. The incoming-rule hypothesis and digit gaps then separate the closed target rectangles. □

Lemma 12.5 (Evacuation, storage and delivery)**.** *Suppose $A_i : D_i \to E_i$, $1 \le i \le N$, are the rational reciprocal rectangle maps of Lemma 12.1, with $N \ge 1$ and every $D_i \subset \{x_1 < 0\}$. There is an effective smooth divergence-free field on $\mathbb{R} \times \mathbb{R}^3$, one-periodic in time, zero near each integer time, and uniformly compactly supported in space, whose one-period map is $(b, z) \mapsto (A_i(b), z)$ on a neighborhood of every $D_i \times \{0\}$. If $E_i \subset \{x_1 < 0\}$, the entire one-period path of each point in $D_i \times \{0\}$ remains in that half-space.*

Proof. Let d_i, b_i be the source and target centers, and let a_{ij}, a'_{ij} be their half-widths. Put $a^*_{ij} = \max(a_{ij}, a'_{ij})$. Choose ϵ to be one quarter of the minimum of 1 and all pairwise maximum-norm distances within the source and target families; omit pair entries when $N = 1$. Define

$$x_{\min} = \min\{x_1 : (x_1, x_2) \in D_i \cup E_i \text{ for some } i\},$$
$$R = \max(1, \max_i a^*_{i1}), \qquad J = 4(R+1), \qquad s_i = (x_{\min} - iJ, 0).$$

The storage rectangle at s_i with half-widths a^*_{ij} lies in $x_1 < 0$. Its first-coordinate projection is separated by more than one from every source and target and from the other storage projections: the relevant gaps are at least $J - R > 1$ and $J - 2R > 1$.

We need only a localized translation and a localized reciprocal scaling. On a time segment $[t_0, t_0 + \delta]$, use $g(t) = \vartheta(3(t - t_0)/\delta - 1)$, whose derivative is supported in the middle third. Using the cutoff κ_ϵ defined in the proof of Theorem 12.3, for half-widths a, b write

$$\chi_{a,b}(y) = \kappa_\epsilon(a, y_1)\kappa_\epsilon(b, y_2)\kappa_\epsilon(0, y_3).$$

A plate translation from center P to Q is produced by

$$C(t) = (1 - g)P + gQ, \qquad \nabla \times \left[\tfrac{1}{2}\chi_{a,b}(x - C(t))\dot{C}(t) \times (x - C(t))\right].$$

On its plateau the velocity is $\dot{C}$ and the exact path is $C(t) + (r, s, 0)$. At storage center $P = (s_i, 0)$, take $q(t) = 1 + (\lambda_i - 1)g(t)$ and the field

$$\nabla \times \left(0, 0, \frac{\dot{q}}{q}(x_1 - P_1)(x_2 - P_2)\chi_{a^*_{i1}, a^*_{i2}}(x - P)\right).$$

It has the exact path $P + (qr, q^{-1}s, 0)$. The denominator has the positive lower bound $\min(1, \lambda_i)$, and the whole path lies in the storage rectangle, because each half-width stays between its source and target values.

Partition a period into $7N$ equal segments and take $Z = 2$. For each i in order, use four segments for

$$(d_i, 0) \longrightarrow (d_i, Z) \longrightarrow (s_i, Z) \longrightarrow (s_i, 0), \qquad \text{then scale at } s_i.$$

Only after all these evacuations, use three segments for each i to deliver its target shape along

$$(s_i, 0) \longrightarrow (s_i, Z) \longrightarrow (b_i, Z) \longrightarrow (b_i, 0).$$

Sum these segment fields and extend periodically. The middle-third time switches give zero neighborhoods of every segment endpoint and every integer time. The result is a smooth finite sum of curls with one compact spatial support.

This is the $7N$ schedule of Lemma 3.3(ii). Its separation hypotheses have explicit margins here. Unpadded source-source and target-target gaps are at least 4ϵ, so their ϵ enlargements leave gap at least 2ϵ. Storage gaps exceed one, leaving more than $1-2\epsilon>0$ after padding. At horizontal travel height the enlarged moving plate and every enlarged stationary plate have vertical gap $2-2\epsilon>0$. Every scaling stays in its reserved storage rectangle. The lemma therefore gives simultaneous noninterference for evacuation and delivery, including when a source meets a target.

The localized potentials above have exactly these affine generators on their plateaux, so the plateau argument of Lemma 3.1 gives endpoint $(d_i+(r,s),0)\mapsto(b_i+(\lambda_i r,\lambda_i^{-1}s),0)$, with normal displacement unchanged. Put $M=\max_i(1,\lambda_i,\lambda_i^{-1})$ and $d_*=\min(2\epsilon,1-2\epsilon,2-2\epsilon)>0$. A neighborhood radius smaller than both $\epsilon/(8M)$ and $d_*/(4M)$ retains the plateaux and all exclusion margins throughout the concatenation. This applies the shared proofs to the displayed potentials and does not identify their fields away from the plateaux.

If E_i is negative, each translation of an individual point joins two negative first coordinates, and its linear interpolation remains negative. Scaling occurs inside the negative storage rectangle. The point is stationary during all other segments. This proves the asserted whole-period exclusion. □

Theorem 12.6 (A periodic force with no loading interval)**.** *Fix a positive computable ν. A deterministic one-tape machine and finite input word in cells $0,\dots,L-1$, with other cells blank and initial head zero, effectively determine a general body force on $\mathbb{R}^3$ extending smoothly and one-periodically to all real times. It has uniformly compact spatial support and bounded mixed derivatives. Its zero-data solution has a unique global smooth velocity in (38), with constructed pressure zero, and*

$$\exists t\ge 0:\quad (\Phi_t((-2,0,0)))_1>0 \quad\Longleftrightarrow\quad \textit{the machine halts.}$$

The fixed label is already the initial code; periodicity begins at zero.

Proof. Insert a fresh nonhalting start state that preserves the read letter, stays put and enters the original start state. Replace missing nonhalting instructions by letter-preserving stay instructions to a halt state. This finite normalization preserves the halting question, including an initially halted machine, and guarantees a nonhalting initial control and at least one rule.

Apply the six-row delayed-head recorder of Lemma 6.1 with initial frontier 1. Initially every other history cell is blank and every mark is zero. The main control records (q,e), where e is the preceding displacement, and the history record is (q,e,a). The old head is marked before the right scan and cleared on return, immediately before displacement into the next main state. At checkpoint k the frontier is $k+1$ and the head i satisfies $|i|\le k$; the next checkpoint uses $2(k+1-i)+3\le 4k+5$ local steps. The incoming-direction and written-symbol properties hold on the full domain, as required by Lemma 12.4.

Give the full blank symbol digit zero and the other full symbols digits $1,\dots,m-1$. Order nonhalting local states with the initial state first and assign offsets $-2h$, $h=1,2,\dots$; assign halting main states offsets $2h$, $h=1,2,\dots$. Let η_{in} be the initial right-stack value and put $y_*=-\eta_{\mathrm{in}}$. Only finitely many full symbols are initially nonblank, so this is a finite rational sum. Every cell left of the initial head is fully blank. The code

$$(x_\sigma+\xi,y_*+\eta,0)$$

therefore places the initial configuration exactly at $(-2,0,0)$. The input changes the common second-coordinate translation, while the observed label and zero initial velocity remain fixed.

Every finitely reached tape still has finite full-symbol support, so its code is rational and can be decoded by rational digit extraction until both tails are zero. At a checkpoint the leftmost nonempty

history site has absolute index 1; its relative position also recovers the head's absolute index. Thus this coding retains the initialized computation.

Translate the rectangles of (44) by the indicated state offsets and y_*. Lemma 12.4 gives their full affine action and separate source and target gaps. Since $\kappa < 1$, every nonhalting state block is strictly negative in the first coordinate and every halting block strictly positive. Halting states have no outgoing rules, so all sources are negative. Apply Lemma 12.5 to obtain its periodic field v. Zero tails can place a code on a rectangle boundary, which is included in that lemma's full-neighborhood conclusion.

Set

$$f = f_0 + \nu f_1, \qquad f_0 = \partial_t v + (v \cdot \nabla)v, \qquad f_1 = -\Delta v.$$

The field and force are smooth, periodic, uniformly compactly supported, and have all mixed derivatives bounded. Since v vanishes near integer times, its initial value is zero. Fixed support and bounded derivatives give $v \in CH^2 \cap C^1 L^2$ and bounded $v, \nabla v$ on every finite interval. Lemma 2.2(i) therefore supplies the exact solution $(v, 0)$, uniqueness in (38), and its global material flow.

At time zero the marked particle is the initial code. If it is a nonhalting code at integer n, its digits select a source rectangle and the next period applies exactly that local instruction. Induction identifies the integer-time codes through the first halt or forever. Finite completion of every recorder scan identifies the original machine at the checkpoint integers. A halt therefore reaches a positive code. If the machine never halts, each used branch has a negative target rectangle. Lemma 12.5 keeps its marked path negative for that whole period, ruling out an intermediate false visit.

Finally the force program uses finite normalization, the finite table, the rational input shift, a finite rectangle list, rational storage and separation data, $7N$ time segments and explicit cutoffs. All scaling denominators have known positive lower bounds. The zero-collar and flat profile estimates following (5) permit evaluation of the repeated formula and every derivative from approximate real arguments without exact seam tests. Every period executes all branch maps, independently of the marked particle's current state; the program never computes the input run. □

12.4 Direction-split marking and a bounded observer

A stay instruction need not clear and reinstall a persistent mark. We separate that case in the next table. Its geometric realization uses a bounded observation box: during horizontal transfers the routing height itself prevents a false visit.

Lemma 12.7 (Direction-split persistent marking)**.** *In the notation of Lemma 10.1, there is an effective partial finite table with a persistent checkpoint mark, pointer at $p_n = n + 2$, and both full-domain incoming properties of Lemma 6.2. A work step ending at w' takes $2(p_n - w') + 4 + \mathbf{1}_{\{d_\tau \neq 0\}}$ local steps.*

Proof. Use full symbols $[g, e, l]$, with work letter g, mark $e \in \{0, 1\}$, and $l \in \{E, P\} \sqcup \{J_\tau\}$. Controls are A_q, B_τ for moving instructions, C_τ, S_τ, and D_q, R_q. For $\tau = (q, a)$ writing b_τ, the complete

table is

control	read and guard	target	write	d
A_q	$[a,1,l],\ d_\tau = 0$	C_τ	$[b_\tau,1,l]$	0
A_q	$[a,1,l],\ d_\tau \neq 0$	B_τ	$[b_\tau,0,l]$	d_τ
B_τ	$[g,0,l]$	C_τ	$[g,1,l]$	0
C_τ	$[g,1,l],\ l \neq P$	S_τ	$[g,1,l]$	1
S_τ	$[g,0,l],\ l \neq P$	S_τ	$[g,0,l]$	1
S_τ	$[g,0,P]$	$D_{q_\tau^+}$	$[g,0,J_\tau]$	1
D_q	$[g,0,E]$	R_q	$[g,0,P]$	-1
R_q	$[g,0,l],\ l \neq P$	R_q	$[g,0,l]$	-1
R_q	$[g,1,l],\ l \neq P$	A_q	$[g,1,l]$	0.

The first two rows use nonhalting q. In those rows and the B_τ row, l is unrestricted, including P; no additional history guard is imposed. Initialize in A_{q_0} at zero, with the input work tape, a unique mark at zero, pointer at two, and empty history elsewhere.

At checkpoint n, the work move ends at $w' = w_n + d_\tau \leq n+1 < p_n$. A stay instruction reaches C_τ in one step with the mark retained. A moving instruction reaches it in two steps after transferring the mark. From C_τ, move right, scan the $p_n - w' - 1$ intervening nonpointer cells, record at p_n, and extend the pointer to $p_n + 1$. The return scans $p_n - w'$ unmarked cells and stops at the persistent mark, entering the correct $A_{q_\tau^+}$. There are five other steps for a stay instruction and six for a moving instruction. This yields the stated count and restores the invariant. Every required guard holds, so the execution continues exactly until halting, including possible initial halting.

For an arbitrary allowed tape, each target determines its incoming move. It is zero into A_q or C_τ, d_τ into B_τ, one into S_τ or D_q, and minus one into R_q. The record τ recovers the source of the work update. A fixed C_τ receives only an A_q entry if $d_\tau = 0$, and only B_τ entries otherwise; the written symbol recovers the old full symbol in either case. At S_τ, the written mark separates entry from scan. At D_q, J_τ identifies the recording instruction and its former pointer. At R_q, the written pointer separates entry from the loop. At A_q, the unique source is R_q and the marked symbol is unchanged. Undoing the indicated shift therefore gives at most one predecessor in every case, on the full stated domain. □

Theorem 12.8 (A bounded-box event with a vertical routing guard)**.** *For every machine and finite input, and fixed positive computable viscosity, there is an effective smooth force on $\mathbb{R}^3$ with uniformly compact spatial support, bounded mixed derivatives and period one for $t \geq 1$, whose zero-data solution satisfies*

$$\exists t \geq 0 :\ X(t;(2,0,0)) \in (0,1)\times(0,1)\times(-1,1) \quad \Longleftrightarrow \quad \textit{the machine halts.}$$

The constructed solution is smooth with pressure zero. It is unique among smooth solutions with $u \in C_t H^2 \cap C_t^1 L^2$, bounded u on each finite interval, and $p \in C_t H^1$. This comparison class requires no bound on the competing velocity's spatial gradient.

Proof. Put $\rho(s) = \sigma(2s-1/2)$, using the effective flat switch from (5). Use the single-halt normalization and Lemma 12.7. For its full alphabet Σ, choose distinct odd digits $e(a) \in \{1,3,\dots,2|\Sigma|-1\}$ and base $B = 2|\Sigma|+1$. For an infinite word $\omega = \omega_0\omega_1\cdots$, define

$$c(\omega) = \sum_{j\geq 0} e(\omega_j) B^{-j-1}.$$

Place A_h at offset zero and the other control squares at first-coordinate offsets $3,6,9,\dots$. The code at height zero is

$$(o_s + c(\text{left stream}),\ c(\text{right stream}),0).$$

Every tail value lies in $[1/(B-1),(B-2)/(B-1)] \subset (0,1)$, so a halt code lies strictly inside the observed box. The two incoming properties give the separated filled rectangles of Lemma 6.2, with minimal left subdivision and first factors $1, B^{-1}, B$ for stay, right and left moves.

Apply the linear version of Lemma 7.1 to the full cuboids with third interval $[-1,1]$, heights $6i$, and switches $\rho(3s), \rho(3s-1), \rho(3s-2)$. Source projections separate lifts, heights separate horizontal motion, and target projections separate descents. Equation (23) gives the exact affine flow on every swept cuboid and a neighborhood.

The initial code x^{in} is rational. For example, if $b = \max(3, \text{input length})$, the first b right-stream symbols include the entire input and the pointer at cell two, after which the stream is constant; the left stream is constant. On the loading interval use

$$C_*(t) = (2,0,0) + \rho(t)(x^{\mathrm{in}} - (2,0,0))$$

and

$$V_*(t,x) = \nabla \times \left\{ \chi(C_*(t),(1,1,1),1;x)\tfrac{1}{2}C_*'(t) \times (x - C_*(t)) \right\}.$$

The cutoff is the centered one defined in Lemma 7.1. Its plateau velocity is C_*', so the fixed particle follows the displayed path. This field vanishes outside $[1/4, 3/4]$. Extend the step field V by zero and set $U = V_* + \sum_{k\geq 1} V(t-k)$, with its residual force at the prescribed ν. The finite union of swept cuboids and the loading tube is bounded; all time joins have zero collars. Consequently U and its force are smooth, uniformly compactly supported, and have bounded mixed derivatives, with period one after loading. Effective evaluation uses the finite formulas and a finite superset of potentially active integer shifts at any approximate time, as in Lemma 7.3.

The precise uniqueness input is Lemma 2.2(i). On each finite interval the constructed U is smooth, has one compact spatial support, and has bounded velocity and gradient. Consequently $U \in CH^2 \cap C^1L^2$, $U(0) = 0$, and the force is exactly $\mathcal{F}_\nu[U]$. The theorem's competitors are smooth classical $v \in CH^2 \cap C^1L^2$ with bounded v and a representative $p \in CH^1$, precisely case (i). No bound on ∇v is added. That case gives $v = U$ and fixes the H^1 pressure representative to zero. The lemma also supplies the unique global material trajectories of the reference field.

Loading and the exact full-box maps put the particle at the code after k local steps at time $1+k$, through the first halt or forever in a nonhalting execution. A halt code, including an initially halted code reached by loading, is inside the observation box. In a nonhalting run, loading has first coordinate at least two. During a later lift, the particle keeps its nonhalting source planar code, whose first coordinate exceeds three. During descent it keeps the nonhalting target planar code with the same property. Throughout the middle third its height is $6i > 1$. These three exclusions cover every real time and prove the event equivalence independently of horizontal excursions in the middle stage. The repeated processor installs every local branch from the finite table; it does not depend on the executed run or its outcome. □

12.5 Separating vertical and planar potentials

Theorem 12.9 (A decaying processor with phasewise planar potentials)**.** *At fixed computable $\nu > 0$ there is an effective force on $\mathbb{R}^3$ with one compact spatial support and all mixed derivatives of time order k bounded by $C_{k,\alpha}(1+t)^{-1-k}$, whose unique smooth zero-data solution satisfies*

$$\exists t \geq 0 : \ (\Phi_t(0))_1 < -1/2 \quad \Longleftrightarrow \quad \textit{halting.}$$

It uses the six-rule recorder with initial frontier 1, full subdivision of the rectangles, and separate vertical-translation and planar Hamiltonian potentials. The constructed pressure is zero and uniqueness is in Lemma 7.3's whole-space class.

Proof. Let Γ be the full alphabet of the six-rule table with $r_0 = 1$. Use odd digits in base $B = 2|\Gamma|+1$ and its fully subdivided maps. Put halt lanes at $-2, -4, \dots$ and the other lanes at $2, 4, \dots$, each of width one. The second coordinate lies in $[0, 1]$. For the resulting rectangles P_i, Q_i, centers p_i, q_i, and first scale λ_i, choose rational $0 < \varepsilon \le 1/4$ smaller than one quarter of all within-source and within-target maximum-norm distances. Empty pair lists impose no condition. Thicken by $[-1, 1]$ and put $H_i = 4i$.

On each third of the cycle use a flat progress function ϑ from zero to one. Lifting uses $z_i = H_i\vartheta$ and potential $A_i = (0, z_i' X_1, 0)$, whose curl is $(0, 0, z_i')$. At height H_i, set $c_i = (1-\vartheta)p_i + \vartheta q_i$ and $\mu_i = 1 - \vartheta + \vartheta\lambda_i$. Use the potential $A_i = (0, 0, \psi_i)$, where

$$\psi_i = c_{i1}'(X_2 - c_{i2}) - c_{i2}'(X_1 - c_{i1}) + \frac{\mu_i'}{\mu_i}(X_1 - c_{i1})(X_2 - c_{i2}).$$

Its curl is $(c_{i1}' + (\mu_i'/\mu_i)(X_1 - c_{i1}), c_{i2}' - (\mu_i'/\mu_i)(X_2 - c_{i2}), 0)$. Thus it realizes reciprocal scaling about the interpolated center. Descending uses the first potential with $z_i = H_i(1-\vartheta)$. Multiply each potential by its current-box ε cutoff before taking the curl. Source gaps, distinct middle heights, and target gaps separate the three stages. On the whole boxes and a neighborhood the uncut formulas are exact. The first coordinate is constant during vertical motion and is the convex interpolation of its endpoints during the planar stage.

For loading put $\theta_0(s) = \sigma(2s - 1/2)$, which is zero near zero and one near one. Let P_* be the rational initial code. Load it in the first unit interval by the planar path $m(s) = \theta_0(s)((P_*)_1, (P_*)_2)$, localized around $(m(s), 0) + [-1, 1]^3$, using potential $(0, 0, m_1'(X_2 - m_2) - m_2'(X_1 - m_1))$. Repeat the three-phase instruction field afterward to obtain W. Code induction and the delayed-head invariant prove the correspondence at times $1 + n$. Nonhalting codes have first coordinate at least 2, so loading and the convex-path bound exclude $X_1 < -1/2$ for every real time. A halt code has first coordinate at most -1, including one reached by loading an initially halted machine. Apply Lemma 9.1; physical code times are $e^{1+n} - 1$. Its support, derivative, event, and uniqueness conclusions give precisely the claimed separate force choice. □

13 Evacuation, columns and obstacle detours

The following refinements choose the occupied regions and the storage order explicitly. Ordered low extraction controls a height observer; simultaneous lifting controls a first-coordinate observer; expanded centers permit three-dimensional detours. Fixed columns and recorded pointers give further ways to keep a whole transition outside a detector. Each proof specifies which parts of the geometry act on solid neighborhoods and which bounds are needed only for the coded central plane.

Unless an individual theorem specifies another class, whole-space comparison means smooth $u \in CH^2 \cap C^1L^2$, bounded $u, \nabla u$ on finite intervals, and a pressure representative in CH^1, as in Section 6.4. The torus comparison class is the classical class of Lemma 2.2. The distinct pressure conditions stated below remain part of their respective results.

13.1 Ordered extraction below the computation

Proposition 13.1 (Ordered full-box transport)**.** *Let D_i, S_i be finite families of closed rational axis-aligned boxes in $\mathbb{R}^3$ with positive side lengths, pairwise disjoint within each family. Suppose the assigned center-to-center affine map F_i satisfies $F_i(D_i) = S_i$ and has positive diagonal linear part A_i of determinant one. There is an effective compactly supported solenoidal unit-time field, zero near the time endpoints, whose time-one map has this action on every full D_i. If both centers of a*

pair have negative third coordinate, every point of D_i whose third coordinate equals that of its center follows a path with negative third coordinate.

Proof. Choose an integer $L > 1$ larger than the absolute values of all endpoint coordinates. The storage columns have first coordinates $U_i = 4(i+1)L$ and common height $Z = -4L$. Write $[\ell_i, r_i]$ for the first-coordinate interval of D_i. Extract sources in decreasing order of r_i. Translate each horizontally right to its column and then vertically to Z, retaining its second coordinate. If an unextracted box meets the active box in both the second and third projections, disjointness forces their first intervals to be separated. That box cannot lie to the right: its right endpoint would then exceed r_i, contrary to the extraction order. It lies strictly to the left of ℓ_i and misses the entire rightward sweep. Stored boxes lie below the original boxes; vertical moves occur in distinct columns outside them.

At storage height, adjust the second center coordinate to its target value and apply A_i^θ. Each intermediate half-width lies between its two endpoint values, hence is less than L. The columns remain disjoint throughout these operations. Deliver boxes in increasing order of the right endpoint of their target first interval: raise a box to its target height and translate it left into its target. The raising column is empty. As above, a previously delivered box whose last two intervals meet the active ones must be separated in the first coordinate. Its smaller right endpoint puts it wholly to the left of the active target interval, so the leftward sweep stops before reaching it. Undelivered boxes remain at storage height. These observations give positive rational clearances for all swept bounding boxes. Choose cutoff padding below each clearance and use Lemma 3.1 in successive flat slots. Induction over the slots proves the full-box assertion. A point initially in the central third-coordinate plane retains zero third-coordinate offset during each diagonal deformation. Its height is interpolated only between the two endpoint heights and Z. If those endpoint heights are negative, every intermediate height is negative. □

The onto hypothesis in the preceding proposition is a useful exact-box specialization. Its target-container version, including the effective input conditions, is given in Section 13.5.

Theorem 13.2 (A positive-height observation)**.** *On $\mathbb{R}^3$, the fixed label 0 and observation $\{x_3 > 0\}$ realize halting effectively by a compactly supported general force with zero initial velocity and pressure, bounded mixed derivatives, and period one after time one. The force has a separate alternative in $L^2(0,\infty;H^j)$ for every j, tending to zero in every spatial supremum norm at rate $O((1+t)^{-1})$.*

Proof. Use Lemma 5.2 and odd digits in base $2m+2$. Give every simulator state a distinct positive integer index i_s and put its third-coordinate center at $z_s = i_s$ for ready halting states and $z_s = -i_s$ otherwise. Thicken the full rectangles of Lemma 5.1 by $[z_s - 1/4, z_s + 1/4]$. The assigned third scale is one, so the maps satisfy Proposition 13.1. The rational initial code Y_0 is loaded from the origin along the straight segment, by a localized translation during $[0,1]$. Repeat the unit-time mechanism after time one. At time $1+k$ the particle is exactly the code after k defined simulator transitions. If the initial machine halts, the loading endpoint already has positive height. If it never halts, loading has nonpositive height and all later transitions have negative endpoint centers; the proposition excludes a positive height at every intervening time. Conversely, a halting checkpoint has positive height. Lemma 6.3 gives the force and uniqueness, and Lemma 9.1 gives the separate temporal alternative. □

13.2 Simultaneous lifting with a convex-coordinate guard

Proposition 13.3 (Simultaneous full-box transport)**.** *Suppose finite planar source and target rectangle families S_i, T_i have positive side lengths and endpoints on the grid $B^{-2}\mathbb{Z}^2$, with integer*

$B \geq 2$, and are separately disjoint. Require their assigned center-to-center maps to satisfy $F_i(S_i) = T_i$ and to have linear part $\operatorname{diag}(\lambda_i, \lambda_i^{-1})$ with $\lambda_i \in \{B^{-1}, 1, B\}$. There is an effective solenoidal motion of the full boxes obtained by taking the product with $[-1, 1]$. For every transported point its first coordinate is a convex combination of its initial and final first coordinates at every time.

Proof. Let p_i^-, p_i^+ be the centers in the plane $z = 0$, set $h_i = 5i$, and choose a smooth Θ equal to zero on $(-\infty, 1/4]$ and one on $[3/4, \infty)$. For $0 \leq t \leq 1$, put

$$\theta = \Theta(3t - 1), \quad l_i = 1 - \theta + \theta\lambda_i, \quad c_i = (1 - \theta)p_i^- + \theta p_i^+ + h_i\{\Theta(3t) - \Theta(3t - 2)\}e_3.$$

The desired path is $g_i(t, y) = c_i(t) + \operatorname{diag}(l_i, l_i^{-1}, 1)(y - p_i^-)$. Use a source padding $\varepsilon = 1/(10B^3)$, transported by this same affine map, for the cutoff neighborhood. During lifting the planar source gap is at least B^{-2}. During lowering the enlarged target padding is at most $B\varepsilon < B^{-2}/2$. During the middle phase the height difference is at least five, whereas the sum of third half-widths is $2 + 2\varepsilon < 5$. Thus these moving cutoff supports can be chosen disjoint. Sum the localized curls of Lemma 3.1. Differentiation and uniqueness give the displayed paths for the full boxes. Finally

$$g_{i,1}(t, y) = (1 - \theta)y_1 + \theta\{p_{i,1}^+ + \lambda_i(y_1 - p_{i,1}^-)\},$$

which proves the asserted guard. Flat time collars hold automatically. □

Theorem 13.4 (A zero-blank half-space test)**.** *The fixed label 0 and event $\{x_1 < -1\}$ in $\mathbb{R}^3$ realize halting with the force and comparison conclusions of Theorem 13.2. This construction uses a zero joint-blank digit, so every initial tape coordinate is a finite radix sum. Its temporal alternative belongs to $L^2(0, \infty; H^j)$ for every j.*

Proof. Use the frontier recorder, zero-blank base $B = 2m$, and state first-coordinate offsets $3j$ for nonterminal controls and $-3j$ for terminal controls, with positive distinct indices. Encode a state by $(o_s + L, R, 0)$. The full branches are on the B^{-2} grid and satisfy the preceding proposition. Load the rational initial code from zero along a straight segment. Nonhalting codes have nonnegative first coordinate; the whole segment and every nonhalting transition therefore avoid $x_1 < -1$. Terminal codes have first coordinate at most -2, including an initially terminal loading endpoint. Exact integer-time simulation gives both implications. The force conclusions follow from the common analytic and logarithmic-clock lemmas, with the compact support needed for all H^j norms. No claim of periodicity is made for the slowed alternative. □

13.3 Expanded centers and obstacle detours

Proposition 13.5 (Full solid boxes with a height guard)**.** *Let the source and target families be finite collections of full rational axis-aligned solid boxes, each family pairwise disjoint. Write their centers as a_i, b_i and their positive half-widths as $r_{i,j}, w_{i,j}$. Require each prescribed affine map to send its entire source box onto its specified target box:*

$$F_i(a_i + h) = b_i + D_i h, \qquad D_i = \operatorname{diag}(w_{i,1}/r_{i,1}, w_{i,2}/r_{i,2}, w_{i,3}/r_{i,3}), \qquad \det D_i = 1.$$

Suppose every source center has height at most -1, and every target center has height at most -1 or exactly 1. They admit an effective compactly supported solenoidal unit-time realization. If a target center has negative height, every point $a_i + h$ of its source box with $h_3 = 0$ remains at negative height throughout. A rational bound for the entire support is computable from the boxes.

Proof. The empty family is realized by zero. Otherwise let $R = 2+$ the largest endpoint half-width, and choose an integer $H > 20R$ so large that the expanded source centers $A_i^* = Ha_i$ and target centers $B_i^* = Hb_i$ are separately more than $20R$ apart in the sup norm. Set

$$Z_0 = 1 + 100R + \max_i\{\|A_i^*\|_\infty, \|B_i^*\|_\infty\}, \qquad P_i = (0, 0, -Z_0 - 100Ri).$$

Each of the unions $\{A_i^*\} \cup \{P_i\}$ and $\{B_i^*\} \cup \{P_i\}$ has separation greater than $20R$. The construction has five phases: expand source centers by a factor from one to H while keeping shapes fixed; move them successively to P_i; change each parked shape by the prescribed positive diagonal power; move them successively to B_i^*; contract target centers to b_i while keeping the final shapes fixed.

The simultaneous expansion is collision free. For each pair of source boxes some coordinate has positive gap $|a_{i,j} - a_{k,j}| - r_{i,j} - r_{k,j}$; multiplication of the center difference by a factor at least one preserves that gap. The same argument applies to contraction toward the disjoint targets. Choose padding less than one quarter of the minimum such gap, and less than one. During a parked deformation the jth half-width is $r_{i,j}^{1-\theta} w_{i,j}^{\theta} \le \max(r_{i,j}, w_{i,j}) < R$. Thus parked deformations have all half-widths less than R and supports of half-width at most $2R$; the storage separation suffices.

For a successive transfer, surround each stationary center by its closed sup-norm cube of radius $4R$. These cubes are pairwise disjoint. Start with the straight segment between the moving centers. Each nontrivial intersection with such a cube can be replaced by a polygonal path on its boundary: join the entrance to a vertex of its face, follow cube edges to an exit-face vertex, and join to the exit. Fixed index orders choose the vertices and edges. All coordinates and intersection parameters are rational. Tangencies need no detour. Since different obstacle cubes are disjoint, the resulting finite polygon stays at distance at least $4R$ from every stationary center. Move along its segments in flat slots, using a cutoff support of half-width at most $2R$. Stationary boxes have half-width less than R, so no such support touches them. Lemma 3.1 realizes each translation and deformation.

For the height assertion, a transfer with negative endpoints has endpoints of height at most $-H$. Its straight segment is negative. It cannot meet an obstacle centered at positive height H, whose bottom is $H - 4R > 0$. Every boundary used around a negative-height obstacle lies below $-H + 4R < 0$. Storage heights are negative as well. Expansion, contraction, and positive diagonal deformation preserve the central-plane condition. Thus all these paths are negative. Finitely many rational vertices, endpoint widths, and cutoff paddings give a rational support bound. □

Theorem 13.6 (A height test inside one torus chart)**.** *On the unit torus, the fixed label* $o = (1/2, 1/2, 1/2)$ *and event*

$$\{x : x_3 \pmod 1 \in (1/2, 1)\} \tag{45}$$

realize halting by a smooth general mean-zero force, periodic after time one, with zero initial velocity and pressure and bounded mixed derivatives. The repeated mechanism depends on the machine alone; only initialization depends on its finite input. The solution is unique in the torus class.

Proof. Use the frontier recorder, odd base $B = 2m + 1$, distinct negative integer heights for nonterminal states and height one for the terminal ready state. Thicken every branch by a third half-width $1/4$. The reciprocal tape map is onto its target rectangle; adjoining the unchanged third interval gives an onto box map with linear part $\mathrm{diag}(B^{-e}, B^e, 1)$ and determinant one. All source centers are negative because terminal controls have no outgoing rules. Apply the preceding proposition. The logical initial code Y_0 is rational and has a uniform machine-dependent bound as the input varies: its two stack coordinates lie in $[0, 1]$ and its height is fixed by the initial control. A moving translation loads the logical origin to Y_0, with a fixed source half-width one and padding one. This gives a machine-dependent rational bound K for the supports of all such loaders and the

repeated mechanism; for example take $K = 1 + 2R + M$ with M an upper bound for all routing vertices and loading coordinates and R enlarged to cover their half-widths. Choose $\lambda = 1/(4K)$ and use the chart $x = o + \lambda Y$. Everything remains strictly inside $(1/4, 3/4)^3$.

Periodize the resulting compactly supported curl fields, loading during the first unit interval and repeating thereafter. Compute the residual at physical viscosity ν, after the chart change. The fields and force have zero mean and all the bounds in Lemma 6.3. For a nonhalting run, loading has nonpositive logical height and each later segment remains negative by the proposition. A terminal code has physical height $1/2 + \lambda$, which lies in the event. There is no chart-boundary crossing that could create a spurious observation. An initially terminal input is detected at the loading endpoint. The initialized correspondence is exact; at checkpoints the number of history records gives n, and the frontier's relative position together with its absolute index $2 + n$ also recovers the absolute work-head position. □

13.4 Fixed columns and separate layers

Proposition 13.7 (Column and layer transport)**.** *Let S_i, T_i be finite rational planar rectangle families of positive side lengths, each separately disjoint, with endpoints on the B^{-2} grid, where $B \geq 2$ is an integer. Let their assigned center-to-center positive reciprocal diagonal affine maps satisfy $F_i(S_i) = T_i$. A finite effective solenoidal field in $\mathbb{R}^3$ realizes all these maps on the rectangles in $z = 0$. Along each tracked path the first coordinate stays between its initial and final values. The field has compact support and flat time collars.*

Proof. Give rectangle i height $h_i = 4i$. A planar cutoff equal to one near its source and supported within padding $1/(4B^2)$ has support disjoint from the other source cutoffs. Multiply it by a third-coordinate cutoff which is one near the entire interval $[0, h_i]$. The curl of the potential $(0, h_i x, 0)$ times these cutoffs equals $(0, 0, h_i)$ along that source column. A common flat schedule lifts all rectangles. The corresponding target cutoffs give a later descent by $-h_i$.

In the middle phase, let p_i, q_i be the centers and $a_i > 0$ the first-coordinate scale. With phase parameter $\mu \in [0, 1]$, set $c_i = p_i + \mu(q_i - p_i)$, $l_i = 1 + \mu(a_i - 1)$ and

$$H_i(\mu, x, y) = d_{i,1} y - d_{i,2} x + \frac{a_i - 1}{l_i}(x - c_{i,1})(y - c_{i,2}), \qquad d_i = q_i - p_i.$$

On a neighborhood of the desired path, the planar Hamiltonian field $(\partial_y H_i, -\partial_x H_i)$ generates $c_i + \operatorname{diag}(l_i, l_i^{-1})(\zeta - p_i)$. The first coordinate is the convex interpolation of its endpoint values. The second half-width obeys

$$\frac{r_{i,2}}{(1 - \mu) + \mu a_i} \leq (1 - \mu) r_{i,2} + \mu r_{i,2}/a_i,$$

by convexity of the reciprocal function. Consequently the swept rectangle lies in the coordinate bounding rectangle of the two endpoints. Localize H_i around that rectangle in the layer $z = h_i$, with third padding less than one, and take the curl of $(0, 0, H_i)$ times that cutoff. For a flat middle ramp $\mu = \eta_2(t)$, multiply the resulting phase field by $\eta_2'(t)$. Layers are disjoint, so the sum has the required action. Lift and descent preserve planar coordinates. The exact paths and ODE uniqueness establish the full-rectangle and intermediate-coordinate assertions. □

13.5 Target containers and prescribed affine images

The named targets in Propositions 13.1, 13.3 and 13.7 can also be rational *containers* for the actual images. More precisely, replace the onto condition by

$$F_i(D_i) \subseteq S_i \quad \text{in Proposition 13.1,} \qquad F_i(S_i) \subseteq T_i \quad \text{in the other two propositions.}$$

Keep the other geometric hypotheses. For ordered transport retain the center-to-center condition. For simultaneous and column transport the actual endpoint center is $F_i(p_i)$, where p_i is the source center; it need not be the center of the container. The maps have the same positive diagonal determinant-one linear parts as before, including the specified three scale choices in Proposition 13.3. All transport and path conclusions remain valid. For arbitrary real map coefficients this is a smooth existence statement. The construction is effective when those coefficients are computable, and otherwise its evaluation is relative to them. An empty list again uses the zero field.

Here are the necessary modifications of the proofs. In ordered transport, write $r_{i,j}$ for a source half-width, $w_{i,j}$ for the corresponding container half-width, and $a_{i,j} > 0$ for the diagonal factor. Containment gives $a_{i,j} r_{i,j} \le w_{i,j}$, hence

$$r_{i,j} a_{i,j}^{\theta} \le \max(r_{i,j}, a_{i,j} r_{i,j}) \le \max(r_{i,j}, w_{i,j}) < L, \qquad 0 \le \theta \le 1.$$

The rational source and container widths therefore bound every parked deformation. After deformation, use translates of the rational target containers as envelopes for the actual images. Deliver in increasing order of the containers' right endpoints. If an earlier delivered image meets the moving image in the last two projections, their containers meet in those projections. Disjointness and the endpoint order force the earlier container wholly to the left of the active target container. Its image therefore misses the entire leftward sweep, which stops in that active container. The original extraction and storage arguments are unchanged. The same rational envelopes give positive cutoff margins. They do not require the actual image endpoints to be rational. The central-cross-section height guard is unchanged because its third offset remains zero.

For simultaneous transport, interpolate toward $F_i(p_i)$. The transported source padding at descent has width at most $B\varepsilon$ outside its actual image, so it lies inside the $B\varepsilon$ enlargement of the target container. The original grid-gap and private-height estimates still separate all supports. The first-coordinate convex identity uses the actual value $F_i(y)$ and remains exact.

For column transport use the same actual endpoint center. The reciprocal convexity estimate bounds each moving second half-width by the linear interpolation of the source and actual-image half-widths. Thus the entire middle sweep lies in the coordinate bounding rectangle of the source and its rational target container. The target-column cutoff equals one on the actual image, so the descent is unchanged. This also proves the stated first-coordinate bound for each point. Finally, a positive computable scale admits computable positive lower and finite upper bounds; its logarithm, powers and the displayed reciprocal interpolations are computable. The containment and determinant identities are hypotheses, not tests performed by the construction.

Theorem 13.8 (A half-space test by fixed columns)**.** *On $\mathbb{R}^3$, the frontier recorder with odd digits in base $2m+1$ realizes halting from the fixed label 0 by the event $x_1 < -1$. It gives a compactly supported general force, periodic after time one, with zero initial velocity and pressure and all mixed derivatives bounded.*

Proof. Put nonterminal state offsets at $3, 6, \dots$ and terminal offsets at $-3, -6, \dots$ in the first coordinate. Lemma 5.1 and Proposition 13.7 give a whole-rectangle phase map. A localized planar Hamiltonian translation loads the origin to the rational initial code in $[0, 1]$ time. A nonterminal loading segment has nonnegative first coordinate, and every later nonterminal segment has first coordinate at least three. Terminal codes have first coordinate below -1. This proves both directions, including initial halting. Repeat the phase map and use Lemma 6.3. Although this route only requires planar coding rectangles, its cutoff plateaus also give a positive neighborhood on which the same affine formulas hold. □

13.6 Delaying the original head movement

Use Lemma 5.3 with singleton passive track removed. Rename $(M(q,s), G(r), A(q,d), B(q,d))$ as $(R_{q,s}, F_r, C_{q,d}, G_{q,d})$ and $(\bot, F)$ as $(\bot, *)$. These bijections identify every row and nonfrontier guard on the full domain, including all q, d controls. The frontier starts at one, and the work movement occurs only on return. At checkpoint n, write i for the work-head position and $j = n+1$ for the frontier position. The exact count is $2(j-i)+3$: the right and left continuing loops have respectively $j-i-1$ and $j-i$ instructions, with four other instructions.

Theorem 13.9 (The delayed-head column realization)**.** *The table of Lemma 5.3 has a realization on $\mathbb{R}^3$ with fixed label 0 and detector $x_1 < -1$, compact spatial support, period one after time one, bounded mixed derivatives and uniformly bounded kinetic energy. The prescribed pressure is zero. Uniqueness holds in the whole-space comparison class, in particular when the competitor and its first spatial derivatives are bounded on every finite time interval.*

Proof. Assign odd digits $1, 3, \dots, 2m-1$ in base $B = 2m+2$, with state offsets positive multiples of three except for negative terminal offsets. The tail of the blank symbol contributes $d_{\mathrm{blank}} B^{-k}/(B-1)$ after a finite prefix, so initialization is rational even though the blank digit is nonzero. Full affine branches follow from the lemma and Lemma 5.1. Use the fixed-column proof with heights $4i$ and the smaller padding $1/(10B^2)$; schedule a slot $[a,b]$ by $\sigma(2(t-a)/(b-a)-1/2)$, which is flat on collars of both endpoints. These choices preserve the same separation proof. A planar Hamiltonian loading segment starts at zero. The convex first- coordinate guard excludes the detector throughout every nonhalting transition and throughout a nonhalting load; a terminal code lies below -1. Finite checkpoint simulation gives the equivalence. The field has one compact support and uniform pointwise bounds, hence uniformly bounded kinetic energy. Lemma 6.3 supplies all remaining assertions. □

13.7 Recorded pointers and seven-slot storage

Theorem 13.10 (A fixed box test with recorded pointers)**.** *On $\mathbb{R}^3$, the fixed label $(2,0,0)$ and open box $(0,1)^2 \times (-1/2, 1/2)$ realize halting by an effective general force with compact spatial support, zero initial velocity and pressure, bounded mixed derivatives, and period one after time one. The construction moves whole solid boxes. Uniqueness holds among smooth competitors with*

$$u \in C^1([0,T]; L^2), \quad \nabla u, \Delta u \in C([0,T]; L^2), \quad p, \nabla p \in C([0,T]; L^2), \quad \sup_{[0,T]\times\mathbb{R}^3} (|u| + |\nabla u|) < \infty$$

for every finite T.

Proof. Normalize to one halt h. For $c = (q,b) \in \mathcal{D} = (Q \setminus \{h\}) \times \Gamma$ write the instruction as $(q^+(c), b^+(c), m(c))$. Use histories

$$\Lambda_0 = \{\varnothing\} \sqcup \{H_e : e \in \mathcal{D} \sqcup \{\star\}\}, \qquad \Lambda = \Lambda_0 \sqcup \{P_e : e \in \mathcal{D} \sqcup \{\star\}\}.$$

The full alphabet is $\mathcal{A} = \Gamma \times \Lambda \times \{0,1\}$, whose tracks are data, history and return mark. Controls

are S_q, L_q, M_c, R_c, F_c, and the complete table is

input	output	move	restriction
$S_q(b,l,0)$	$M_c(b^+(c),l,0)$	$m(c)$	$c=(q,b),\ l\in\Lambda_0$
$M_c(b,l,0)$	$R_c(b,l,1)$	1	$l\in\Lambda_0$
$R_c(b,l,0)$	$R_c(b,l,0)$	1	$l\in\Lambda_0$
$R_c(b,P_e,0)$	$F_c(b,H_e,0)$	1	$e\in\mathcal{D}\sqcup\{\star\}$
$F_c(b,\varnothing,0)$	$L_{q^+(c)}(b,P_c,0)$	-1	
$L_q(b,l,0)$	$L_q(b,l,0)$	-1	$l\in\Lambda_0$
$L_q(b,l,1)$	$S_q(b,l,0)$	0	$l\in\Lambda_0$.

Initially put $P_\star$ at absolute cell three, empty history elsewhere, marks zero, and control S_{q_0}. At ready checkpoint k, the pointer is at $r=3+k$, all cells to its right are empty, and the work configuration is the original one. After the first row the work head satisfies $j'\le k+1\le r-2$. The scan marks it, reaches the pointer, changes P_e into H_e, writes P_c in the next empty cell, and returns to erase the mark. Both scans are finite and positive. This proves the initialized simulation. The initial pointer tag is retained as $H_\star$ after the first cycle, so the absolute cell-three reference is permanent.

The inverse works on all allowed tapes. Into M_c the tag fixes the old instruction and move; into R_c mark one distinguishes entry from a loop; into F_c the written H_e identifies the old pointer P_e; into L_q a pointer output identifies entry, including the old F_c, whereas a loop writes a nonpointer; into S_q only mark erasure applies. Incoming shifts are fixed in every target control. The remaining written components retain all unchanged symbols, proving the two properties of Lemma 5.1.

Let m_Q be the number of controls, and number them bijectively by $\iota(s)\in\{0,\dots,m_Q-1\}$ with $\iota(S_h)=0$. Use odd digits in base $B=2|\mathcal{A}|+1$ and state offsets $o_s=3\iota(s)$. The physical code is $(o_s+L,R,0)$. Terminal codes lie strictly in $(0,1)^2\times\{0\}$; nonterminal first coordinates are at least three. Form the full branch boxes by taking the product with $[-1,1]$. Their planar sides have length at most one. Choose $H_x=3m_Q+1$ and parking centers $\pi_i=(H_x+10+4i,0)$, all at height zero.

Here is a storage construction distinct from ordered low extraction. Evacuate each source in three successive operations: lift its center to height ten, translate horizontally there to π_i, and lower it to zero. Once every box is parked, process each in four operations: deform its planar shape at the parking center, lift it to height ten, translate it there to its target center, and lower it into the target. This uses $7N$ slots for N branches. During deformation use the positive scale $l=1+\theta(\lambda_i-1)$ and its reciprocal, not an exponential scale. Both side lengths remain bounded by the maximum of their endpoint lengths, which is at most one. The third half-width remains one.

Let δ be the minimum of one and all positive source-source and target-target planar distances, omitting empty sets of pairs. Use cutoff plateaus with padding $\varepsilon=\delta/8$ and support padding 2ε. During a lift or descent the planar projection separates the moving box from the current sources or delivered targets. Parked boxes lie outside their entire first-coordinate range, and parking centers have spacing four. During a horizontal translation the box lies in $9\le z\le 11$, whereas all stationary boxes lie in $-1\le z\le 1$. These positive gaps also separate the padded supports. Lemma 3.1 therefore realizes every operation without disturbing a stationary box. The full assigned affine map follows by composition.

For a branch whose target control is nonterminal, every full point has first coordinate at least three, including the storage and transfer paths. Load the fixed label $(2,0,0)$ to the rational initial code by a localized straight translation. If the code is nonterminal, the loading segment has first coordinate at least two; if it is terminal, the endpoint is in the observation box. At later integer samples the exact recorder code gives terminal entry, and the preceding bounds exclude the observation at all

other times of a nonhalting run. Repeat the finite transport field and apply Lemma 6.3. Finally the displayed comparison class implies $u \in CH^2$, since $\|u\|_{H^2} \le C(\|u\|_2 + \|\Delta u\|_2)$ by the Fourier multiplier identity; its pressure assumptions give $p \in CH^1$. Thus the stated uniqueness follows from the common whole-space comparison, without changing this route's original hypotheses. □

13.8 Planar detours behind a vertical observation guard

Theorem 13.11 (A guarded planar observation with unrestricted pressure)**.** *On $\mathbb{R}^3$, the fixed label $(-4,0,0)$ and fixed box $(-1,2)^2 \times (-1,1)$ realize halting by a compactly supported general force with zero initial velocity and prescribed pressure and all mixed derivatives bounded. One choice is periodic after time one. A separate choice satisfies $\|f(t)\|_\infty = O((1+t)^{-2/3})$ and $f \in L^2([0,\infty) \times \mathbb{R}^3)$. Uniqueness holds among smooth competitors with $u \in C([0,T]; H^2) \cap C^1([0,T]; L^2)$ and bounded $u, \nabla u$ on each finite interval, without any spatial integrability condition on pressure. Pressure is unique modulo a function of time.*

Proof. Use the frontier recorder after normalization to one halting state, and write $\mathcal{A}$ for its full cell alphabet and m_Q for its number of controls. Number the controls bijectively by $\iota(s) \in \{0, \dots, m_Q - 1\}$, giving the terminal control index zero. Use odd base $B = 2|\mathcal{A}| + 1$ and first-coordinate offsets $o_s = 4\iota(s)$. Code the configuration by $(o_s + L, R)$. The origin of the original tape can be recovered from the number of history records and the frontier position, so this head-relative coding loses no initialized information. The full planar branches have reciprocal positive diagonal maps.

We first realize finite separately disjoint source and target rectangle families S_i, T_i with rational endpoints and positive side lengths, equipped with assigned positive reciprocal diagonal affine maps F_i satisfying $F_i(S_i) = T_i$, by compactly supported planar Hamiltonian motion. Write c_i, q_i for the source and target centers. With no branches use zero. With one branch, interpolate its center directly and use a positive linear first-coordinate scale and its reciprocal; there are no stationary obstacles. For $N \ge 2$, let $R = 1+$ the largest endpoint half-width and let $m_0 > 0$ be the minimum sup-norm separation within the two center families. Put $K = 1 + 32R/m_0$. Expand all source centers by factors from one to K, leaving their shapes unchanged. Let $M = \max_i\{|(Kc_i)_1|, |(Kq_i)_1|\}$ and choose parking centers $P_i = (M + 32Ri, 0)$. Each union of expanded endpoint centers and parking centers has separation at least $32R$.

Move expanded sources successively to their parking centers, deform them there using the linear reciprocal scale, move them successively to the expanded target centers, then contract those centers to their targets. For each successive transfer put a square of sup-radius $4R$ around every stationary center. Replace each nontrivial segment intersection with an obstacle square by a counterclockwise path along its boundary. The intersection points and corners are rational; the squares are disjoint, so the resulting finite polygon stays at distance at least $4R$ from all stationary centers. Flat schedules on its segments give smooth center motion. The simultaneous expansion and contraction preserve a positive coordinate gap for every pair, exactly as in the solid-box argument. If g_0, g_1 are their minimum positive gaps, choose padding $\varepsilon = \min(g_0, g_1, 2R)/4$ for these phases. During individual transfers use padding below R; the stationary half-widths are below R. Thus all required cutoff supports avoid stationary rectangles.

For a moving center c and horizontal scale l, the potential

$$H = \chi\left[\dot{c}_x(y - c_y) - \dot{c}_y(x - c_x) + \frac{\dot{l}}{l}(x - c_x)(y - c_y)\right]$$

gives the required motion on the cutoff plateau. The other current rectangles are fixed. This proves every full planar branch, including its boundary, by induction through the operations. A one-square

loading translation with half-width $1/4$ takes $(-4,0)$ to the initial rational code. All these supports lie in a square $[-L,L]^2$, where one can choose $L = 10 + S_0 + 4m_Q$ with S_0 a finite bound for routing vertices and cutoff widths. The same L covers every input of a fixed machine, since initial codes are uniformly bounded.

The planar detours need not avoid the horizontal observation box. We instead make them at height three. Choose a switch η with collars and set $h(s) = 3\{\eta(3s) - \eta(3s-2)\}$ on a unit interval. It lifts in the first third, stays at three in the middle, and lowers in the last third. Choose Γ equal to one near $[-L,L]^2 \times [-1,4]$, with compact support, and $\beta(z)$ compactly supported and equal to one near three. For any one of the planar fields w, zero outside its phase interval, put

$$Z(s,x) = h'(s)\nabla \times (0, x_1\Gamma(x), 0), \qquad V_w(s,x) = Z(s,x) + 3\beta(x_3)(w(3s-1,x_1,x_2),0).$$

Along each tracked path the first term is exact vertical translation. The second term acts only in the middle third, at height three, and runs the full planar phase. It is divergence free because w is planar solenoidal and β is independent of the planar coordinates. Thus the trajectory lifts at its exact source, performs every detour outside the vertical detector interval, and lowers at its exact target. Use this construction for the loader and for each repeated phase. While its height can lie in $(-1,1)$, a nonterminal computational particle has first coordinate at least four; the initial loading source has first coordinate -4. A terminal code lies in $(0,1)^2 \times \{0\}$. These facts prove the all-real-time observation equivalence, including an initially terminal input.

For the comparison assertion, the constructed field has compact support and bounded mixed derivatives, so it satisfies the reference hypotheses of Lemma 2.2. The displayed competitor class is exactly case (v) of that lemma. Its proof obtains $\nabla p \in L^2$ from the equation and cancels the pressure term by Lemma 2.1; it does not require an integrable pressure representative. Hence the velocity is unique, and pressure is determined up to a function of time.

Finally let $U(s)$ denote the eventually periodic field just constructed and set $\rho(t) = (1+t)^{1/3} - 1$, $\widehat{U} = \rho' U(\rho(t))$. The new residual is

$$\widehat{f} = \rho'' U + (\rho')^2\{U_s + (U\cdot\nabla)U\} - \nu\rho'\Delta U, \quad \rho' = \tfrac{1}{3}(1+t)^{-2/3}, \quad \rho'' = -\tfrac{2}{9}(1+t)^{-5/3}.$$

Uniform derivative bounds and the fixed compact support give the stated decay and space-time L^2 estimate. Chain differentiation also gives all mixed bounds. The clock is effective on $[0,\infty)$, starts at zero, is finite at finite time and is onto; hence it preserves precisely the same material event. The initial zero collar is preserved as well. This slower force is an alternative to, not a periodic version of, the first force. □

14 Torus charts and phasewise localization

The next constructions keep the whole motion inside a specified torus chart. A scaled linear interpolation preserves a convex first coordinate; an exponential interpolation in a fixed chart instead uses endpoint-width bounds. A phasewise construction then exhibits the vertical and planar potentials separately, and the final variant retains a permanent origin track. These choices give different path estimates for their observers. In every case the force is recomputed at the prescribed physical viscosity after any spatial rescaling.

14.1 Physical chart scaling and thickened torus boxes

Proposition 14.1 (Linear scaling into the torus)**.** *The direction recorder without its passive track has a unit-torus realization periodic after $t = 1$, with fixed label $(1/8, 0, 0)$ and halting event $x_1 \in (-1/4, 0) \pmod 1$. Each period acts on full solid boxes; the pressure is zero and the force has zero mean.*

Proof. Use the controls of Lemma 5.3. Let $\Sigma = \Gamma \times (\{\perp, F\} \sqcup \mathcal{R}) \times \{0,1\}$ be its full cell alphabet after deleting the singleton passive track. Let S be the finite set of recorder controls and choose a bijection $j : S \to \{1, \dots, |S|\}$. Use odd digits with $B = 2|\Sigma| + 1$, and offsets $o_s = -2j(s)$ for halting main controls and $o_s = 2j(s)$ for the others. Number the resulting reciprocal branches $i = 1, \dots, N$, and write k_i for branch i's first-coordinate scale. Thicken each reciprocal branch by $[-1, 1]$. Its horizontal family gaps are at least B^{-2}. Lift branch i to $4i$, use the linear first scale $\kappa_i = 1 + \theta(k_i - 1)$, and descend. Padding smaller than $(4B^2)^{-1}$ separates all phases: in the middle the half-height is one and height gaps are four; in the other phases horizontal gaps suffice. The two horizontal half-widths are at most $1/2$. Use the true-plateau cutoff of Lemma 3.1; it supplies at least the full-box guarantee, together with a neighborhood extension. The exact first coordinate obeys

$$X_1(\tau) = (1 - \theta)X_1(0) + \theta X_1(1). \tag{46}$$

Let V be the period-one extension of the unit-time field just constructed. All codes, paths and padded supports lie in $(-L, L)^3$ for $L = 10 + 2|S| + 4N$. Put $a = 1/(16L)$ and define in the torus patch $(-1/2, 1/2)^3$ the repeated field $U(t, x) = aV(t, x/a)$. Its support lies in $(-1/16, 1/16)^3$ and its curl potential is the scaled potential $a^2 A(t, x/a)$. In particular its mean is zero. The residual must be recomputed at the physical viscosity:

$$\mathcal{F}_\nu[U](t, x) = a\{V_t + (V \cdot \nabla)V - (\nu/a^2)\Delta V\}(t, x/a). \tag{47}$$

This keeps the specified viscosity rather than silently rescaling it. During the first unit interval load from $(1/8, 0, 0)$ to the scaled initial code using a curl translation with a box of half-width $1/32$ and padding $1/32$. Its support lies inside $(-1/4, 1/4)^3$.

For a nonhalting run both loading endpoints have positive first coordinate, and every subsequent pair of coded endpoints is positive. Equation (46) preserves that sign throughout. A halting checkpoint has negative first coordinate in $(-1/16, 0)$. All paths stay in the stated patch, so passage modulo one introduces no extra visit. Checkpoint times are $1 + \sum_{l<n}(5 + 2(l - h_l))$. This proves the all-time equivalence, including initial halting; Proposition 3.2 finishes the proof. □

Proposition 14.2 (Exponential deformation in a fixed chart)**.** *The move-first recorder has a unit-torus realization periodic after one loading interval, with fixed label* $(1/4, 1/4, 1/4)$ *and halting event* $x_1 \in (2/3, 5/6) \pmod 1$*. The repeated motion acts on thickened closed rectangles and their neighborhoods. Its mean-zero residual has the form* $f_0 + \nu f_1$ *with coefficients independent of* ν*.*

Proof. First normalize to one halting work state and use the move-first recorder of Lemma 5.2; let Σ be its full three-track cell alphabet. Initial halting is preserved by this normalization. For m controls take $\ell = 1/[64(m + 1)]$. Nonhalt squares have lower left corners $(1/4 + 2j\ell, 1/4)$, $0 \le j \le m - 2$; the halt square has lower left corner $(3/4, 1/4)$. Use odd digits and $b = 2|\Sigma| + 1$, with coding height $1/4$. There are N reciprocal branches with gaps at least ℓb^{-2}. Thicken them by half-height $h_0 = 1/[64(N + 1)]$ and choose heights $z_i = 1/2 + i/[4(N + 1)]$. Use exponential scale k_i^θ, center interpolation, and $\epsilon = \min(h_0/2, \ell/(8b^2))$. The first and last phases have horizontal gaps exceeding 2ϵ; middle height gaps equal $16h_0$ and exceed $2(h_0 + \epsilon)$. Widths never exceed their endpoint values. A concrete plateau cutoff for a box centered at C_i, with half-widths w_{ij}, is

$$\chi_i = \prod_{j=1}^{3} \sigma\left(\frac{(w_{ij} + \epsilon)^2 - (x_j - C_{ij})^2}{(w_{ij} + \epsilon)^2 - (w_{ij} + \epsilon/2)^2}\right).$$

Use its product with $\frac{1}{2}\dot{C}_i \times (x - C_i) + (0, 0, (\dot{a}_i/a_i)(x_1 - C_{i1})(x_2 - C_{i2}))$, where $a_i = k_i^\theta$. Its curl gives the exact affine motion on an open neighborhood of each box. All supports remain inside the unit cube.

Load the initial rational code along its segment from the fixed label, using a radial cutoff of radius $1/32$ and a flat switch $\sigma(4t-1)$. Then repeat the unit motion, with phase switches $\sigma(8\tau-1), \sigma(8\tau-3), \sigma(8\tau-5)$. These choices give spatial zero collars of width $1/8$ and temporal zero collars of width $1/16$ at nonnegative integer joins, including loading. The force follows from $f_0 = U_t + (U\cdot\nabla)U$, $f_1 = -\Delta U$.

Nonhalt centers have first coordinate at most $1/4 + 1/32$ and half-width at most $\ell/2$, hence their entire trajectories satisfy $x_1 \le 1/4 + 1/32 + \ell/2 < 2/3$. This is a width bound, not a convexity claim for exponential scaling. Loading also avoids the slab for a nonhalting input. Halt codes enter it at a finite integer sample, including time one for an initial halt. The record block length and the known frontier site $n+2$ recover absolute head position at each work checkpoint. All force and uniqueness assertions now follow from Proposition 3.2. □

14.2 Thin-sheet curls and a slab observer

The next realization stays directly inside the unit cube. It uses a vertical translation potential in the first and last phases and a planar Hamiltonian potential in the middle. Because its observer ignores height, the proof must keep the first coordinate safe throughout all three phases.

Theorem 14.3 (A direct unit-cube implementation)**.** *At each fixed positive computable viscosity, every machine and finite input has an effective smooth mean-zero force on the unit flat three-torus, with bounded mixed derivatives and period one for $t \ge 1$, whose smooth zero-data solution satisfies*

$$\exists t \ge 0 :\ X(t;(1/4,1/4,1/4)) \in (2/3,5/6)\times\mathbb{T}^2 \quad\Longleftrightarrow\quad \textit{the machine halts.}$$

The normalized pressure is zero and the solution is unique in the classical torus comparison class of Section 2.

Proof. For the cutoffs in this proof, recall

$$E(r) = \begin{cases} e^{-1/r}, & r>0,\\ 0, & r\le 0,\end{cases} \qquad \Theta(r) = \frac{E(r)}{E(r)+E(1-r)}.$$

Use the single-halt normalization and the seven-row recorder in Lemma 5.2, with frontier initially at two and all marks initially zero. In the present notation its controls are S_q, U_r, W_r, C_q, D_q (ready, work-move completed, right scan, frontier extension, left scan), and its full symbols are (a,g,m) with history $g \in \{E,F\}\sqcup\{\widehat{r}\}$. The first row writes and moves, the second marks and starts the scan, and the last clears the mark and stays. These are exactly the seven rows already proved, after renaming. Thus the new work head j' is less than $n+2$, a work step takes $2(n+2-j')+4$ local steps, and both full-domain incoming properties hold.

Let M count the partial controls, let $H = S_h$, and put

$$\ell = \frac{1}{100(M+1)}, \qquad o_H = (3/4,1/2), \qquad o_{s_i} = (1/4+2i\ell,1/2)\quad (0\le i<M-1), \qquad z_* = 1/4.$$

In the state frames $o_s + \ell[0,1]^2$, use odd digits in base $B = 2|\Sigma|+1$ for the full alphabet Σ. The minimal left-split subdivision in Lemma 6.2 gives separately separated rectangles E_i^-, E_i^+, centers c_i^-, c_i^+, and maps

$$y \longmapsto c_i^+ + \operatorname{diag}(a_i, a_i^{-1})(y - c_i^-), \qquad a_i \in \{B^{-1}, 1, B\}.$$

The common frame scale leaves these linear parts unchanged; all rectangle side lengths are at most ℓ.

For N branches set $z_i = 1/2 + i/(4(N+1))$, $1 \le i \le N$. Let δ be one tenth of the minimum of $1/100$, all pairwise source gaps, all pairwise target gaps and all $|z_i - z_j|$; omit empty pair lists. For a box $D = \prod_j [l_j, r_j]$, allowing zero widths, define

$$\chi_D^\delta(x) = \prod_{j=1}^{3} \Theta(2(x_j - l_j + \delta)/\delta)\, \Theta(2(r_j + \delta - x_j)/\delta)\,.$$

It equals one on the $\delta/2$ padding and is supported in the δ padding. Put $\theta(s) = \Theta(3s-1)$. On the first third of a unit step use

$$z_i(s) = z_* + (z_i - z_*)\theta(3s), \qquad D_i(s) = E_i^- \times \{z_i(s)\}, \qquad P_i = \chi^\delta_{D_i(s)}(0, \dot z_i(s) x_1, 0).$$

Its curl on the plateau is $(0, 0, \dot z_i)$; source gaps separate these supports. Hence every point of the whole source plate lifts with planar coordinates unchanged.

On the middle third set

$$\eta = \theta(3s-1), \quad c_i = (1-\eta)c_i^- + \eta c_i^+, \quad \mu_i = 1 - \eta + \eta a_i,$$

$$D_i(s) = [c_i + \operatorname{diag}(\mu_i, \mu_i^{-1})(E_i^- - c_i^-)] \times \{z_i\}.$$

With $\gamma_i = \dot\mu_i/\mu_i$, use

$$P_i = \chi^\delta_{D_i(s)}(0, 0, \Psi_i), \qquad \Psi_i = \dot c_{i1}(x_2 - c_{i2}) - \dot c_{i2}(x_1 - c_{i1}) + \gamma_i (x_1 - c_{i1})(x_2 - c_{i2}).$$

Its plateau curl is $(\dot c_{i1} + \gamma_i(x_1 - c_{i1}), \dot c_{i2} - \gamma_i(x_2 - c_{i2}), 0)$, which is precisely the reciprocal-scale affine velocity. The distinct heights separate the supports during this phase. In the last third use the vertical potential again, with E_i^+ and $z_i(s) = z_i + (z_* - z_i)\theta(3s-2)$; target gaps separate the descents. Summing the curls gives the required field in each phase, with zero collars at every join.

All supports lie strictly inside the unit cube: horizontal centers lie in $[1/4, 3/4+\ell] \times [1/2, 1/2+\ell]$, half-widths are at most $\ell/2$, heights lie between $1/4$ and $3/4$, and $\delta \le 10^{-3}$. Thus the field extends smoothly and with zero mean to the torus. The construction acts on neighborhoods, not only plates. An initial perturbation of max norm less than $\delta/(4B)$ stays within $\delta/4$ of its affine plate path: vertical phases preserve the perturbation, and both horizontal scale factors are at most B. It remains on the plateau, so uniqueness of the smooth ODE gives that same affine continuation throughout.

Write the rational initial code as (b, z_*). During one unit of time load the fixed particle using

$$c_0(t) = (1 - \theta(t))(1/4, 1/4) + \theta(t) b$$

and the curl of

$$\chi^{1/100}_{\{(c_0(t), z_*)\}}(0, 0, \dot c_{01}(x_2 - c_{02}) - \dot c_{02}(x_1 - c_{01})).$$

This has plateau velocity $(\dot c_0, 0)$, support inside the cube, and zero endpoint collars. Repeat the step field thereafter and define its residual force at ν. The finite compact phase formulas give effective mixed-derivative bounds; the only scale denominators satisfy $\mu_i \ge \min(1, a_i) > 0$. Curls have zero mean, and incompressibility gives zero mean for the residual. Lemma 7.3 supplies the unique zero-data PDE solution and its global trajectories.

At times $1+k$, induction through the full branch maps gives the exact local code through the first halt, or for all k in the nonhalting case. A halt code has first coordinate in $[3/4, 3/4+\ell] \subset (2/3, 5/6)$, including a code reached solely by loading an initially halted machine. For a nonhalting input the loader stays below $1/2$ in first coordinate. Lift and descent retain the nonhalting planar endpoint

codes. In the middle phase the centers are convex combinations of nonhalting centers and the first half-width is at most $\ell/2$, so

$$X_1 \le 1/4 + 2M\ell + \ell/2 < 1/2.$$

This excludes the height-independent observer at every intermediate time. Only the loader depends on the input; the repeated field depends on the finite machine table. □

14.3 A permanent origin marker and moving cutoff neighborhoods

Theorem 14.4 (A height-separated mechanism with an origin track)**.** *On the unit torus, the fixed label* $(1/8, 3/8, 1/4)$ *and strip* $2/3 < x_1 < 7/8$ *realize halting by a smooth mean-zero general force, periodic after time one, with zero initial velocity and pressure and bounded mixed derivatives. A permanent tape track identifies the original absolute origin. For a fixed machine the repeated field is independent of the input.*

Proof. Normalize the work machine to one halting state as in Section 5, preserving initial halting. Use four tracks: work, head mark, history, and an origin bit. The history is empty except for a frontier at cell two; the origin bit is one only at absolute cell zero. In a ready control S_q, the head mark is one at the work head. At intermediate controls use the seven-row recorder unchanged, with the origin track copied in every row. The first row now requires ready mark one and writes mark zero at the old head before the work move. The final return row leaves mark one when entering S_q. These are the only two changes to the seven rows.

For precision, this table is the conjugate of the frontier table on its full configuration domain by the following involution: toggle the mark at the current head exactly when the control is ready, and leave it unchanged at every other control. Also take the direct product with the passive origin track. This is a bijection of all configurations, not merely of initialized ones. It proves the full-domain inverse. The incoming-rule properties also persist: into the work-tag control the first-row output mark is zero; into the right scan the marking row and its loop still write one and zero; into the return scan frontier and nonfrontier outputs remain distinct; into the ready control only the last row applies. The initial configuration is the conjugate of the frontier initialization. Consequently its cycle length is $2(2 + n - h') + 4$, its absolute frontier is $2 + n$, and its origin bit remains fixed at the absolute origin. This verifies the distinct initialization without another appeal to an existential compiler.

Denote the resulting full four-track alphabet by $\mathcal{A}$, and let $K = 2|\mathcal{A}| + 1$ be its odd-digit radix. For m controls take state-square side $r = 1/[16(m + 1)]$, with nonterminal lower corners $(1/8 + 2(i - 1)r, 1/8)$ and terminal lower corner $(3/4, 1/8)$. Their common coding height is $z_0 = 1/4$. Lemma 5.1 supplies separated full planar rectangles; let N be the number of branches. Give branch i height $h_i = 1/2 + i/[4(N + 1)]$. During a first phase lift its center from z_0 to h_i, keeping its planar shape; during a middle phase interpolate its planar center, with width $w_i = (1 - \mu)w_i^- + \mu w_i^+$ and height $v_i = w_i^- v_i^-/w_i$; during a final phase lower it to z_0. The planar first coordinate is again an endpoint convex combination. Choose a fixed positive padding less than one tenth of all source and target gaps, all chart-face margins and all height separations. The moving padded rectangles are disjoint at every phase. During the middle phase this follows from the different heights; during the first and last phases it follows from endpoint planar separation.

Here is a direct potential for these moving neighborhoods. With moving center $c = (c_x, c_y, c_z)$ and $\lambda = \dot{w}/w = -\dot{v}/v$, set

$$A = (0, \dot{c}_z(x - c_x), \dot{c}_x(y - c_y) - \dot{c}_y(x - c_x) + \lambda(x - c_x)(y - c_y)).$$

Its curl on the plateau is $(\dot{c}_x + \lambda(x - c_x), \dot{c}_y - \lambda(y - c_y), \dot{c}_z)$. Multiply it by a cutoff supported in the moving padded neighborhood and sum the curls. All supports remain in the open cube. The displayed center-and-shape paths solve this field on the full planar rectangles; uniqueness proves their exact endpoint maps. The positive padding and bounded planar scales also give a uniform initial neighborhood that stays on every plateau, by the perturbation argument of Lemma 3.1. Each component of the total field has zero mean after periodization, since the entire field is a curl.

For initialization, choose equal small squares about the fixed label and the rational input code at height z_0, with half-side less than one quarter of their minimum chart-face margin. Move one to the other by the same localized potential for translation during $[0, 1]$. A nonterminal loading first coordinate lies between $1/8$ and $1/4$, and every subsequent nonterminal phase lies below $1/4$. A terminal endpoint lies in the detector. The initialized symbolic equivalence, including initial halting, therefore holds at all real times. Repeating the finite middle mechanism and applying Lemma 6.3 gives the asserted force and comparison properties. Only the loading translation and its input code depend on the finite word. □

15 Clock profiles for selected processors

Lemma 9.1 gives the common logarithmic change of time. We now apply it to two Euclidean processors with different pressure classes, and then use a companion's autonomous spatial clock to obtain stationary or decaying forcing on the torus. In every case the clock traverses every finite computational time; periodic, stationary and decaying profiles remain separate choices.

15.1 Preserving the declared comparison classes

The two templates below use different whole-space pressure assumptions. Slowing their trajectories preserves those declared comparison classes because the new reference velocity retains the required regularity on every finite physical-time interval.

Proposition 15.1. *The template of Theorem 11.3 admits a force in $L^2([0,\infty)\times\mathbb{R}^3)$ with the same material label and event. The template of Theorem 11.4 admits a force in $L^2([0,\infty);H^j(\mathbb{R}^3))$ for every integer $j \geq 0$. Both retain uniform compact support, zero initial velocity, bounded mixed derivatives, their stated pressure and velocity comparison classes, and finite effective prescriptions.*

Proof. Apply Lemma 9.1 to each template, including its loading interval if present. Both are initially zero, have one compact support, and have uniformly bounded mixed derivatives. The lemma gives the exact residual and all mixed bounds; fixed support also gives $\|f(t)\|_{H^j} \leq C_j(1+t)^{-1}$ for every fixed j.

On each finite interval the slowed reference field retains every Sobolev regularity and bounded derivative required by its original comparison class. Case (i) of Lemma 2.2 therefore applies to the direction shuttle's continuous H^1 pressure representative, and case (iii) applies to the guarded-history processor's continuous L^2 pressure gradient. Both constructed pressures are zero.

The new path is $X(\log(1+t); a_0)$. The clock lemma preserves the event and the effective prescription. In particular, loaded samples $1+k$ occur at physical times $e^{1+k}-1$, whereas the aligned construction's samples k occur at $e^k - 1$. These times have no finite accumulation. The new forces have the stated temporal integrability; no periodicity in physical time is asserted for them. □

15.2 The input-offset alternative

A fixed material label can also be retained by shifting the state charts by the initial tape code. The complete planar routing and spatial-clock construction for this choice belongs to [30, Theorem 4.1 and Section 5]. The following application fixes its match with the present conventions.

Corollary 15.2 (Two profiles of the input-offset processor)**.** *On $\mathbb{T}^3$, the fixed label $P = (1/4, 1/4, 0)$ and strip $O = \{1/32 < Y < 1/8\}$ realize halting from zero velocity by two effective mean-zero general forces. One is stationary after time one. The other tends to zero uniformly and has every mixed-derivative spatial supremum norm in $L^2(0, \infty)$. Both have bounded mixed derivatives, zero prescribed pressure, uniformly bounded kinetic energy, and uniqueness in the torus comparison class.*

Proof. Add the companion's fresh nonhalting dummy start and use its seven-row compiler. The permutation of (work, mark, history) into our track order identifies its table with Lemma 5.2, including the full-domain inverse. Let K be the full compiler alphabet size, m the number of controls, and N the number of full branches. With odd digits in base $B = 2K + 2$ and initial rational stack coordinates x_0, y_0, its state-square parameters are

$$\lambda = \frac{1}{100(m + N + 1)}, \qquad a_* = 1/4 - \lambda x_0, \qquad b_{\mathrm{start}} = 1/4 - \lambda y_0. \tag{48}$$

The halt control has lower height $1/16$; enumerate the controls other than start and halt by ℓ and give them lower heights $1/3 + \ell/[12(m + 1)]$, as in that theorem. Thus the initial code is $(1/4, 1/4)$, and the common full branch formulas of Lemma 5.1 are precisely its inputs. The cited theorem gives the full closed-rectangle routing, positive cutoff neighborhoods, and the all-time guard in $J_* = [1/8, 7/8] \times [1/32, 7/8]$. Write $V(s, X, Y)$ for its planar velocity, one-periodic in s. Choose χ_* smooth, compactly supported in the open unit-square chart and equal to one near J_*. The autonomous field from the companion's Section 5 is

$$W = (V_1(Z, X, Y), V_2(Z, X, Y), \partial_X(X\chi_*(X, Y))), \tag{49}$$

with the compactly supported products extended periodically. It is solenoidal and mean zero; its phase speed is one along the tracked path. Its trajectory from P enters O exactly upon halting. These are the specific geometric imports, rather than an identification based only on the headline conclusion.

Take the stationary and slow profiles from [30, Section 5 and Proposition 6.1], respectively:

$$\alpha_0 = \sigma(2t - 1), \qquad \alpha_1 = \sigma(2t - 1)/(1 + t). \tag{50}$$

Their exact force formula at the fixed physical viscosity is

$$U_\alpha = \alpha W, \qquad f_\alpha = \alpha' W + \alpha^2 (W \cdot \nabla) W - \nu \alpha \Delta W. \tag{51}$$

Lemma 2.2 supplies the stated comparison conclusion. For $s_\alpha(t) = \int_0^t \alpha(r)\, dr$, symmetry of σ gives $s_{\alpha_0}(1) = 1/4$ and $s_{\alpha_0}(t) = t - 3/4$ for $t \geq 1$. The other clock satisfies $s_{\alpha_1}(t) = s_{\alpha_1}(1) + \log((1 + t)/2)$ there. Both are continuous, finite at finite time and onto $[0, \infty)$, so both preserve the complete trajectory event, including original initial halting through the dummy start. The first force is stationary after one. In the second, the coefficient orders are respectively $(1 + t)^{-2}, (1 + t)^{-2}$ and $(1 + t)^{-1}$; each time derivative gives an additional inverse power. The spatial factors in (51) have effective bounds at every order. Thus for every $h, r \geq 0$,

$$\|\partial_t^h U_{\alpha_1}(t)\|_{C^r} + \|\partial_t^h f_{\alpha_1}(t)\|_{C^r} \leq C_{h,r}(1 + t)^{-1-h}. \tag{52}$$

The constant is effective; increasing it over $[0,1]$ makes the estimate valid for all $t \geq 0$. The exponent is common to all spatial orders, and the square of the envelope has integral $(1+2h)^{-1}$. This proves the asserted bounded and temporal L^2 norms.

For the stationary profile, Proposition 2.4 preserves stationarity after time one and every mixed-derivative bound. For the slow profile, (2) and (4) apply to (52) through spatial order $r+2$. They retain the rate $(1+t)^{-1-h}$ for the projected force through order r and for its pressure through order $r+1$, with effective constants and temporal L^2 bounds. In either projection choice the velocity and event are unchanged and the mean-zero pressure is $-\phi$. The stationary and slow profiles remain separate choices. □

References

[1] Donald G. Aronson. Non-negative solutions of linear parabolic equations. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*, 22(4):607–694, 1968.

[2] Holger Bock Axelsen and Robert Glück. What do reversible programs compute? *Lecture Notes in Computer Science*, 6604:42–56, 2011.

[3] Augustin Banyaga. Sur la structure du groupe des difféomorphismes qui préservent une forme symplectique. *Commentarii Mathematici Helvetici*, 53(1):174–227, 1978.

[4] Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.

[5] Robert Cardona. Hydrodynamic and symbolic models of computation with advice. *Revista Matemática Iberoamericana*, 41(1):313–338, 2025.

[6] Robert Cardona, Eva Miranda, and Daniel Peralta-Salas. Turing universality of the incompressible Euler equations and a conjecture of Moore. *International Mathematics Research Notices*, 2022(22):18092–18109, 2022.

[7] Robert Cardona, Eva Miranda, and Daniel Peralta-Salas. Computability and Beltrami fields in Euclidean space. *Journal de Mathématiques Pures et Appliquées*, 169:50–81, 2023.

[8] Robert Cardona, Eva Miranda, and Daniel Peralta-Salas. Towards a fluid computer. *Foundations of Computational Mathematics*, 25:1921–1937, 2025.

[9] Robert Cardona, Eva Miranda, Daniel Peralta-Salas, and Francisco Presas. Constructing Turing complete Euler flows in dimension 3. *Proceedings of the National Academy of Sciences*, 118(19):e2026818118, 2021.

[10] Robert Cardona, Eva Miranda, Daniel Peralta-Salas, and Francisco Presas. Universality of Euler flows and flexibility of Reeb embeddings. *Advances in Mathematics*, 428:109142, 2023.

[11] Peter Constantin and Ciprian Foias. *Navier–Stokes Equations*. University of Chicago Press, Chicago, 1988.

[12] Søren Dyhr, Ángel González-Prieto, Eva Miranda, and Daniel Peralta-Salas. Turing complete Navier–Stokes steady states via cosymplectic geometry. *PNAS Nexus*, 5(5):pgag131, 2026.

[13] John Etnyre and Robert Ghrist. Contact topology and hydrodynamics: I. Beltrami fields and the Seifert conjecture. *Nonlinearity*, 13(2):441–458, 2000.

[14] Giovanni P. Galdi. *An Introduction to the Mathematical Theory of the Navier–Stokes Equations: Steady-State Problems*. Springer, New York, 2 edition, 2011.

[15] T. H. Gronwall. Note on the derivatives with respect to a parameter of the solutions of a system of differential equations. *Annals of Mathematics*, 20(4):292–296, 1919.

[16] Pascal Koiran, Michel Cosnard, and Max Garzon. Computability with low-dimensional dynamical systems. *Theoretical Computer Science*, 132(1–2):113–128, 1994.

[17] A. B. Krygin. Continuation of diffeomorphisms retaining volume. *Functional Analysis and Its Applications*, 5(2):147–150, 1971.

[18] O. A. Ladyzhenskaya. *The Mathematical Theory of Viscous Incompressible Flow*. Gordon and Breach, New York, 2 edition, 1969.

[19] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

[20] Jean Leray. Sur le mouvement d'un liquide visqueux emplissant l'espace. *Acta Mathematica*, 63:193–248, 1934.

[21] Marvin L. Minsky. Recursive unsolvability of Post's problem of "tag" and other topics in theory of Turing machines. *Annals of Mathematics*, 74(3):437–455, 1961.

[22] Marvin L. Minsky. *Computation: Finite and Infinite Machines*. Prentice-Hall, Englewood Cliffs, NJ, 1967.

[23] Cristopher Moore. Unpredictability and undecidability in dynamical systems. *Physical Review Letters*, 64(20):2354–2357, 1990.

[24] Cristopher Moore. Generalized shifts: unpredictability and undecidability in dynamical systems. *Nonlinearity*, 4(2):199–230, 1991.

[25] Kenichi Morita. *Theory of Reversible Computing*. Springer, Tokyo, 2017.

[26] Jürgen Moser. On the volume elements on a manifold. *Transactions of the American Mathematical Society*, 120:286–294, 1965.

[27] John Nash. Continuity of solutions of parabolic and elliptic equations. *American Journal of Mathematics*, 80(4):931–954, 1958.

[28] OpenAI. Finite Instructions and Solenoidal Shear Flows. OpenAI Math Release preprint OAI:Finite-Instructions-and-Solenoidal-Shear-Flows-September-27-2026, 2026.

[29] OpenAI. Prefix Instructions and Incompressible Flows. OpenAI Math Release preprint OAI:Prefix-Instructions-and-Incompressible-Flows-September-27-2026, 2026.

[30] OpenAI. Universal Computation with Eventually Stationary Navier–Stokes Forcing. OpenAI Math Release preprint OAI:Universal-Computation-with-Eventually-Stationary-Navier-Stokes-Forcing-September-27-2026, 2026.

[31] OpenAI. Velocity-Field Detection of Computation in Forced Navier–Stokes Flows. OpenAI Math Release preprint OAI:Velocity-Field-Detection-of-Computation-in-Forced-Navier-Stokes-Flows-September-27-2026, 2026.

[32] Marian B. Pour-El and J. Ian Richards. *Computability in Analysis and Physics*. Springer-Verlag, Berlin, 1989.

[33] Giovanni Prodi. Un teorema di unicità per le equazioni di Navier–Stokes. *Annali di Matematica Pura ed Applicata*, 48:173–182, 1959.

[34] James Serrin. On the interior regularity of weak solutions of the Navier–Stokes equations. *Archive for Rational Mechanics and Analysis*, 9:187–195, 1962.

[35] James Serrin. The initial value problem for the Navier–Stokes equations. In Rudolph E. Langer, editor, *Nonlinear Problems*, pages 69–98. University of Wisconsin Press, Madison, 1963.

[36] Hava T. Siegelmann and Eduardo D. Sontag. On the computational power of neural nets. *Journal of Computer and System Sciences*, 50(1):132–150, 1995.

[37] Terence Tao. Finite time blowup for an averaged three-dimensional Navier–Stokes equation. *Journal of the American Mathematical Society*, 29(3):601–674, 2016.

[38] A. M. Turing. On computable numbers, with an application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society*, 42:230–265, 1936–1937.

[39] Hermann von Helmholtz. Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen. *Journal für die reine und angewandte Mathematik*, 55:25–55, 1858.

[40] Klaus Weihrauch. *Computable Analysis: An Introduction*. Springer-Verlag, Berlin, 2000.

[41] Hermann Weyl. The method of orthogonal projection in potential theory. *Duke Mathematical Journal*, 7(1):411–444, 1940.